GSSSB · STAFF NURSE EXAM

Medical-Surgical Nursing (MSN)

Complete Colourful Notes · Diagrams · Mnemonics · MCQs



20 Chapters | 15 Coloured Diagrams | Nursing Management | Normal Values | Drugs & Antidotes | Practice MCQs + Answers

Hinglish explanation + English exam terms



Sister Asha
Aapki MSN teacher

Index – Kya padhenge?

1 **Introduction to MSN & Nursing Process**
Health-illness, roles of nurse, nursing process (ADPIE), NANDA, documentation, positions

2 **Fluid, Electrolyte & Acid-Base Balance**
Body fluids, IV fluids, Na, K, Ca, Mg imbalances, ABG analysis, I/O chart

3 **Shock, Pain, Inflammation & Infection Control**
Types & stages of shock, pain scales & WHO ladder, inflammation, wound healing, asepsis, BMW

4 **Perioperative Nursing**
Pre-op preparation, anaesthesia, OT team, intra-op safety, post-op care & complications

5 **Cardiovascular System Disorders**
Heart anatomy, ECG, CAD, angina, MI, heart failure, HTN, valvular & inflammatory diseases, DVT, cardiac drugs

6 **Respiratory System Disorders**
Anatomy, assessment, O2 therapy, COPD, asthma, pneumonia, TB, pleural problems, chest tube, ARDS, lung cancer

7 **Gastrointestinal System Disorders**
GI anatomy, GERD, peptic ulcer, GI bleed, appendicitis, IBD, obstruction, hernia, colorectal cancer, stoma care, NG tube

8 **Liver, Biliary System & Pancreas**
LFT, jaundice, hepatitis A-E, cirrhosis, varices, hepatic encephalopathy, gallstones, pancreatitis

9 **Renal, Urinary & Male Reproductive Disorders**
Nephron, RFT, UTI, calculi, AKI, CKD, dialysis, glomerulonephritis, nephrotic syndrome, BPH, catheter care

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Glands & hormones, diabetes (DKA, HHS, hypoglycaemia, insulin), thyroid, parathyroid, adrenal, pituitary

11 **Neurological System Disorders**
Brain lobes, cranial nerves, GCS, ICP, head injury, stroke, seizures, meningitis, spinal cord injury, Parkinson, MG, GBS

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18 **Emergency, Critical Care & Disaster Nursing**
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19 **Geriatric Nursing & Quick Revision**
Ageing changes, falls, delirium vs dementia; master tables – lab values, drugs, tests, signs, days, 100 one-liners

20 **MSN Grand Test – 60 Mixed MCQs**
Full-syllabus mock test in GSSSB pattern with answer key

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – exam me baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Introduction to MSN & Nursing Process

1.1 Medical-Surgical Nursing kya hai?

D Definition

Medical-Surgical Nursing (MSN) = Adult patients (18+ years) ki nursing care jo **medical illness** (dawai se treat) ya **surgical condition** (operation se treat) ke kaaran hospital me hain. Isme health promotion, illness prevention, cure aur rehabilitation sab aata hai.

- **Medical** = Diagnosis & treatment by drugs, diet, rest (e.g. Diabetes, Hypertension, Pneumonia)
- **Surgical** = Treatment by operation (e.g. Appendectomy, Cholecystectomy, Fracture fixation)
- Founder of modern nursing: **Florence Nightingale** (Lady with the Lamp, born **12 May 1820**) → 12 May = **International Nurses Day**.

Health & Illness – important definitions

Term	Meaning
Health (WHO 1948)	"A state of complete physical, mental and social well-being and not merely the absence of disease or infirmity"
Disease	Body ke structure/function me abnormal change (objective)
Illness	Patient ki apni feeling ki wo bimar hai (subjective)
Acute illness	Achanak, short duration (< 6 months), e.g. appendicitis
Chronic illness	Dheere, long duration (> 6 months), e.g. DM, CKD
Incidence	**Naye** cases in a time period
Prevalence	**Total** (old + new) cases at a time
Prognosis	Disease ka expected outcome
Etiology	Disease ka cause
Iatrogenic	Treatment / health worker ki wajah se hui problem
Nosocomial	Hospital acquired infection (48 hrs after admission)
Idiopathic	Cause unknown

Levels of Prevention (Leavell & Clark)

Level	Goal	Example
Primordial	Risk factor hi na aaye	Healthy lifestyle in children
Primary	Disease hone se pehle rokna	Immunization, health education, chemoprophylaxis
Secondary	Early diagnosis & prompt treatment	Screening (Pap smear, BP check, mammography)
Tertiary	Disability limitation & rehabilitation	Physiotherapy after stroke, prosthesis

1.2 Roles of Medical-Surgical Nurse

- **Care giver** – direct bedside care
- **Educator** – patient & family ko health teaching
- **Advocate** – patient ke rights ki raksha
- **Counsellor**, **Manager/Coordinator**, **Researcher**, **Change agent**, **Leader**

1.3 Nursing Process – ADPIE

ADPIE (5 steps)

Assessment → **D**iagnosis → **P**lanning → **I**mplementation → **E**valuation
Nursing process ka concept: **Lydia Hall (1955)** ne diya. 5 steps me nursing diagnosis step **NANDA** (1973) se aaya.

Step 1 – Assessment (data collection)

- **Subjective data** = patient jo **bolta** hai (symptoms) – e.g. "mujhe dard ho raha hai"
- **Objective data** = nurse jo **dekhti/measure** karti hai (signs) – e.g. BP 150/90, wound redness
- **Primary source** = patient khud (best source); **Secondary** = family, records, lab reports
- Methods: Interview, Observation, Physical examination

Physical Examination ka order

Inspection → **P**alpation → **P**ercussion → **A**uscultation (IPPA)
Abdomen me exception: Inspection → **Auscultation** → Percussion → Palpation (taaki palpation se bowel sounds change na ho).

Step 2 – Nursing Diagnosis (NANDA-I)

- Nursing diagnosis = patient ka **response** to health problem (doctor ki tarah disease ka naam nahi).
- **PES format**: **P**roblem + **E**tiology ("related to") + **S**igns/Symptoms ("as evidenced by")
- Example: **Acute pain related to surgical incision as evidenced by pain score 7/10 and facial grimacing.**

Type	Example
Actual	Impaired gas exchange r/t … AEB …
Risk	Risk for infection r/t invasive lines (no AEB part)
Health promotion / Wellness	Readiness for enhanced nutrition
Syndrome	Rape-trauma syndrome

Doctor vs Nurse diagnosis

Medical diagnosis = **disease** (e.g. Pneumonia) – treatment ke saath badalta nahi.
Nursing diagnosis = **patient ka response** (e.g. Ineffective airway clearance) – din me bhi badal sakta hai.

Step 3 – Planning

- **Priority setting** – **Maslow's Hierarchy** use karo: Physiological → Safety → Love/belonging → Esteem → Self-actualization.
- **ABC rule**: **A**irway → **B**reathing → **C**irculation (sabse pehle airway!)
- Goals **SMART** hone chahiye: **S**pecific, **M**easurable, **A**chievable, **R**ealistic, **T**ime-bound.

Step 4 – Implementation

- **Independent** actions – nurse khud (positioning, health education, turning)
- **Dependent** – doctor ke order par (medicine, IV fluid)
- **Interdependent / Collaborative** – team ke saath (physiotherapist, dietician)

Step 5 – Evaluation

Goal met / partially met / not met → plan revise karo. Nursing process **cyclic & dynamic** hai.



Maslow & ABC ka exam me use

Jab question pooche "**Nurse ki first / priority action kya hai?**" – to hamesha socho: (1) Airway, (2) Breathing, (3) Circulation, (4) Safety, phir comfort, phir teaching. Assessment hamesha intervention se pehle aata hai – jab tak emergency (jaise cardiac arrest) na ho.

1.4 Vital Signs – Normal Values (Adult)

Vital sign	Normal	Abnormal terms
Temperature (oral)	36.5 – 37.5 °C (97.7–99.5 °F)	Pyrexia >37.5 °C; Hyperpyrexia >41 °C; Hypothermia <35 °C
Pulse	60 – 100 /min	Tachycardia >100; Bradycardia <60
Respiration	12 – 20 /min	Tachypnoea >20; Bradypnoea <12; Apnoea = no breathing
BP	<120/<80 mmHg	Hypertension ≥140/90 (JNC-7)
SpO_2	95 – 100 %	Hypoxaemia <90 %
Pain	"5th vital sign"	Scale 0–10

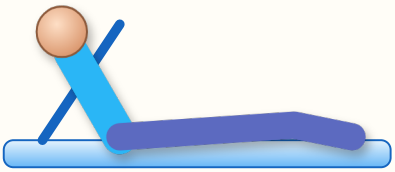
- **Pulse pressure** = Systolic − Diastolic (normal ~40 mmHg)
- **MAP (Mean Arterial Pressure)** = (SBP + 2 × DBP) ÷ 3 → normal **70–100 mmHg**; organ perfusion ke liye ≥ 65 chahiye.
- Pulse lene ke liye best site: **Radial**; cardiac arrest me **Carotid**; infants me **Brachial**; apex beat: **Left 5th intercostal space, mid-clavicular line** (1 full minute).
- Rectal temperature oral se **~0.5 °C zyada**, axillary **~0.5 °C kam**.

Exam me aksar

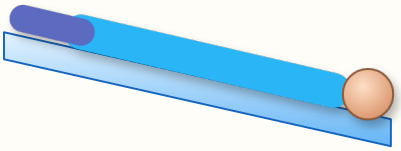
- **Pulse deficit** = Apical rate − Radial rate (Atrial fibrillation me milta hai). Do nurses ek saath count karti hain.
- **Cheyne-Stokes breathing** = deep-shallow-apnoea cycle → near death, heart failure, ↑ICP.
- **Kussmaul breathing** = deep, rapid → **DKA** (metabolic acidosis).
- **Biot's breathing** = irregular with apnoea → meningitis, brain damage.

1.5 Important Patient Positions

Important Patient Positions



Fowler's (45-60°) / High (90°)
Dyspnoea, HF, COPD, post-thyroid



Trendelenburg (head low)
Pelvic surgery; (shock me ab NOT preferred)



Sim's (left lateral, R knee bent)
Enema, rectal exam, unconscious



Supine / Dorsal recumbent
Spinal injury, post spinal anaes.



Prone (pet ke bal)
ARDS (proning), back surgery



Lithotomy (legs in stirrups)
Vaginal exam, delivery, TURP, cysto

More yaad rakho:
Knee-chest → sigmoidoscopy, cord prolapse · Orthopneic (table pe jhukna) → severe dyspnoea
Left lateral → enema, liver biopsy ke baad RIGHT side · Semi-Fowler (30°) → ↑ICP, NG feed, post-op
Modified Trendelenburg (sirf legs up) → shock · Pneumonectomy → operated side down / supine

Important positions – kab kaunsa use karein

Positions – quick list

- **Fowler's** (45–60°) / **High Fowler's** (90°) → breathlessness, heart failure, after thyroidectomy, NG insertion
- **Semi-Fowler's** (30°) → raised ICP, post-abdominal surgery, enteral feeding (aspiration rokne ke liye)
- **Supine** → post spinal anaesthesia (headache rokne ke liye flat), spinal injury
- **Prone** → ARDS ("proning"), after lumbar spine surgery
- **Left lateral / Sim's** → enema, rectal examination, unconscious patient (aspiration rokta hai)
- **Right lateral** → **after liver biopsy** (pressure se bleeding rukti hai)
- **Lithotomy** → vaginal exam, delivery, TURP, cystoscopy
- **Knee-chest** → sigmoidoscopy, cord prolapse
- **Trendelenburg** (head down) → pelvic surgery; shock me ab **modified Trendelenburg** (sirf legs 20–30° up)
- **Reverse Trendelenburg** → head up – hiatus hernia, GERD

1.6 Documentation & Records

- Rule: **"Not documented = Not done."**
- Ink pen se likho, error par **single line** kheecho, "error" likh ke sign karo; **white-ink / overwriting nahi**.
- **SOAP** = Subjective, Objective, Assessment, Plan (SOAPIE me + Intervention, Evaluation)

- **SBAR** (handover) = **S**ituation, **B**ackground, **A**ssessment, **R**ecommendation
- **Kardex** = quick reference card of patient care plan
- **Incident report** = koi accident/error hone par (patient file me nahi lagate)
- **Informed consent** – doctor ki responsibility, nurse **witness** banti hai. Patient adult (18+), conscious, sedation-free hona chahiye; minor me parent/guardian.

? Practice MCQs – Chapter 1

1. Nursing process ka correct order hai –
(a) Planning, Assessment, Diagnosis, Evaluation, Implementation
(b) Assessment, Diagnosis, Planning, Implementation, Evaluation
(c) Diagnosis, Assessment, Planning, Evaluation, Implementation
(d) Assessment, Planning, Diagnosis, Implementation, Evaluation

2. Abdomen examination ka correct sequence hai –
(a) IPPA (b) Auscultation first (c) Inspection, Auscultation, Percussion, Palpation (d) Palpation first

3. "Patient says she feels nauseated" – ye kis type ka data hai?
(a) Secondary (b) Tertiary (c) Objective (d) Subjective

4. Normal MAP kitna hota hai?
(a) 40–60 (b) 70–100 (c) 110–130 (d) 140–160 mmHg

5. Pap smear screening kis level of prevention me aata hai?
(a) Secondary (b) Tertiary (c) Primary (d) Primordial

6. Liver biopsy ke baad patient ko kis position me rakhte hain?
(a) Prone (b) Left lateral (c) Fowler's (d) Right lateral

7. Kussmaul breathing kis condition me milta hai?
(a) Metabolic alkalosis (b) Stroke (c) Asthma (d) DKA

8. Informed consent lene ki primary responsibility kiski hai?
(a) Nurse (b) Relative (c) Doctor/Surgeon (d) Ward boy

9. "Risk for infection" type of nursing diagnosis me kaunsa part nahi hota?
(a) Problem (b) Label (c) Signs & symptoms (AEB) (d) Etiology

10. Nursing process ka concept sabse pehle kisne diya?
(a) Florence Nightingale (b) Dorothea Orem (c) Lydia Hall (d) Virginia Henderson

✓ Answer Key – Chapter 1

1-b · 2-c · 3-d · 4-b · 5-a · 6-d · 7-d · 8-c · 9-c · 10-c

2 Fluid, Electrolyte & Acid–Base Balance

2.1 Body Fluid Compartments

Body water – yaad rakho

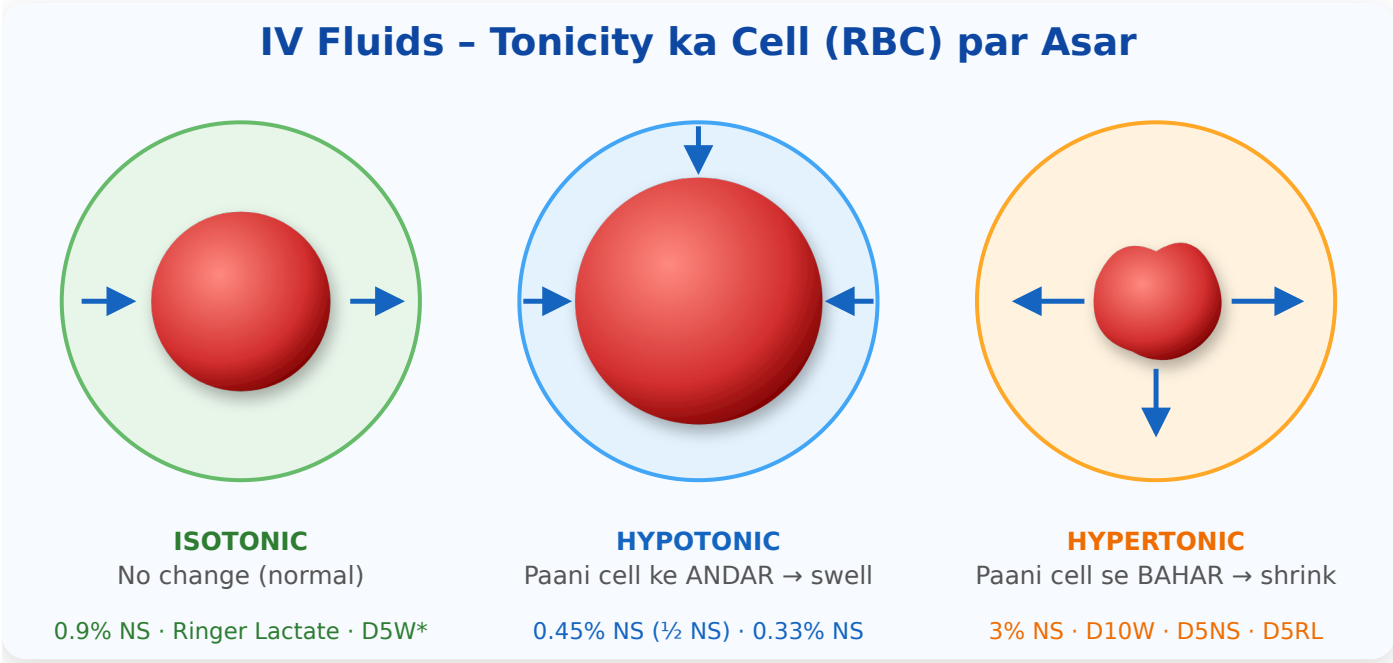
- Adult body weight ka **~60 %** paani (newborn ~75–80 %, elderly ~45–50 %; female me fat zyada → paani kam).
- **ICF (Intracellular fluid)** = 2/3 (40 % body wt) → main ion **K^+** (and PO_4^{3-}, Mg^{2+})
- **ECF (Extracellular fluid)** = 1/3 (20 % body wt) → main ion **Na^+** (and Cl^-, HCO_3^-)
  - ECF = Interstitial (15 %) + Intravascular/plasma (5 %) + Transcellular (CSF, pleural, synovial)
- Normal **daily intake ≈ 2500 ml** = Output (Urine 1500 ml + insensible loss via skin & lungs ~900 ml + faeces 100 ml).
- Normal urine output = **0.5–1 ml/kg/hr** (minimum **30 ml/hr**). <400 ml/day = **Oliguria**; <100 ml/day = **Anuria**; >3000 ml/day = **Polyuria**.

Fluid movement mechanisms

Process	Meaning
Diffusion	Solute high → low concentration (no energy)
Osmosis	**Water** low solute → high solute concentration (semi-permeable membrane)
Filtration	Hydrostatic pressure se fluid ka movement (capillary, glomerulus)
Active transport	Energy (ATP) use, low → high; e.g. **Na^+-K^+ pump** (3 Na out, 2 K in)

- Normal **serum osmolality = 275–295 mOsm/kg**.
- **Oncotic (colloid osmotic) pressure** – mainly **albumin** se; albumin kam → oedema (cirrhosis, nephrotic syndrome).

2.2 IV Fluids



Isotonic, Hypotonic, Hypertonic fluid ka RBC par asar

Type	Examples	Use	Caution
Isotonic (≈ 290 mOsm)	0.9 % NaCl (NS), Ringer's Lactate (RL), D5W*	Hypovolaemia, shock, blood transfusion (**only NS**), burns (RL)	Fluid overload in HF/renal failure
Hypotonic	0.45 % NaCl, 0.33 % NaCl, 2.5 % dextrose	Cellular dehydration, hypernatraemia, DKA (later)	**↑ICP me contraindicated** (cerebral oedema), burns, trauma
Hypertonic	3 % / 5 % NaCl, D10W, D5NS, D5RL, D5½NS	Severe hyponatraemia, cerebral oedema (3 % NS)	Fluid overload, pulmonary oedema; 3 % NS **slowly** & in ICU
Colloids	Albumin 5 %/25 %, Dextran, Hetastarch, Plasma	Plasma volume expander	Allergy, bleeding (dextran)



D5W ka confusion

D5W bag me **isotonic** hota hai, lekin body me dextrose metabolise ho jata hai aur sirf free water bachta hai – isliye **physiologically hypotonic** ki tarah behave karta hai. Isliye ↑ICP / head injury me D5W nahi dete.

! Blood transfusion ke saath

Blood ke saath **sirf 0.9 % Normal Saline** chalao. Dextrose → RBC haemolysis & clumping; RL ka calcium → clotting.

IV flow rate calculation

- **Drops/min** = (Volume in ml × Drop factor) ÷ Time in minutes
- Macro-drip set = **15 or 20 drops/ml**; Micro-drip (paediatric) = **60 microdrops/ml** (micro drops/min = ml/hr)

- **ml/hr** = Total volume ÷ hours
- Example: 1000 ml in 8 hrs, drop factor 20 → (1000 × 20) ÷ 480 = **41.6 ≈ 42 drops/min**

! IV therapy complications

Complication	Signs	Action
Infiltration	Cool, pale, swollen site, flow slow	Stop, remove cannula, **elevate limb**, warm/cold compress as per fluid
Extravasation	Vesicant drug (e.g. chemo, dopamine) leak → tissue necrosis	Stop, **aspirate**, antidote (e.g. phentolamine for dopamine/norepinephrine)
Phlebitis	Red, warm, tender along vein	Remove, **warm compress**, re-site in other limb
Air embolism	Sudden dyspnoea, chest pain, hypotension	Clamp, **LEFT side + Trendelenburg** (Durant's position), O_2
Fluid overload	Crackles, distended neck veins, ↑BP, dyspnoea	Slow IV, **High Fowler's**, diuretic, inform

2.3 Sodium (Na^+) – Normal 135–145 mEq/L

Hyponatraemia (< 135)

Causes: Vomiting, diarrhoea, diuretics, **SIADH**, excess water, adrenal insufficiency, heart failure.
S/S: Headache, confusion, nausea, muscle cramps, **seizures**, coma (brain oedema).
Management:

- Fluid restriction (SIADH)
- Hypertonic saline (3 % NS) in severe cases – **slow correction** (max 8–12 mEq/L in 24 hr)
- Seizure precautions

Hypernatraemia (> 145)

Causes: Dehydration, **Diabetes insipidus**, excess Na intake, hyperventilation.
S/S: Thirst, dry sticky mucosa, restlessness, fever, agitation, seizures.
Management:

- Hypotonic fluids slowly (0.45 % NS / D5W)
- Treat cause (DI → Desmopressin)
- Daily weight, I/O chart

! Fast correction ka khatra

Hyponatraemia ko tezi se theek karna → **Osmotic demyelination syndrome** (Central pontine myelinolysis). Hypernatraemia ko tezi se theek karna → **cerebral oedema**.

2.4 Potassium (K^+) – Normal 3.5–5.0 mEq/L

Hypokalaemia (< 3.5)

Causes: Vomiting, NG suction, **loop/thiazide diuretics** (furosemide), diarrhoea, insulin therapy, alkalosis.
S/S: Muscle weakness, leg cramps, **flat T wave, prominent U wave**, ST depression,

Hyperkalaemia (> 5.0)

Causes: Renal failure, K-sparing diuretics (spironolactone), ACE inhibitors, burns/crush injury, **acidosis**, Addison's disease, haemolysed sample.
S/S: Tall peaked T wave, wide QRS, bradycardia,

constipation/paralytic ileus, ↑risk of **digoxin toxicity**.
Management: K-rich diet (banana, orange, potato, coconut water, spinach); oral KCl; IV KCl.

VF/cardiac arrest, muscle weakness, paresthesia.
Management: Calcium gluconate 10 % IV (**heart protect**), **insulin + dextrose**, salbutamol neb, sodium bicarbonate, **Kayexalate** (sodium polystyrene sulfonate), **dialysis**.

! IV Potassium – Golden rules

- **NEVER IV push / bolus** → cardiac arrest!
- Always **diluted** in IV fluid; peripheral line me max **10 mEq/hr** (ICU monitored up to 20 mEq/hr).
- Check **urine output** first (≥ 30 ml/hr) – "**No pee, no K**".
- Infusion pump use karo; cardiac monitoring.

2.5 Calcium (Ca^{2+}) – Normal 8.5–10.5 mg/dL (ionised 4.5–5.5)

	Hypocalcaemia (< 8.5)	Hypercalcaemia (> 10.5)
Causes	**Post-thyroidectomy / parathyroidectomy**, hypoparathyroidism, Vit D deficiency, pancreatitis, massive blood transfusion (citrate), CKD	Hyperparathyroidism, bone **malignancy**, prolonged immobilization, thiazides, excess Vit D
S/S	**Tetany**, numbness/tingling around mouth, **Chvostek's sign**, **Trousseau's sign**, laryngospasm, seizures, prolonged QT	Muscle weakness, **"Bones, stones, groans, moans"** – bone pain, renal stones, abdominal pain, confusion; shortened QT
Management	**IV Calcium gluconate 10 %** (slowly), oral Ca + Vit D, seizure precautions	**Fluids (NS)** + loop diuretic, calcitonin, bisphosphonates, mobilize patient

★ Chvostek vs Trousseau

- **C**hvostek = **C**heek tap (facial nerve, in front of ear) → facial muscle twitch
- **T**rousseau = BP cuff inflate (above SBP for 3 min) → **carpal spasm** (hand)
  Dono = **hypocalcaemia** (and hypomagnesaemia).

2.6 Magnesium (Mg^{2+}) – Normal 1.5–2.5 mEq/L

- **Hypomagnesaemia**: alcoholism, malnutrition, diarrhoea → tremors, tetany, seizures, arrhythmia (Torsades de pointes). Rx: **Magnesium sulphate** IV.
- **Hypermagnesaemia**: renal failure, excess $MgSO_4$ (e.g. pre-eclampsia treatment) → **loss of deep tendon reflexes**, respiratory depression, hypotension. Antidote: **Calcium gluconate**.

✎ $MgSO_4$ toxicity – monitoring (very common question)

Before each dose check: **Patellar (knee) reflex present**, **RR ≥ 12–16/min**, **Urine output ≥ 30 ml/hr**.
First sign of toxicity = **loss of patellar reflex**. Antidote = **10 % Calcium gluconate 10 ml IV**.

Other values

- **Chloride** 96–106 mEq/L · **Phosphate** 2.5–4.5 mg/dL (Ca ke ulta chalta hai) · **Bicarbonate** 22–26 mEq/L

2.7 Fluid Volume Deficit vs Excess

Fluid Volume DEFICIT (Dehydration / Hypovolaemia)

- Thirst, dry mucosa, **poor skin turgor**
- **Tachycardia, weak thready pulse, hypotension**, postural hypotension
- ↓ urine output, **concentrated urine (↑ specific gravity > 1.030)**
- ↑ Hct, ↑ BUN
- Flat neck veins, weight loss
- Rx: Oral/IV isotonic fluids, I/O, daily weight

Fluid Volume EXCESS (Hypervolaemia)

- **Oedema**, weight gain (1 kg = 1 L fluid)
- **Bounding pulse, hypertension**
- **Distended neck veins (JVD)**, **crackles** in lungs, dyspnoea
- Dilute urine, ↓ Hct, ↓ BUN
- S3 heart sound
- Rx: Fluid & Na restriction, diuretics, **Fowler's**, daily weight



Daily weight – best indicator

Fluid status ka **sabse accurate indicator = daily weight** (same time, same scale, same clothes, preferably morning after voiding). **1 kg weight gain ≈ 1 litre fluid retention.**

2.8 Acid-Base Balance & ABG

Normal ABG values

Parameter	Normal
pH	**7.35 – 7.45**
$PaCO_2$ (respiratory component, acid)	**35 – 45 mmHg**
HCO_3^- (metabolic component, base)	**22 – 26 mEq/L**
PaO_2	80 – 100 mmHg
SaO_2	95 – 100 %
Base excess	−2 to +2

- Body ke 3 buffer systems: (1) **Chemical buffers** (bicarbonate-carbonic acid – fastest, seconds), (2) **Lungs** (minutes – CO_2 exhale), (3) **Kidneys** (hours–days, slowest but most powerful – HCO_3^- retain / H^+ excrete).
- Normal ratio **$HCO_3^- : H_2CO_3 = 20 : 1$**.

ROME – ABG ka trick

Respiratory = **O**pposite (pH ↑ to $PaCO_2$ ↓, pH ↓ to $PaCO_2$ ↑)
Metabolic = **E**qual (pH aur HCO_3^- same direction me)

Disorder	pH	Primary change	Compensation	Common causes	Key signs
Respiratory Acidosis	↓	**$PaCO_2$ ↑**	Kidney ↑HCO_3^-	**COPD**, hypoventilation, opioid/sedative overdose, pneumonia, chest injury	Headache, drowsiness, confusion
Respiratory Alkalosis	↑	**$PaCO_2$ ↓**	Kidney ↓HCO_3^-	**Hyperventilation, anxiety/panic**, high altitude, fever, early salicylate	Light-headed, tingling, **tetany**
Metabolic Acidosis	↓	**HCO_3^- ↓**	Lungs ↓$PaCO_2$ (**Kussmaul**)	**DKA**, renal failure, **diarrhoea**, lactic acidosis (shock), aspirin poisoning	Kussmaul breathing, hyperkalaemia
Metabolic Alkalosis	↑	**HCO_3^- ↑**	Lungs ↑$PaCO_2$ (slow breathing)	**Vomiting, NG suction**, diuretics, antacid excess	Hypokalaemia, muscle cramps

ABG interpretation – 3 steps

1. **pH dekho** → < 7.35 Acidosis, > 7.45 Alkalosis
2. **$PaCO_2$ dekho** → kya pH ke hisaab se **opposite** gaya? → Respiratory cause
3. **HCO_3^- dekho** → kya pH ke saath **same direction** me hai? → Metabolic cause
4. **Compensation**: dusra value bhi abnormal ho aur pH normal range me aa jaye = **fully compensated**; pH abhi bhi abnormal = **partially compensated**; dusra value normal = **uncompensated**.

Solved example

pH 7.30, $PaCO_2$ 55, HCO_3^- 25 → pH ↓ (acidosis), $PaCO_2$ ↑ (opposite = respiratory), HCO_3^- normal → **Uncompensated Respiratory Acidosis** (e.g. COPD exacerbation).
pH 7.48, $PaCO_2$ 40, HCO_3^- 32 → **Uncompensated Metabolic Alkalosis** (e.g. repeated vomiting).



Diarrhoea vs Vomiting

Vomiting → stomach ka acid (HCl) jata hai → **Metabolic Alkalosis**.
Diarrhoea → intestine ka bicarbonate jata hai → **Metabolic Acidosis**.

- **Allen's test** – ABG ke liye radial artery puncture se pehle **ulnar artery ki patency** check karne ke liye. After ABG: **5 min pressure** (anticoagulant par 10 min), sample **ice me** aur 15–30 min me lab bhejo, air bubbles hatao.

Practice MCQs – Chapter 2

1. ICF ka main cation kaunsa hai?
   (a) Ca^{2+} (b) K^+ (c) Na^+ (d) Cl^-
2. Blood transfusion ke saath kaunsa IV fluid compatible hai?
   (a) D10W (b) Ringer's lactate (c) D5W (d) 0.9 % NaCl
3. Hyperkalaemia ka ECG change hai –
   (a) Flat T wave (b) U wave (c) ST depression (d) Tall peaked T wave
4. Trousseau's sign kis imbalance me positive hota hai?
   (a) Hypernatraemia (b) Hypocalcaemia (c) Hypercalcaemia (d) Hyperkalaemia

5. pH 7.50, $PaCO_2$ 30, HCO_3^- 24 – interpretation?
(a) Respiratory alkalosis (b) Respiratory acidosis (c) Normal (d) Metabolic alkalosis

6. $MgSO_4$ toxicity ka first sign hai –
(a) Hypotension (b) Loss of patellar reflex (c) Oliguria (d) Respiratory arrest

7. Head injury patient me kaunsa IV fluid avoid karna chahiye?
(a) Mannitol (b) 0.45 % NS (c) 0.9 % NS (d) 3 % NS

8. 1000 ml fluid 10 ghante me, drop factor 15 – drops/min?
(a) 15 (b) 25 (c) 32 (d) 40

9. Fluid status ka sabse accurate indicator hai –
(a) Skin turgor (b) Pulse (c) Daily weight (d) BP

10. Repeated vomiting se hota hai –
(a) Respiratory acidosis (b) Metabolic acidosis (c) Respiratory alkalosis (d) Metabolic alkalosis

11. Air embolism me patient ko kis position me rakhte hain?
(a) High Fowler's (b) Right lateral, head up (c) Prone (d) Left lateral with Trendelenburg

12. IV potassium ke baare me galat statement kaunsa hai?
(a) Urine output check (b) Diluted dena (c) Infusion pump se dena (d) IV push de sakte hain

Answer Key – Chapter 2

1-b · 2-d · 3-d · 4-b · 5-a · 6-b · 7-b · 8-b (1000×15÷600 = 25) · 9-c · 10-d · 11-d · 12-d

3 Shock, Pain, Inflammation & Infection Control

3.1 Shock

Definition

Shock = Ek life-threatening condition jisme **tissue perfusion inadequate** ho jata hai → cells ko O_2 & nutrients nahi milte → anaerobic metabolism → **lactic acidosis** → organ failure.

Types of Shock

Type	Cause	Key feature
Hypovolaemic (most common)	Haemorrhage, burns, severe vomiting/diarrhoea, dehydration	↓ blood volume; **cold clammy skin**, tachycardia
Cardiogenic	**MI** (most common), arrhythmia, cardiomyopathy	Heart pump fail; **crackles, JVD**, ↑CVP
Obstructive	Cardiac tamponade, tension pneumothorax, massive PE	Blood flow me rukawat
Distributive – Septic	Infection (Gram –ve bacteria most common)	Early: **warm flushed skin, fever**; late: cold
Distributive – Anaphylactic	Allergy (penicillin, bee sting, food)	Bronchospasm, stridor, urticaria, angioedema
Distributive – Neurogenic	**Spinal cord injury** (T6 ke upar), spinal anaesthesia	**Hypotension + BRADYCARDIA**, warm dry skin

Neurogenic shock ki pehchaan

Sab shocks me **tachycardia** hoti hai – **sirf Neurogenic shock me BRADYCARDIA** (sympathetic tone loss). Skin **warm & dry**.

Stages of Shock

Stage	Features
1. Initial	Cellular level changes, no clinical signs
2. Compensatory	BP normal (body compensate karta hai), **tachycardia**, tachypnoea, cold pale skin, ↓ urine, restlessness/anxiety (**earliest sign = restlessness/change in mental status**)
3. Progressive	**BP gir jata hai** (SBP < 90), MAP < 65, oliguria, metabolic acidosis, confusion, weak pulse
4. Refractory (Irreversible)	Treatment ka response nahi, MODS (multi-organ failure), death

Management of Shock

- **ABC** – airway, high-flow **O_2**

- Position: **Modified Trendelenburg** (supine with **legs raised 20–30°**) – head-down nahi
- **2 large bore IV cannula (14–16 G)** – fluids: crystalloids (**NS/RL**), blood for haemorrhage
- Vasopressors: **Noradrenaline** (septic shock – drug of choice), Dopamine, Dobutamine (cardiogenic – inotrope)
- **Anaphylaxis: Adrenaline (Epinephrine) 1:1000, 0.3–0.5 mg IM** (anterolateral thigh) – **drug of choice**
- Septic: blood culture **before** antibiotics, broad-spectrum antibiotics within 1 hour
- Monitor: vitals every 5–15 min, **urine output (Foley, hourly ≥ 30 ml/hr)**, CVP (normal **2–8 mmHg** / 5–10 cmH_2O), mental status, ABG, lactate

! Shock me kya NAHI karna

- Hypovolaemic shock me pehle **fluids** – vasopressor nahi (tank khali hai to pump kya karega).
- Cardiogenic shock me **fluids carefully** – overload → pulmonary oedema.
- Patient ko garam kambal se **over-heat mat karo** (vasodilation) – sirf normal warmth.

3.2 Pain

D Definition (IASP)

"Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage." – **Pain is whatever the patient says it is** (Margo McCaffery). Pain = **5th vital sign**.

Type	Description
Acute	< 3–6 months, sudden, protective, e.g. post-op
Chronic	> 3–6 months, persistent, e.g. arthritis, cancer
Referred	Kisi aur jagah mehsoos, e.g. MI → left arm/jaw; gallbladder → right shoulder; appendix → periumbilical
Phantom limb	Amputated part me dard (real pain hai)
Neuropathic	Nerve damage, burning/shooting – e.g. diabetic neuropathy
Visceral	Internal organs – diffuse, cramping

Pain assessment

- **PQRST**: **P**rovoking factors, **Q**uality, **R**egion/Radiation, **S**everity (0–10), **T**iming
- Scales: **Numeric Rating Scale (0–10)**, Visual Analogue Scale (10 cm line), **Wong-Baker FACES** (children ≥ 3 yrs), **FLACC** (infants, non-verbal), Behavioural pain scale (ICU)
- **Gate Control Theory** – **Melzack & Wall (1965)**: massage, TENS se "gate" band → pain kam.

WHO Analgesic Ladder (cancer pain)

1. **Step 1 – Mild pain**: Non-opioid (Paracetamol, NSAIDs) ± adjuvant
2. **Step 2 – Moderate pain**: Weak opioid (**Codeine, Tramadol**) + non-opioid ± adjuvant
3. **Step 3 – Severe pain**: Strong opioid (**Morphine**, Fentanyl) + non-opioid ± adjuvant
   - Dawa **"by the clock"** (regular interval) – PRN nahi; **by mouth** preferred.

Opioids – nursing care

- Side effects: **respiratory depression** (most serious – RR < 12 par hold), **constipation** (most common – laxative do), nausea, sedation, urinary retention, pinpoint pupils.
- **Antidote = Naloxone (Narcan)**.
- **PCA** (Patient Controlled Analgesia) – sirf patient button dabaye, family nahi.

3.3 Inflammation & Wound Healing

- **5 cardinal signs of inflammation**: **Rubor** (redness), **Calor** (heat), **Tumor** (swelling), **Dolor** (pain), **Functio laesa** (loss of function – Virchow).
- First 4 = **Celsus**.

Phases of wound healing

1. **Haemostasis** – minutes (platelet plug, clot)
2. **Inflammatory phase** – day 0–4 (neutrophils, macrophages; redness, swelling)
3. **Proliferative phase** – day 4–21 (**granulation tissue**, collagen, epithelialization)
4. **Maturation / Remodelling** – 21 days to 1–2 years (scar strong hota hai)

Intention	Example
Primary intention	Clean surgical wound, edges approximated (sutured) – minimal scar
Secondary intention	Wound open chhoda, granulation se bharta hai – pressure ulcer, burns, infected wound – bada scar
Tertiary (delayed primary)	Pehle open rakhte hain (infection control), phir 3–5 din baad suture

Wound healing ke liye zaroori nutrients

Protein (most important), **Vitamin C** (collagen synthesis), **Vitamin A**, **Zinc**, adequate calories & hydration. Delay karne wale: diabetes, steroids, smoking, infection, malnutrition, old age, poor circulation.

Wound complications

- **Dehiscence** = wound edges ka khulna (usually 5–10 post-op day)
- **Evisceration** = wound se **intestine/organs bahar aa jaye** → **EMERGENCY**: patient ko **low Fowler's with knees bent**, organs ko **sterile saline-soaked dressing** se dhako, NPO, surgeon ko turant inform; pushing back mat karo.
- Prevention: **splinting** the incision with a pillow while coughing.

3.4 Infection Control

Chain of Infection (6 links)

Infectious agent → Reservoir → Portal of exit → Mode of transmission → Portal of entry → Susceptible host. Kisi bhi ek link ko todo → infection ruk jayega. **Hand hygiene** = single most effective method.

WHO – 5 Moments of Hand Hygiene

1. **Before** touching a patient
2. **Before** clean/aseptic procedure
3. **After** body fluid exposure risk
4. **After** touching a patient
5. **After** touching patient surroundings

Hand wash duration: soap-water **40–60 sec**; alcohol rub **20–30 sec**. Steps: **6 (WHO)**.

Asepsis

- **Medical asepsis** (clean technique) – microorganisms ki sankhya kam karna (hand washing, gloves).
- **Surgical asepsis** (sterile technique) – **saare** microorganisms + spores khatam (OT, catheterization, IV).
- **Sterilization** – all microbes + spores destroy: **Autoclave (moist heat) 121 °C, 15 psi, 15–20 min** (or 134 °C for 3–4 min); dry heat (hot air oven) 160 °C × 2 hr; ETO gas (heat sensitive items); gamma radiation (disposables).
- **Disinfection** – spores ko chhod ke baaki microbes: glutaraldehyde 2 % (Cidex – endoscopes), sodium hypochlorite 1 % (blood spills), alcohol 70 %.

Sterile field rules

- Sterile field ke **1 inch border = non-sterile**.
- **Waist level ke neeche** & peeth ke peeche = non-sterile.
- Sterile object doubt ho to **non-sterile maano**.
- Sterile field ko kabhi **peeth mat dikhao** / **unattended mat chhodo**.
- Wet sterile drape = contaminated (strike-through).

Isolation precautions (CDC)

Precaution	Diseases	PPE / Room
Standard	**Sab patients**	Hand hygiene, gloves, mask/gown as needed
Airborne (< 5 µm)	**TB, Measles, Chickenpox**, disseminated herpes zoster	**N95 respirator**, **negative pressure room** (6–12 air changes/hr), door closed
Droplet (> 5 µm, 3 feet/1 m)	**Influenza, Mumps, Rubella, Diphtheria, Pertussis, Meningococcal** meningitis	Surgical mask, private room
Contact	**MRSA, VRE**, C. difficile, scabies, wound infection, herpes simplex	Gloves + gown, dedicated equipment

Airborne = "MTV"

Measles, **T**B, **V**aricella (chickenpox) → Airborne + N95.

- PPE **donning** order: Gown → Mask/N95 → Goggles/face shield → **Gloves** (last).
- PPE **doffing** order: **Gloves** (first, most contaminated) → Goggles → Gown → Mask → hand hygiene.

Biomedical Waste Management (BMW Rules 2016, amended 2018)

Colour	Waste
Yellow	Human anatomical waste, soiled dressings/cotton, expired medicines, chemical waste, micro-lab waste → **incineration**
Red	**Contaminated recyclable plastic** – IV sets, catheters, tubing, syringes (without needle), gloves, urine bags → autoclave & recycle
White (translucent, puncture-proof)	**Sharps** – needles, blades, scalpels
Blue (cardboard box)	**Glassware** – broken/discarded vials, ampoules, metallic implants

- **Needle stick injury** → wash with soap & running water (do NOT squeeze/scrub), report, check source status, **PEP within 2 hours (max 72 hrs)** for HIV; Hep B vaccine status check.
- Needle ko **recap mat karo** (if needed, one-hand scoop technique).

? Practice MCQs – Chapter 3

1. Kaunse shock me bradycardia milti hai?
(a) Hypovolaemic (b) Neurogenic (c) Septic (d) Cardiogenic

2. Anaphylactic shock ka drug of choice –
(a) Dopamine (b) Adrenaline IM (c) Antihistamine (d) Hydrocortisone

3. Shock ka earliest sign hai –
(a) Hypotension (b) Cyanosis (c) Oliguria (d) Restlessness / change in mental status

4. Opioid ka antidote –
(a) Protamine (b) Flumazenil (c) Atropine (d) Naloxone

5. Gate control theory kisne diya?
(a) Freud (b) Melzack & Wall (c) Maslow (d) Selye

6. Wound evisceration me first action –
(a) Patient ko chalne do (b) Sterile saline-soaked dressing se dhako (c) Organs andar push karo (d) Dry gauze lagao

7. Measles patient ko kaunsa isolation chahiye?
(a) Airborne (b) Contact (c) Standard only (d) Droplet

8. Used needles kis colour ke container me?
(a) Yellow (b) Blue (c) White translucent puncture-proof (d) Red

9. Autoclave ka standard –
(a) 100 °C, 30 min (b) 121 °C, 15 psi, 15–20 min (c) 160 °C, 2 hr (d) 60 °C, 1 hr

10. PPE doffing me sabse pehle kya utarte hain?
(a) Mask (b) Goggles (c) Gloves (d) Gown

11. Collagen synthesis ke liye kaunsa vitamin zaroori hai?
(a) Vit K (b) Vit D (c) Vit C (d) Vit B12

12. Cancer pain me WHO ladder step 2 me kaunsi dawa aati hai?
(a) Fentanyl (b) Tramadol / Codeine (c) Only paracetamol (d) Morphine

Answer Key – Chapter 3

1-b · 2-b · 3-d · 4-d · 5-b · 6-b · 7-a · 8-c · 9-b · 10-c · 11-c · 12-b

4 Perioperative Nursing

D Perioperative = 3 phases

Pre-operative (surgery ka decision → OT table par transfer) → **Intra-operative** (OT table → recovery room/PACU) → **Post-operative** (PACU → full recovery/follow-up).

Classification of surgery

By urgency	Meaning	Example
Emergency	Turant, life-saving	Ruptured spleen, gunshot, obstructed hernia
Urgent	24–48 hrs me	Acute cholecystitis, kidney stone
Required/Planned	Few weeks–months	Cataract, BPH
Elective	Patient ki marzi, delay se harm nahi	Hernia repair, cosmetic
Optional	Personal choice	Cosmetic surgery

By purpose	Meaning
Diagnostic	Biopsy, exploratory laparotomy
Ablative	Diseased part remove – "-ectomy" (appendectomy)
Palliative	Symptoms relieve, cure nahi (colostomy in inoperable ca)
Reconstructive	Function/appearance restore (skin graft)
Transplant, **Constructive** (congenital defect repair – cleft lip)	

★ Surgical suffixes

-ectomy = removal · **-otomy** = cutting into · **-ostomy** = artificial opening · **-plasty** = repair · **-scopy** = looking inside · **-pexy** = fixation · **-orrhaphy** = suturing/repair (herniorrhaphy)

4.1 Pre-operative Care

Physical preparation checklist

- **Informed consent** signed (surgeon explain kare; nurse witness) – sedation/pre-medication se **pehle**.
- **NPO**: solids **6–8 hrs**, clear fluids **2 hrs** before anaesthesia (ASA guidelines) – aspiration rokne ke liye.
- **Investigations**: CBC, blood group & cross-match, BT/CT, PT/INR, blood sugar, RFT, electrolytes, ECG (> 40 yrs), chest X-ray, HIV/HBsAg.
- **Skin preparation**: bath with antiseptic (chlorhexidine) previous night; **hair removal by clipper, just before surgery** (razor → nicks → infection).
- **Bowel preparation**: enema/laxatives for GI surgery.
- **Bladder**: void before going to OT (or Foley catheter).

- Remove **jewellery, dentures, contact lenses, hairpins, nail polish/make-up** (nail colour se SpO_2 & cyanosis check nahi hota), prosthesis; hearing aid can stay if needed.
- **Medications**: anticoagulants (warfarin 5 days, aspirin 7 days pehle stop – as per order), insulin dose adjust, antihypertensives usually continue with sip of water.
- **ID band**, site marking (by surgeon), allergy band.
- **Pre-medication**: given **30–60 min before** – atropine/glycopyrrolate (secretions ↓), midazolam (anxiety ↓), opioid, antacid/ranitidine, antiemetic. Pre-med ke baad **side rails up**, bed par rehne do.

Pre-operative teaching (post-op complications rokne ke liye)

- **Deep breathing & coughing** exercises, **incentive spirometry** (atelectasis/pneumonia prevention)
- **Leg exercises**, early ambulation (DVT prevention)
- **Splinting incision** with pillow while coughing
- Turning in bed, pain scale use, PCA use



Smoking & surgery

Surgery se kam se kam **4–8 weeks pehle smoking chhodne** ki salah – lung complications & wound healing problem kam hote hain.

4.2 Anaesthesia

Type	Details	Nursing points
General	Complete loss of consciousness; IV (Propofol, Thiopentone, Ketamine) + inhalation (Sevoflurane, Isoflurane, N_2O) + muscle relaxant (Succinylcholine, Vecuronium)	Airway, aspiration risk; **Malignant hyperthermia** risk
Spinal (Subarachnoid block)	Drug (bupivacaine) in **subarachnoid space**, **L3–L4 / L4–L5** level	Post-spinal **headache** → keep **flat/supine 6–8 hrs**, fluids; hypotension watch
Epidural	Drug in **epidural space** (catheter – continuous)	Labour analgesia, post-op pain
Local / Regional / Nerve block	Lignocaine (Xylocaine) – max 3 mg/kg (7 mg/kg with adrenaline)	**Lignocaine with adrenaline** – fingers, toes, nose, ear, penis me **NOT** (necrosis)
Conscious (moderate) sedation	Midazolam + fentanyl – patient airway maintain karta hai	Endoscopy, minor procedures

Stages of General Anaesthesia (Guedel's)

1. **Stage I – Analgesia/Induction**: start → loss of consciousness
2. **Stage II – Excitement/Delirium**: irregular breathing, struggling – **stimulus avoid karo (quiet OT)**
3. **Stage III – Surgical anaesthesia**: surgery isi stage me hoti hai
4. **Stage IV – Medullary paralysis/Overdose**: respiratory arrest – **danger**

Malignant Hyperthermia

Genetic reaction to **succinylcholine** & inhalation anaesthetics → sudden **↑ temperature (up to 43 °C)**, muscle rigidity (jaw/masseter), tachycardia, **↑ $ETCO_2$ (earliest sign)**, hyperkalaemia.
Drug: Dantrolene sodium IV; stop triggering agent, 100 % O_2, cooling.

4.3 Intra-operative Care

OT Team

- **Sterile team (scrubbed)**: Surgeon, assistant surgeon, **scrub nurse** (instruments handle, count, passes instruments)
- **Unsterile team**: Anaesthetist, **circulating nurse** (OT manage, documentation, specimen, outside supplies, count verify), technicians

OT zones

Unrestricted (street clothes allowed) → **Semi-restricted** (scrubs + cap) → **Restricted** (scrubs, cap, **mask**, sterile procedures – theatre itself).

- OT temperature **20–24 °C**, humidity **30–60 %**, positive pressure, ≥ 15–20 air changes/hr.

Surgical safety

- **WHO Surgical Safety Checklist (2008)**: **Sign In** (before induction) → **Time Out** (before skin incision) → **Sign Out** (before patient leaves OT).
- **Swab/sponge, needle & instrument count**: before surgery, before closure of cavity, at skin closure – scrub & circulating nurse dono milke.
- Surgical scrub: **2–6 min** (as per policy), hands **elbows se upar** rakhte hain, fingertips → elbow.
- **Electrosurgery**: grounding pad (dispersive electrode) – fleshy, dry, hairless area par (thigh/buttock) – burns rokne ke liye.

Intra-op positions

- **Supine** – most abdominal surgeries; **Lithotomy** – gynaec/urology/perineal; **Prone** – spine; **Lateral (kidney position)** – nephrectomy; **Trendelenburg** – pelvic/lower abdominal; **Reverse Trendelenburg** – head & neck, gallbladder.
- Pressure points ko padding; extremities ko 90° se zyada abduct na karo (**brachial plexus injury**).

4.4 Post-operative Care

Immediate (PACU / Recovery room)

- **Airway first**: patient ko **lateral / side-lying (recovery) position** with head turned until gag reflex returns (aspiration rokne ke liye); oropharyngeal airway tab tak rakho jab tak patient khud na nikal de.
- Vitals **every 15 min** (first hour), then 30 min, then hourly.
- **Aldrete score ≥ 9** → PACU se ward transfer (Activity, Respiration, Circulation, Consciousness, O_2 saturation – each 0–2).

- Watch: bleeding (dressing, drains), urine output (≥ 30 ml/hr), pain, temperature, nausea/vomiting.
- Patient ko pehli baar void karna: **6–8 hrs** ke andar.

Post-operative complications – Timeline ("Wind, Water, Wound, Walk, Wonder drugs")

Day	Complication	Prevention / Management
Day 1–2 – Wind	**Atelectasis** (most common, first 48 hrs – fever), pneumonia	Deep breathing, coughing, **incentive spirometry**, early ambulation
Day 3–5 – Water	**UTI** (catheter)	Remove catheter early, fluids
Day 5–7 – Wound	**Wound infection**, dehiscence	Asepsis, nutrition, splinting
Day 7–10 – Walk	**DVT / Pulmonary embolism**	Early ambulation, leg exercises, TED stockings, low-molecular-weight heparin
Any time – Wonder drugs	Drug fever, reactions	Review drugs

♫ **5 W of post-op fever**

Wind (lungs, 24–48 h) · **W**ater (UTI, 3–5 d) · **W**ound (5–7 d) · **W**alking (DVT, 7–10 d) · **W**onder drugs (any time)

Other post-op problems

- **Haemorrhage** – Primary (during surgery), **Reactionary** (within 24 hrs – BP ↑ → ligature slip), **Secondary** (7–14 days – infection).
- **Hypovolaemic shock** – tachycardia, ↓BP, cold clammy skin.
- **Paralytic ileus** – absent bowel sounds, distension; NPO + **NG tube** + IV fluids; ambulation. Bowel sounds return → **"flatus pass"** → oral fluids start.
- **Urinary retention** – first void within 8 hrs; bladder scan, warm water, privacy, catheter if needed.
- **Hiccups** (phrenic nerve irritation), **post-op nausea & vomiting** (ondansetron).
- **Pulmonary embolism** – sudden **dyspnoea, chest pain, tachycardia, anxiety** → High Fowler's, O_2, inform, anticoagulants.
- **DVT** – calf pain, swelling, warmth; **Homan's sign** (not reliable now) – **massage mat karo**.

! **Drains & tubes**

- **Drain output** chart karo; **> 100 ml/hr** bright red blood → inform surgeon.
- **T-tube** (after CBD exploration) – bile drain, normal 300–500 ml first day; bag gallbladder level ke neeche.
- Hemovac/Jackson-Pratt – suction maintain (compress & close).

? **Practice MCQs – Chapter 4**

1. Pre-operative hair removal ka best method –
   (a) Clipper just before surgery (b) Depilatory cream 1 week before (c) Hair removal not needed (d) Razor previous night

2. Spinal anaesthesia ke baad headache rokne ke liye position –
(a) Prone (b) Fowler's (c) Flat/supine (d) Trendelenburg

3. Malignant hyperthermia ki drug –
(a) Paracetamol (b) Dantrolene (c) Atropine (d) Naloxone

4. "Time Out" kab hota hai?
(a) Before skin incision (b) After surgery (c) Before induction (d) In ward

5. Instruments pass karne ka kaam kaun karta hai?
(a) Circulating nurse (b) Anaesthetist (c) Scrub nurse (d) Ward nurse

6. Post-op 1st–2nd day fever ka common cause –
(a) Wound infection (b) DVT (c) UTI (d) Atelectasis

7. Lignocaine with adrenaline kahan contraindicated hai?
(a) Fingers & toes (b) Abdomen (c) Back (d) Thigh

8. Reactionary haemorrhage kab hota hai?
(a) During surgery (b) 7–14 days (c) Within 24 hrs (d) After 1 month

9. PACU se discharge ke liye Aldrete score –
(a) ≥ 5 (b) ≥ 7 (c) ≥ 9 (d) 10 only

10. Guedel's stage me surgery kis stage me hoti hai?
(a) III (b) II (c) I (d) IV

11. Electro-surgery ka grounding pad kahan lagate hain?
(a) Hairy wet area (b) Bony prominence (c) Fleshy dry area like thigh (d) Over scar

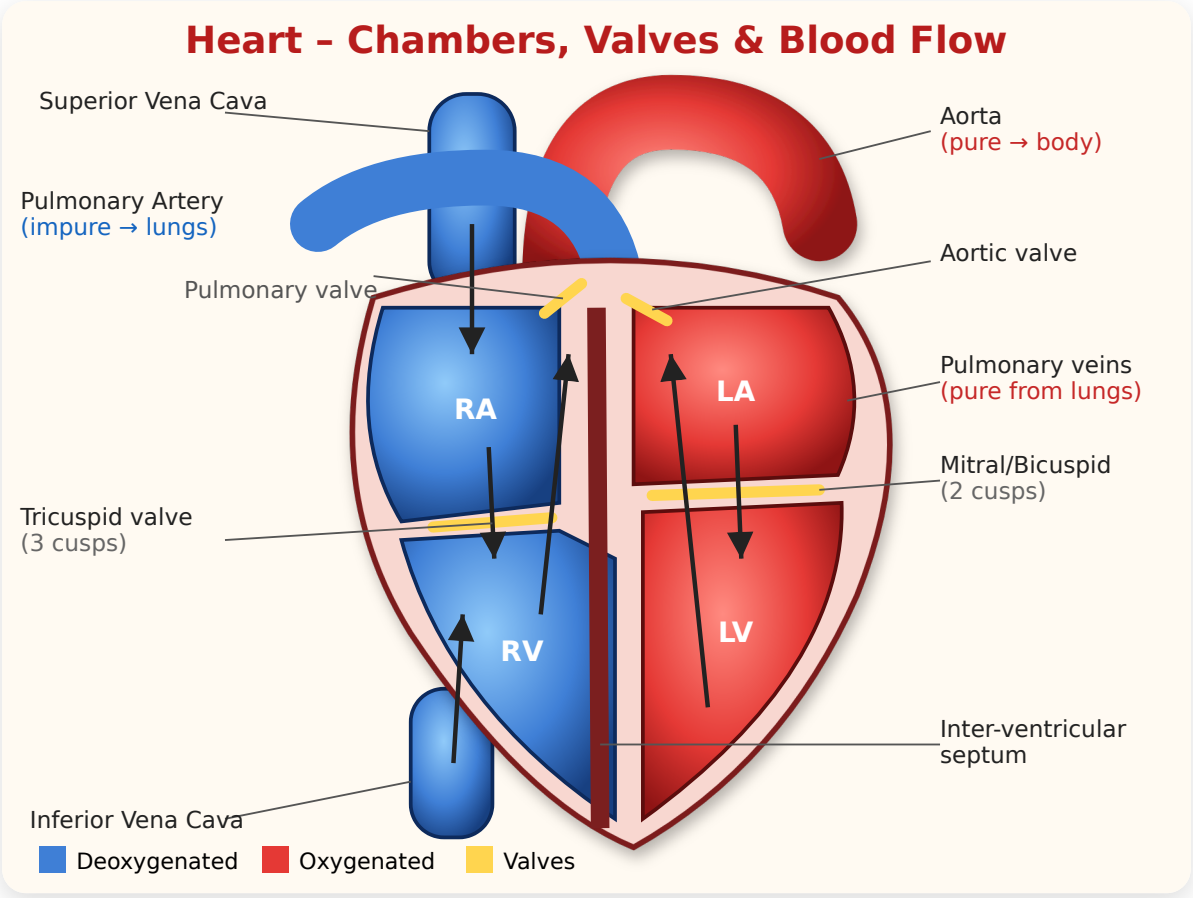
12. Post-op paralytic ileus me –
(a) Enema daily (b) Oral feeding start (c) NG tube decompression + NPO (d) Laxatives

Answer Key – Chapter 4

1-a · 2-c · 3-b · 4-a · 5-c · 6-d · 7-a · 8-c · 9-c · 10-a · 11-c · 12-c

5 Cardiovascular System Disorders

5.1 Applied Anatomy & Physiology



Heart ke 4 chambers, 4 valves aur blood flow ka rasta

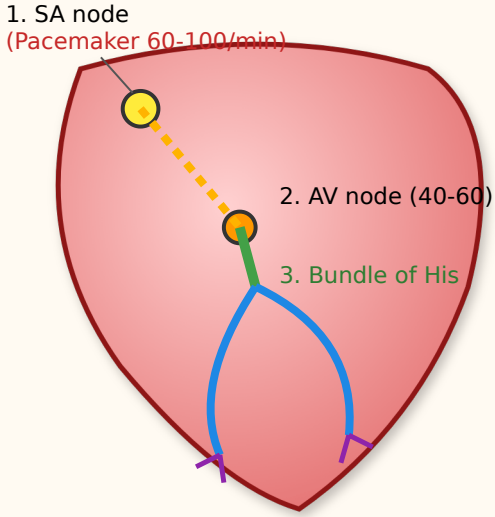
Heart – quick facts

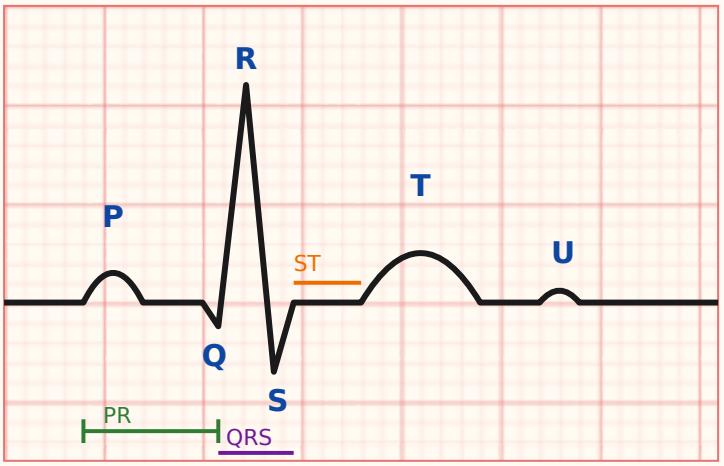
- Weight ~**300 g**, size = apni **closed fist** jitna; **mediastinum** me, 2/3 left side.
- Layers: **Pericardium** (outer sac, 10–50 ml fluid) → **Myocardium** (muscle) → **Endocardium** (inner).
- Valves: **AV valves** – Tricuspid (right), **Mitral/Bicuspid** (left); **Semilunar** – Pulmonary, Aortic.
- **S1 "Lub"** = AV valves close (start of systole); **S2 "Dub"** = semilunar valves close. **S3** = heart failure (ventricular gallop); **S4** = stiff ventricle (HTN, MI).
- Coronary arteries aorta ki base se nikalti hain; **LAD (Left anterior descending) = "widow maker"**. Coronary filling **diastole** me hoti hai.
- **Cardiac output (CO) = Heart rate × Stroke volume** = ~**5 L/min** (SV ~70 ml). **Cardiac index** 2.5–4 L/min/m^2.
- **Preload** = ventricle me end-diastolic volume (stretch) · **Afterload** = resistance jiske against ventricle pump karta hai.
- **Frank-Starling law**: zyada stretch → zyada force (ek limit tak).

Blood flow trick

"Try Pulling My Aorta" → **T**ricuspid → **P**ulmonary → **M**itral → **A**ortic (valve sequence).
Body → SVC/IVC → RA → RV → Pulmonary artery → **Lungs** → Pulmonary veins → LA → LV → Aorta → Body.
Pulmonary artery = only artery carrying impure blood; Pulmonary vein = only vein carrying pure blood.

Conduction System & Normal ECG (PQRST)





P wave = Atrial depolarization
QRS = Ventricular depolarization (0.06-0.10 s)
T wave = Ventricular repolarization
PR interval = 0.12 – 0.20 s
1 small box = 0.04 s · **1 big box** = 0.20 s

Atrial repolarization QRS ke andar chhup jata hai (dikhta nahi).
ST elevation = MI (STEMI) · Tall peaked T = Hyperkalemia · U wave = Hypokalemia

Conduction system aur normal ECG waves

ECG basics

- **12-lead ECG**: 6 limb leads (I, II, III, aVR, aVL, aVF) + 6 chest leads (V1–V6).
- Chest lead placement: **V1** – 4th ICS right sternal border; **V2** – 4th ICS left sternal border; **V3** – between V2 & V4; **V4** – 5th ICS mid-clavicular; **V5** – anterior axillary line; **V6** – mid-axillary line (same level as V4).
- Limb lead colour (AHA): **RA – White, LA – Black, RL – Green, LL – Red** ("**White on right, smoke (black) over fire (red), green is ground**").
- **Heart rate** from ECG = **300 ÷ number of big boxes** between two R waves (or 1500 ÷ small boxes).

Common arrhythmias

Arrhythmia	ECG / Feature	Treatment
Sinus tachycardia	> 100, normal P	Treat cause (fever, pain, hypovolaemia)
Sinus bradycardia	< 60	**Atropine** if symptomatic, pacing
Atrial fibrillation (most common chronic arrhythmia)	**No P waves, irregularly irregular**, fibrillatory waves	Rate control (beta-blocker, diltiazem, digoxin), **anticoagulant (stroke risk)**, cardioversion

Arrhythmia	ECG / Feature	Treatment
Atrial flutter	**Saw-tooth** F waves	Cardioversion, rate control
PVC	Wide bizarre QRS, no P	Treat cause (hypokalaemia, hypoxia)
Ventricular tachycardia (VT)	Wide QRS, rate > 100	With pulse: amiodarone/synchronized cardioversion; **pulseless → defibrillation**
Ventricular fibrillation (VF)	Chaotic, no QRS – **no pulse, no cardiac output**	**Immediate defibrillation** + CPR
Asystole	Flat line	CPR + **adrenaline**; **NOT shockable**
Heart block (3rd degree / complete)	P & QRS independent	**Pacemaker**

Shockable vs non-shockable

Shockable: VF, pulseless VT → Defibrillation (**biphasic 120–200 J**, monophasic 360 J).
Non-shockable: Asystole, PEA → CPR + Adrenaline 1 mg IV every 3–5 min.
Synchronized cardioversion – AF, flutter, unstable VT with pulse (R wave par shock).

5.2 Coronary Artery Disease (CAD) & Atherosclerosis

- **Atherosclerosis** = arteries me **fatty plaque** (cholesterol) jamna → narrowing → ischaemia.
- **Risk factors** – Non-modifiable: age (M > 45, F > 55), male gender, family history, race. Modifiable: **smoking (most important modifiable)**, hypertension, **hyperlipidaemia**, diabetes, obesity, sedentary life, stress.
- Lipid targets: Total cholesterol **< 200 mg/dL**, **LDL < 100** ("bad"), **HDL > 40 (M) / > 50 (F)** ("good"), Triglycerides **< 150**.

Angina Pectoris

Angina

Myocardium ko O_2 demand > supply → **temporary ischaemia** → chest pain. **No permanent damage (no necrosis).**

Type	Feature
Stable angina	Exertion se, **rest / nitroglycerin se 3–5 min me relief**
Unstable angina (pre-infarction)	Rest me bhi, increasing frequency – **emergency (ACS)**
Variant / Prinzmetal	**Coronary spasm**, rest me, early morning; ST elevation transient – calcium channel blockers

Management: Rest, **Nitroglycerin (NTG) sublingual**, O_2 if SpO_2 < 90 %, aspirin, beta-blockers, statins; PCI/CABG.

! Nitroglycerin (sublingual) – teaching

- **1 tablet SL every 5 min, max 3 doses**. 1st dose ke 5 min baad relief nahi → **call emergency (108)**.
- Side effect: **headache** (most common), hypotension, flushing, dizziness → **sit/lie down** ke lena.
- Store in **dark/amber glass bottle**, tightly closed; replace every 6 months (or as per label); tongue par **burning/tingling** = potent.
- **Sildenafil (Viagra)** ke saath **contraindicated** (severe hypotension).
- NTG patch: **hairless area**, rotate sites, **remove at night for 10–12 hrs** (tolerance rokne ke liye); wear gloves while applying.

5.3 Myocardial Infarction (MI)

D MI = Heart Attack

Coronary artery ka **complete blockage** (thrombus on plaque) → myocardium ki **necrosis (death)**. Most common artery = **LAD**. Most common time = **early morning**.

Clinical features

- **Severe crushing/squeezing chest pain**, retrosternal, **> 20–30 min**, radiates to **left arm, jaw, neck, back**; **NOT relieved by rest or NTG**.
- Sweating (diaphoresis), nausea/vomiting, dyspnoea, anxiety, **fear of death**, pallor.
- **Silent MI** – diabetics & elderly & women (atypical: fatigue, indigestion).

Diagnosis

Marker	Rises	Peaks	Returns normal
Troponin I / T (most specific & sensitive)	3–6 hr	24 hr	**7–14 days**
CK-MB	3–6 hr	12–24 hr	2–3 days (re-infarction detect)
Myoglobin (earliest)	1–2 hr	6–8 hr	24 hr

- **ECG**: Hyperacute T → **ST elevation** (injury) → **pathological Q wave** (necrosis – permanent) → T inversion (ischaemia).
- **STEMI** vs **NSTEMI** (ST elevation nahi, troponin positive).

♫ MONA – Initial MI management

Morphine (pain & anxiety; preload ↓) · **O**xygen (if SpO_2 < 90 %) · **N**itroglycerin · **A**spirin (**325 mg chewable**, non-enteric coated)

Correct order of giving: **Oxygen → Aspirin → Nitrate → Morphine** ("**OANM**")

Management

- **Thrombolytics** (Streptokinase, Alteplase/tPA, Tenecteplase) – **within 12 hrs** (best < 3 hrs; "**door-to-needle < 30 min**"). Main risk = **bleeding**; contraindicated: recent stroke, active bleeding, recent surgery.
- **PCI (angioplasty + stent)** – "**door-to-balloon < 90 min**" – preferred.

- **CABG** – multi-vessel disease (graft: saphenous vein, internal mammary artery).
- Drugs: **antiplatelets** (aspirin + clopidogrel), **heparin**, **beta-blockers** (O_2 demand ↓), **ACE inhibitors** (remodelling ↓), **statins**.
- **Nursing**: complete **bed rest first 12–24 hrs**, semi-Fowler's, cardiac monitoring, IV access, **stool softener (avoid Valsalva)**, low-fat low-salt diet, small meals, anxiety ↓, cardiac rehabilitation.
- Sexual activity: usually after **4–6 weeks** / when patient can **climb 2 flights of stairs** without symptoms.

! MI complications

Arrhythmias (most common – **VF most common cause of death in first hours**), cardiogenic shock, heart failure, pericarditis, **Dressler syndrome** (pericarditis 2–10 weeks later), ventricular aneurysm/rupture, papillary muscle rupture, thromboembolism.

5.4 Heart Failure (HF / CCF)

D Heart Failure

Heart body ki need ke hisaab se **sufficient blood pump nahi kar pata**. Most common cause = **CAD/MI** & **hypertension**.

LEFT-sided failure → LUNGS
(blood backs up into lungs)

- **Dyspnoea**, **orthopnoea** (lying down breathless – pillows count), **PND** (paroxysmal nocturnal dyspnoea)
- **Crackles/rales**, cough with **pink frothy sputum** (pulmonary oedema)
- Fatigue, restlessness, confusion
- S3 gallop, tachycardia, ↓ urine output
- Cyanosis

RIGHT-sided failure → BODY (systemic)
(blood backs up into venous system)

- **Pedal / dependent oedema** (pitting), sacral oedema in bed-bound
- **Jugular vein distension (JVD)**, ↑ CVP
- **Hepatomegaly**, **ascites**, RUQ pain
- **Weight gain**, anorexia, nausea
- Nocturia
- Cause: **Left HF (most common)**, **cor pulmonale (COPD)**

♫ L = Lungs, R = Rest of body

Left failure → **L**ungs (dyspnoea, crackles). **R**ight failure → **R**est of body (oedema, JVD, liver).

- **Diagnosis**: **BNP (B-type natriuretic peptide) ↑** (> 100 pg/ml), echo (**ejection fraction**; normal **55–70 %**; HFrEF ≤ 40 %), chest X-ray (cardiomegaly).
- **NYHA classification**: Class I (no limitation) → II (slight, ordinary activity se symptoms) → III (marked, less than ordinary activity) → IV (symptoms **at rest**).

Management

- **Position: High Fowler's** with legs dependent (venous return ↓) – acute pulmonary oedema me.
- O_2, **Diuretics** (**Furosemide** – loop), ACE inhibitors/ARB/ARNI, **beta-blockers** (carvedilol, metoprolol), spironolactone, **Digoxin**, vasodilators.

- **Sodium restriction (< 2 g/day)**, fluid restriction (1.5–2 L), **daily weight** (gain **> 1–1.5 kg/day or 2 kg/week → report**), I/O chart.
- Rest + gradual activity, small frequent meals.

! Digoxin – very important

- Action: **+ve inotrope** (force ↑), **–ve chronotrope** (HR ↓).
- **Apical pulse 1 full minute** before giving – **< 60/min (adult)** ya **< 70 (child)**, **< 90–110 (infant)** → **hold** & inform.
- Therapeutic level: **0.5–2.0 ng/ml**.
- **Toxicity signs**: **anorexia, nausea, vomiting (earliest)**, **visual disturbances – yellow-green halos**, blurred vision, bradycardia, arrhythmias, confusion.
- **Hypokalaemia** → digoxin toxicity ↑ (furosemide ke saath K^+ monitor).
- **Antidote: Digibind (Digoxin immune Fab)**.

5.5 Hypertension

Category (ACC/AHA 2017)	Systolic		Diastolic
Normal	< 120	and	< 80
Elevated	120–129	and	< 80
Stage 1 HTN	130–139	or	80–89
Stage 2 HTN	≥ 140	or	≥ 90
Hypertensive crisis	**> 180**	and/or	**> 120**

(JNC-7: HTN = ≥ 140/90; Prehypertension 120–139/80–89 – Indian exams me dono pooche jaate hain.)

- **Primary/Essential HTN** (90–95 %) – no known cause. **Secondary** – renal disease, pheochromocytoma, Cushing's, coarctation of aorta, pregnancy, drugs (OCPs, steroids).
- **"Silent killer"** – usually asymptomatic; occipital headache (morning), epistaxis, dizziness.
- Target organ damage: **Heart** (LVH, HF, MI), **Brain** (stroke), **Kidney** (CKD), **Eyes** (retinopathy).
- **Hypertensive emergency** (with organ damage) → IV **labetalol, nitroprusside, nicardipine**; BP ko **first hour me 25 % se zyada nahi** girana.

★ Lifestyle – DASH diet

- **Salt < 5 g/day** (WHO) / Na < 2.3 g · DASH diet (fruits, vegetables, low-fat dairy) · weight loss · **exercise 30 min/day, 5 days/week** · limit alcohol · quit smoking · stress reduction.

Antihypertensive drugs

Group	Example	Key side effect / nursing point
Diuretics – Thiazide	Hydrochlorothiazide	Hypokalaemia, hyperglycaemia, hyperuricaemia
Loop	Furosemide	**Hypokalaemia**, ototoxicity, give **morning**
K-sparing	Spironolactone	**Hyperkalaemia**, gynaecomastia

Group	Example	Key side effect / nursing point
ACE inhibitors (-pril)	Enalapril, Lisinopril, Captopril	**Dry cough** (most common), **angioedema**, hyperkalaemia, first-dose hypotension; **contraindicated in pregnancy**
ARBs (-sartan)	Losartan, Telmisartan	No cough; hyperkalaemia
Beta-blockers (-olol)	Atenolol, Metoprolol, Propranolol	**Bradycardia**, bronchospasm (**avoid in asthma** – non-selective), masks hypoglycaemia, don't stop suddenly
Calcium channel blockers	Amlodipine, Nifedipine, Diltiazem	**Ankle oedema**, headache, flushing, gingival hyperplasia; avoid grapefruit
Alpha-blockers (-zosin)	Prazosin	**First-dose orthostatic hypotension** – bedtime dose
Central alpha agonist	Clonidine, **Methyldopa (pregnancy)**	Rebound HTN if stopped suddenly
Vasodilator	Hydralazine, **Sodium nitroprusside**	Nitroprusside – **light sensitive (cover bottle)**, **cyanide toxicity**



Orthostatic hypotension ki teaching

Antihypertensive lene wale patient ko bolo – bed se **dheere-dheere utho** (lying → sitting → standing), pehle pair latka ke baitho. Hot bath, alcohol, long standing avoid.

5.6 Valvular & Inflammatory Heart Diseases

Rheumatic Fever & Rheumatic Heart Disease (RHD)

- Cause: **Group A β-haemolytic Streptococcus** throat infection (pharyngitis) ke **2–3 weeks baad** autoimmune reaction; age **5–15 yrs**.
- Most commonly affected valve: **Mitral** (then aortic).
- **Jones criteria** – Major (**J♥NES**): **J**oints (migratory polyarthritis – most common), **♥** Carditis, **N**odules (subcutaneous), **E**rythema marginatum, **S**ydenham's chorea. Minor: fever, arthralgia, ↑ESR/CRP, prolonged PR interval. Diagnosis = evidence of strep infection (**ASO titre ↑**) + **2 major** or **1 major + 2 minor**.
- Treatment: Penicillin, aspirin (arthritis), bed rest; **Secondary prophylaxis: Benzathine penicillin 1.2 million units IM every 3–4 weeks** (≥ 5 yrs/till age 21–40 depending on carditis).

Valve disease	Key features
Mitral stenosis (most common RHD lesion)	Dyspnoea, **haemoptysis**, **AF**, **mid-diastolic rumbling murmur**, malar flush, thrombo-embolism
Mitral regurgitation	Pan-systolic murmur, fatigue, dyspnoea
Mitral valve prolapse	Mid-systolic click, mostly young women, asymptomatic
Aortic stenosis	**Triad: Angina, Syncope, Heart failure (dyspnoea)**; ejection systolic murmur; avoid strenuous exercise
Aortic regurgitation	Wide pulse pressure, **water-hammer (Corrigan) pulse**, head bobbing (de Musset sign)

- Valve replacement: **Mechanical valve** – lifelong **warfarin** (INR 2.5–3.5); **Biological/tissue valve** – lasts 10–15 yrs, no long-term anticoagulation.

Infective Endocarditis

- Cause: **Streptococcus viridans** (sub-acute; after dental procedure), **Staph aureus** (acute; IV drug users – tricuspid valve).
- Signs: fever, new murmur, **Osler's nodes** (painful, fingers), **Janeway lesions** (painless, palms/soles), **splinter haemorrhages** (nails), **Roth spots** (retina), petechiae, clubbing, emboli.
- Diagnosis: **blood cultures (3 sets)**, echo (**vegetations**). Rx: IV antibiotics **4–6 weeks**; **antibiotic prophylaxis before dental procedures** in high-risk patients.

Pericarditis & Cardiac Tamponade

- **Pericarditis**: **sharp chest pain ↑ on lying down/deep breath, relieved by sitting up & leaning forward**; **pericardial friction rub** (hallmark); ECG – diffuse ST elevation. Rx: NSAIDs, colchicine.
- **Cardiac tamponade** – pericardial sac me fluid/blood → heart compress. **Beck's triad**: **Hypotension + Muffled/distant heart sounds + JVD**; **pulsus paradoxus** (SBP falls > 10 mmHg on inspiration). Rx: **Pericardiocentesis** (emergency).

Cardiomyopathy

Dilated (most common; alcohol, viral), **Hypertrophic** (genetic; sudden death in young athletes; **avoid dehydration & strenuous exercise**, no digoxin/nitrates), **Restrictive** (amyloidosis).

5.7 Vascular Disorders

Peripheral Arterial vs Venous Disease

Feature	Arterial (PAD)	Venous (CVI)
Pain	**Intermittent claudication** (walking se calf pain, rest se relief); rest pain in severe	Aching, heaviness, **elevation se relief**
Pulses	**Weak/absent**	Normal
Skin	Pale, cool, **shiny, hairless**, thick nails	**Brown pigmentation**, warm, **oedema**
Ulcer	**Toes/feet, deep, punched-out**, painful, pale base	**Medial malleolus (ankle), shallow**, irregular, red base
Position	**Legs dependent (neeche)**	**Legs elevated (upar)**
Rx	Walking program, stop smoking, cilostazol, bypass	Compression stockings, elevation, Unna boot



Position trick

Arterial → **A**ir me nahi (legs DOWN – gravity se blood pahunche). **V**enous → **V**ertical up (legs ELEVATED).

- **Buerger's disease (Thromboangiitis obliterans)** – young **male smokers**; **absolute smoking cessation** = only effective treatment.

- **Raynaud's phenomenon** – cold/stress se fingers **white → blue → red**; mostly young women; keep warm, avoid caffeine & smoking, CCB.
- **Aortic aneurysm** – abdominal (most common, below renal arteries) – **pulsatile abdominal mass**, bruit; rupture → severe back pain, shock. Avoid straining; BP control.
- **Varicose veins** – dilated tortuous superficial veins (long saphenous); **Trendelenburg test**; Rx: elastic stockings, sclerotherapy, stripping; avoid long standing & crossing legs.

DVT (Deep Vein Thrombosis)

- **Virchow's triad**: **Venous stasis** + **Vessel wall injury** + **Hypercoagulability**.
- Risk: immobility, post-op (esp. hip/knee surgery), pregnancy, OCPs, obesity, cancer, long flights.
- S/S: unilateral **calf pain, swelling, warmth, redness**; Homan's sign (not reliable). Diagnosis: **Duplex Doppler USG**, D-dimer.
- Rx: **Heparin** (initial) → **Warfarin** (overlap 4–5 days); LMWH (enoxaparin – SC in abdomen, don't rub, don't expel air bubble); bed rest initially with leg elevation.
- **Do NOT massage** the leg (embolism). Complication: **Pulmonary embolism**.

Anticoagulants – Heparin vs Warfarin

	Heparin	Warfarin
Route	IV / SC	Oral
Onset	Immediate	3–5 days
Monitoring	**aPTT** (1.5–2.5 × control; normal 30–40 s)	**PT/INR** (target **INR 2–3**; mechanical valve 2.5–3.5)
Antidote	**Protamine sulphate**	**Vitamin K** (phytonadione) + FFP
Pregnancy	**Safe** (doesn't cross placenta)	Teratogenic – avoid
Diet	–	**Consistent** intake of green leafy vegetables (Vit K)

5.8 Cardiac Investigations & Procedures – Nursing points

- **Cardiac catheterization/angiography** – femoral or radial; check allergy (**iodine/seafood**), NPO 6–8 hrs, consent. **After**: affected **leg straight 4–6 hrs** (femoral), pressure on site, check **distal pulses**, colour, temperature every 15 min; fluids to flush dye; watch bleeding/haematoma.
- **Echocardiography** – non-invasive, ejection fraction. TEE (transoesophageal) – NPO, gag reflex check after.
- **Stress test (TMT)** – avoid caffeine, smoking, heavy meal before; stop if chest pain/ST changes.
- **Holter monitoring** – 24–48 hr ECG; patient diary maintain kare; no bath.
- **CVP** – normal **2–8 mmHg (3–8 cmH_2O)**; zero point = **phlebostatic axis** (4th ICS, mid-axillary line). ↑CVP = fluid overload/RHF; ↓CVP = hypovolaemia.
- **Pacemaker**: avoid MRI (non-compatible), keep mobile phone > 15 cm, **arm on pacemaker side – no raising above shoulder for few weeks**, check pulse daily, carry ID card.

5.9 CPR / BLS (AHA)

Adult BLS – remember numbers

- Sequence: **C-A-B** (Compressions → Airway → Breathing) – (newborn me A-B-C).
- Check responsiveness & breathing + **carotid pulse ≤ 10 seconds**.
- **Compression rate 100–120/min**, **depth 5–6 cm (2–2.4 inch)**, allow **full chest recoil**, minimize interruptions (< 10 s).
- **Ratio 30 : 2** (1 or 2 rescuers, adult); child/infant 2 rescuers = **15 : 2**.
- Hand position: **lower half of sternum** (centre of chest). Infant: 2 fingers / 2-thumb encircling just below nipple line; depth **4 cm** infant, **5 cm** child.
- Airway: **Head tilt–chin lift**; suspected spinal injury → **jaw thrust**.
- With advanced airway: **1 breath every 6 sec (10/min)** + continuous compressions.
- **AED** – as soon as available; switch compressors every **2 min** (5 cycles).
- **Drowning victim / asphyxia** – breathing important; lay rescuer for child alone – **2 min CPR then call**.

- **Chain of survival (IHCA)**: Early recognition & prevention → activate emergency response → high-quality CPR → defibrillation → post-arrest care → recovery.
- ACLS drugs: **Adrenaline 1 mg IV every 3–5 min**; **Amiodarone 300 mg** first dose (then 150 mg) for VF/pVT; Lidocaine alternative.

Practice MCQs – Chapter 5

1. MI ka most specific cardiac marker –
   (a) CK-MB (b) Troponin (c) Myoglobin (d) LDH
2. Digoxin dene se pehle kya check karte hain?
   (a) Temperature (b) Radial pulse 30 sec (c) Apical pulse 1 min (d) BP
3. Digoxin toxicity ka sign –
   (a) Yellow-green halos (b) Tachypnoea (c) Polyuria (d) Hypertension
4. Left-sided heart failure ka sign –
   (a) JVD (b) Pedal oedema (c) Crackles & orthopnoea (d) Hepatomegaly
5. Heparin ka antidote –
   (a) Vitamin K (b) Protamine sulphate (c) Atropine (d) Naloxone
6. Sublingual NTG maximum kitni baar?
   (a) 1 (b) 2 (c) 3 doses, 5 min interval (d) 5
7. ACE inhibitor ka common side effect –
   (a) Ankle oedema (b) Bradycardia (c) Dry cough (d) Hypokalaemia
8. Beck's triad kis condition me milta hai?
   (a) Pericarditis (b) HF (c) MI (d) Cardiac tamponade
9. Pericarditis ka pain kis position me kam hota hai?
   (a) Prone (b) Sitting & leaning forward (c) Supine (d) Left lateral
10. Adult CPR me compression depth –
    (a) 2–3 cm (b) 4 cm (c) 5–6 cm (d) 8 cm

11. Arterial insufficiency me legs ki position –
(a) Elevated (b) Dependent (neeche) (c) Crossed (d) Flexed at hip

12. Asystole ka treatment –
(a) Cardioversion (b) Defibrillation (c) CPR + Adrenaline (d) Atropine only

13. Rheumatic heart disease me most commonly affected valve –
(a) Tricuspid (b) Aortic (c) Pulmonary (d) Mitral

14. Warfarin therapy me kaunsa test monitor karte hain?
(a) Platelet count (b) BT (c) PT/INR (d) aPTT

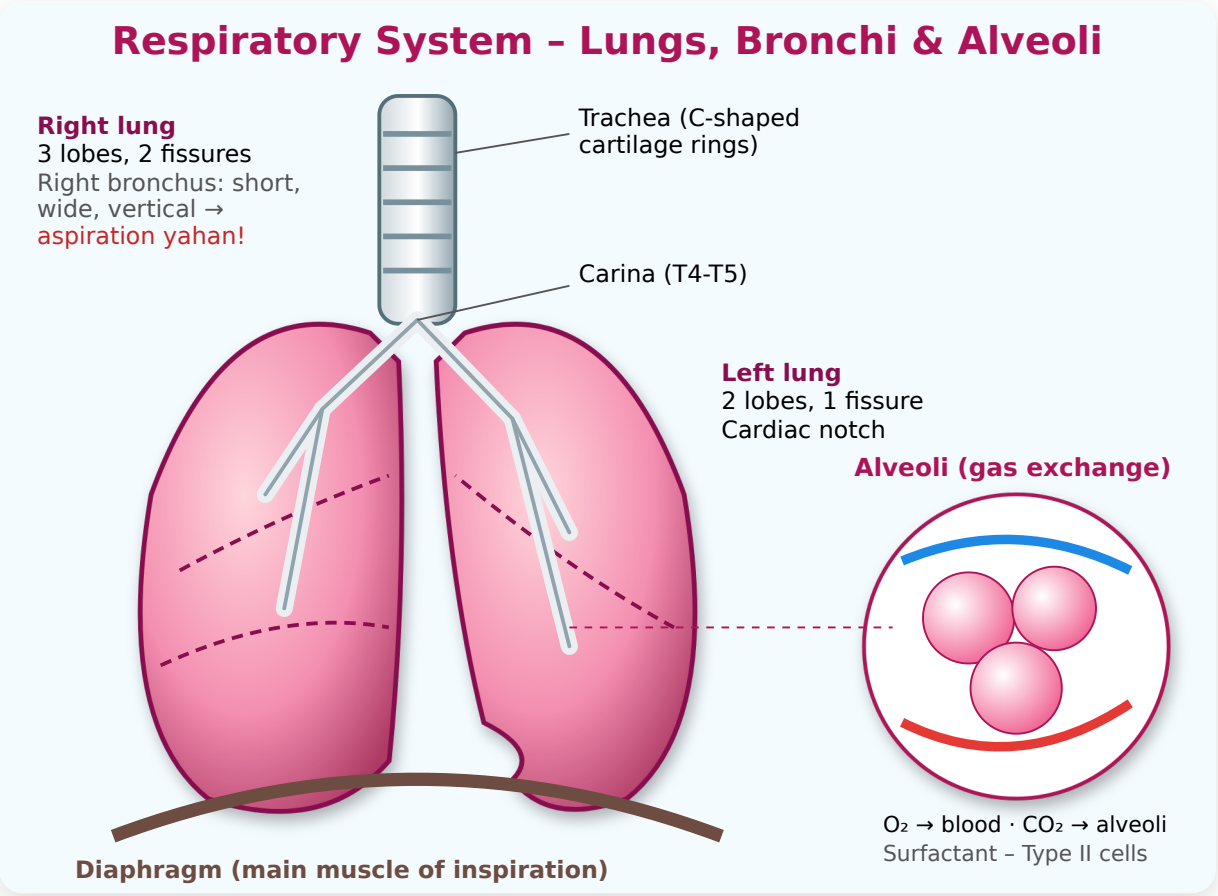
15. Sodium nitroprusside ki infusion ko –
(a) Fridge me rakhte hain (b) Garam karte hain (c) Light se bachate hain (cover) (d) Shake karte hain

Answer Key – Chapter 5

1-b · 2-c · 3-a · 4-c · 5-b · 6-c · 7-c · 8-d · 9-b · 10-c · 11-b · 12-c · 13-d · 14-c · 15-c

6 Respiratory System Disorders

6.1 Applied Anatomy & Physiology



Respiratory tract – trachea se alveoli tak

Quick facts

- Upper tract: nose, pharynx, larynx · Lower tract: trachea, bronchi, bronchioles, alveoli.
- **Trachea** ~10–12 cm, **C-shaped cartilage rings**, bifurcates at **carina (T4–T5 / angle of Louis)**.
- **Right main bronchus** – shorter, wider, more vertical → **foreign body aspiration & ET tube misplacement** right side me.
- Right lung **3 lobes**, left lung **2 lobes** (+ lingula, cardiac notch).
- **Alveoli** ~300 million; **Type I** cells – gas exchange; **Type II** cells – **surfactant** (surface tension ↓, alveoli collapse nahi hote).
- **Main muscle of respiration = Diaphragm** (phrenic nerve, **C3-C4-C5** – "C3, 4, 5 keeps the diaphragm alive").
- Respiratory centre: **Medulla & Pons**. Main stimulus to breathe = **↑ CO_2 (H^+)**; COPD me **hypoxic drive (↓ O_2)**.
- **Tidal volume** ~500 ml · **Vital capacity** ~4800 ml · **Residual volume** ~1200 ml · **Total lung capacity** ~6000 ml · Dead space ~150 ml.

Respiratory assessment

- **Breath sounds**: Vesicular (normal, peripheral), Bronchial (over trachea).
- **Adventitious sounds**: **Crackles/rales** (fluid – pneumonia, HF; inspiration), **Wheeze** (narrow airway – asthma; expiration), **Rhonchi** (secretions in large airways – clears on coughing), **Stridor** (upper airway obstruction – **emergency**), **Pleural friction rub** (grating – pleurisy).
- **Clubbing** – chronic hypoxia (COPD, bronchiectasis, lung cancer, CHD); angle > 180°, **Schamroth window test**.
- **Barrel chest** (AP : lateral = 1:1) → emphysema. Normal AP : lateral = **1 : 2**.
- **Cyanosis**: central (lips, tongue – reduced Hb > 5 g/dL) vs peripheral (extremities).
- **Pulse oximetry** – normal 95–100 %; inaccurate with nail polish, cold extremities, anaemia, CO poisoning (falsely high).

Diagnostic tests – nursing points

- **Sputum culture** – **early morning**, deep cough, before antibiotics, rinse mouth with water (no mouthwash/toothpaste); **AFB for TB – 2 samples** (spot + early morning).
- **Bronchoscopy** – NPO 6–8 hrs; after: **NPO till gag reflex returns** (2 hrs), watch for bleeding, laryngospasm.
- **Thoracentesis** – position: **sitting upright, leaning forward over over-bed table**; max **1000–1500 ml** at a time; after: chest X-ray (pneumothorax check), lie on **unaffected side**.
- **PFT/Spirometry** – **FEV_1/FVC ratio < 0.7 (70 %)** = obstructive disease (COPD, asthma).
- **Mantoux test (TST)** – **0.1 ml (5 TU) PPD intradermal**, forearm; read at **48–72 hrs**; **induration** (not redness) measure: **≥ 10 mm positive** (≥ 5 mm in HIV/immunocompromised).

6.2 Oxygen Therapy

Device	Flow	FiO_2	Note
Nasal cannula	**1–6 L/min**	**24–44 %** (each L ≈ +4 %)	Most common; > 4 L → humidify
Simple face mask	**5–10 L/min** (min 5 to flush CO_2)	40–60 %	
Partial rebreather	6–11 L/min	60–75 %	Bag 1/3 – 1/2 full
Non-rebreather mask	**10–15 L/min**	**up to 90–100 %** (≈ 80–95 %)	Bag **never fully deflate**; highest O_2 without intubation
Venturi mask	4–12 L/min	**Precise 24–60 %**	**Best for COPD** (exact FiO_2)
High-flow nasal cannula	up to 60 L/min	21–100 %	Heated humidified

! Oxygen safety

- **COPD patient → low-flow O_2 (1–2 L/min, SpO_2 target 88–92 %)** – high O_2 se **hypoxic drive** khatam → CO_2 narcosis/apnoea.
- O_2 supports combustion → **"No Smoking"** sign, no candles/matches, cotton clothes (static), no oil/grease on equipment.
- O_2 toxicity: FiO_2 > 50 % for > 24–48 hrs.
- Cylinder colour (India): **O_2 – black body, white shoulder**; **N_2O – blue**; CO_2 – grey; Air – grey with black/white shoulder.

6.3 COPD (Chronic Obstructive Pulmonary Disease)

D COPD

Progressive, **irreversible** airflow limitation. Includes **Chronic bronchitis** + **Emphysema**. Most common cause = **cigarette smoking** (also biomass/chulha smoke, **α_1-antitrypsin deficiency** – young emphysema).

	Chronic Bronchitis – "Blue Bloater"	Emphysema – "Pink Puffer"
Definition	**Productive cough ≥ 3 months/year for 2 consecutive years**	Permanent **alveolar wall destruction**, air trapping
Look	Cyanosed, obese, oedema (cor pulmonale)	Thin, pink, **barrel chest**, **pursed-lip breathing**
Main symptom	Cough + **copious sputum**	**Dyspnoea**, minimal cough
ABG	Hypoxaemia, hypercapnia early	Near-normal till late

Management

- **Stop smoking** – most important intervention.
- **Bronchodilators**: SABA (salbutamol), LABA (salmeterol, formoterol), anticholinergic (**ipratropium, tiotropium**), theophylline (level 10–20 µg/ml; toxicity – tachycardia, seizures).
- Inhaled corticosteroids (budesonide) – **rinse mouth after** (oral candidiasis).
- **Low-flow O_2** (88–92 %), pulmonary rehab, vaccines (**influenza yearly, pneumococcal**).
- Breathing: **Pursed-lip breathing** (inhale nose 2 count, exhale lips 4 count → prevents airway collapse, CO_2 out) & **diaphragmatic/abdominal breathing**.
- Position: **Tripod / Orthopneic** (sit, lean forward on table).
- Diet: **high-calorie, high-protein, small frequent meals**; avoid gas-forming foods; fluids 2–3 L/day (unless HF) – secretions thin.
- **Chest physiotherapy**: postural drainage (before meals or 1–2 hr after), percussion, vibration.

6.4 Bronchial Asthma

- Chronic **reversible** airway inflammation → bronchospasm + mucus + oedema.
- Triggers: allergens (dust, pollen, mites), cold air, exercise, smoke, **aspirin/NSAIDs**, **beta-blockers**, stress, infection.

- S/S: **expiratory wheeze**, dyspnoea, chest tightness, cough (worse at night/early morning).
- **Status asthmaticus** – severe attack not responding to usual therapy → **"silent chest" (no wheeze) = ominous sign** (very little air movement) → respiratory failure.

Asthma drugs

- **Reliever (rescue)**: **SABA – Salbutamol (Albuterol)** – drug of choice for acute attack.
- **Controller**: Inhaled corticosteroids (most effective long-term), LABA, **Montelukast** (leukotriene antagonist), cromolyn.
- Status asthmaticus: O_2, nebulized salbutamol + ipratropium, **IV hydrocortisone/methylprednisolone**, **IV magnesium sulphate**, aminophylline.



MDI (Inhaler) technique

Shake → **exhale fully** → mouthpiece in mouth → press & **inhale slowly & deeply** → **hold breath 10 sec** → wait **1 min** before second puff. **Bronchodilator pehle, steroid baad me** (steroid deep jaye), steroid ke baad **mouth rinse**. Spacer use better.
Peak flow meter: Green 80–100 % (OK), Yellow 50–80 % (caution), Red < 50 % (emergency).

6.5 Pneumonia

- Lung parenchyma ki **inflammation/infection** – alveoli me exudate (consolidation).
- Types: **Community-acquired (CAP)** – most common organism **Streptococcus pneumoniae**; **Hospital-acquired (HAP)** – ≥ 48 hrs after admission (Pseudomonas, Klebsiella, MRSA); **VAP** (ventilator, ≥ 48 hrs after intubation); **Aspiration pneumonia** – right lower lobe, stroke/unconscious patients.
- S/S: high fever with chills, **productive cough (rusty sputum – pneumococcal)**, **pleuritic chest pain**, tachypnoea, crackles, **bronchial breath sounds**, ↑ tactile fremitus, **dull percussion** over consolidation. Elderly → **confusion** may be first sign.
- Diagnosis: chest X-ray (consolidation), sputum culture, CBC.
- Rx: antibiotics, O_2, fluids **2.5–3 L/day**, antipyretics, **semi/high Fowler's**, deep breathing, incentive spirometry, turn every 2 hrs, ambulation.
- Prevention of VAP: **HOB 30–45°**, oral care with chlorhexidine, daily sedation vacation, hand hygiene.

6.6 Pulmonary Tuberculosis

- Organism: Mycobacterium tuberculosis (acid-fast bacillus, **aerobic** – lung apex me), discovered by **Robert Koch (1882)** – **24 March = World TB Day**. Transmission: **airborne droplet nuclei**.
- S/S: **cough > 2 weeks** (presumptive TB), **evening rise of temperature**, **night sweats**, **weight loss**, anorexia, haemoptysis, fatigue.
- Diagnosis: **Sputum smear (AFB, Ziehl-Neelsen stain)**, **CBNAAT/GeneXpert** (detects TB + **rifampicin resistance in 2 hrs**) / TrueNat, chest X-ray, culture (LJ medium – gold standard, 6–8 weeks), Mantoux, IGRA.
- **NTEP (National TB Elimination Programme)** – earlier RNTCP; India goal: **End TB by 2025** (SDG 2030). **Nikshay Poshan Yojana** – ₹500/month → **₹1000/month (from Nov 2024)** nutrition support. DOTS = Directly Observed Treatment Short-course.

Drug (HRZE)	Key side effect	Nursing tip
Isoniazid (H)	**Peripheral neuropathy**, hepatotoxicity	Give **Pyridoxine (Vit B6)**
Rifampicin (R)	**Orange-red urine, sweat, tears** (harmless), hepatotoxicity, ↓ OCP effect	Soft contact lenses stain; use other contraception
Pyrazinamide (Z)	**Hyperuricaemia (gout)**, hepatotoxicity	Fluids
Ethambutol (E)	**Optic neuritis – red-green colour blindness**, ↓ visual acuity	Monthly eye check; avoid in young children
Streptomycin (S)	**Ototoxicity (CN VIII)**, nephrotoxicity	Hearing check

- Regimen (drug-sensitive, new): **Intensive phase 2 months HRZE + Continuation phase 4 months HRE** (daily, weight-band FDCs) = 6 months.
- **MDR-TB** = resistant to **INH + Rifampicin**; XDR-TB = MDR + fluoroquinolone + Bedaquiline/linezolid resistance. Shorter oral **BPaLM** regimen (Bedaquiline, Pretomanid, Linezolid, Moxifloxacin) – 6 months.
- Nursing: **airborne isolation, N95** for staff, surgical mask for patient during transport; cover mouth while coughing; sputum disposal (burn / disinfect); **treatment adherence** (most important); nutritious diet; non-infectious after ~**2–3 weeks** of effective treatment.
- **BCG vaccine** – at birth, 0.05 ml (infant < 1 month) / 0.1 ml, **intradermal, left upper arm** – protects mainly against severe childhood TB (TB meningitis, miliary).

6.7 Pleural Disorders & Chest Injuries

- **Pleural effusion** – pleural space me fluid (> 15 ml normal). Transudate (HF, cirrhosis) vs Exudate (TB, cancer, pneumonia). S/S: dyspnoea, ↓ breath sounds, **stony dull percussion**. Rx: **Thoracentesis**, treat cause, pleurodesis.
- **Empyema** – pus in pleural space → chest tube drainage.
- **Pneumothorax** – air in pleural space → lung collapse. S/S: sudden **sharp chest pain**, dyspnoea, **absent breath sounds** on affected side, **hyper-resonant** percussion.
  - **Spontaneous** – tall thin young males, smokers.
  - **Tension pneumothorax** – air enters but can't exit → **mediastinal & tracheal shift to OPPOSITE side**, severe hypotension, distended neck veins → **EMERGENCY: needle decompression at 2nd ICS mid-clavicular line** (or 4th/5th ICS anterior-axillary) then chest tube.
  - **Open pneumothorax / sucking chest wound** → cover with **3-sided occlusive dressing** (air bahar jaa sake, andar na aaye).
- **Haemothorax** – blood in pleural space.
- **Flail chest** – **2 or more adjacent ribs fractured at 2 or more places** → **paradoxical chest movement** (inspiration me andar, expiration me bahar). Rx: O_2, analgesia, mechanical ventilation (internal splinting).
- **Rib fracture** – avoid tight strapping (atelectasis); analgesics, deep breathing.

Chest Tube & Underwater Seal Drainage (UWSD)

Chest tube – sabse important points

- Site: **Air (pneumothorax) → 2nd ICS mid-clavicular** (air upar jati hai); **Fluid/blood → 5th–6th ICS mid-axillary** (fluid neeche).
- **3 chambers**: (1) **Collection chamber**, (2) **Water-seal chamber** (2 cm water – one-way valve), (3) **Suction control chamber** (usually 20 cm H_2O).
- **Tidaling (fluctuation)** in water-seal chamber – inspiration par upar, expiration par neeche (spontaneous breathing) = **normal, tube patent**. Tidaling band = lung re-expanded OR tube blocked/kinked.
- **Continuous bubbling** in water-seal = **air leak** → check connections. Intermittent bubbling (cough/exhale) in pneumothorax = normal. **Gentle continuous bubbling in suction chamber = normal.**
- Drainage bottle **always below chest level** (60–90 cm); never lift above chest.
- **No routine clamping** (tension pneumothorax risk). Clamp only briefly to locate leak / change bottle / as ordered.
- **Tube accidentally pulls out → cover site immediately with sterile occlusive (petroleum) gauze** → inform.
- **Tube disconnects from bottle → put the end in sterile water (2 cm below surface)** → reconnect new system.
- Drainage > **100–200 ml/hr** or bright red → inform. Milking/stripping – only if ordered.
- Removal: patient does **Valsalva / holds breath at end of full inspiration**; occlusive dressing; chest X-ray after.
- Keep **2 haemostats (clamps)** & sterile water/vaseline gauze at bedside.

6.8 Other Important Conditions

- **ARDS** – acute non-cardiogenic pulmonary oedema; causes: sepsis (most common), trauma, aspiration, pancreatitis. Hallmark: **refractory hypoxaemia** (not responding to ↑ O_2); **"white-out" / ground-glass** on X-ray; $PaO_2/FiO_2 \leq 300$. Rx: mechanical ventilation with **low tidal volume (6 ml/kg)**, **PEEP**, **prone positioning**, treat cause.
- **Pulmonary embolism** – mostly from **DVT of legs**. **Sudden dyspnoea (most common), pleuritic chest pain, tachycardia, anxiety, haemoptysis**, ↓SpO_2. Diagnosis: **CT pulmonary angiography** (gold standard now), D-dimer, V/Q scan. Rx: **High Fowler's, O_2**, heparin, thrombolysis (massive), IVC filter.
- **Bronchiectasis** – permanent dilation of bronchi; **copious foul-smelling sputum (3-layered)**, haemoptysis, clubbing → **postural drainage**.
- **Lung abscess** – foul sputum; aspiration; antibiotics, postural drainage.
- **Occupational lung diseases**: **Silicosis** (stone/quarry), **Asbestosis** (asbestos – mesothelioma), **Anthracosis/Black lung** (coal), **Byssinosis** (cotton – Monday fever), **Bagassosis** (sugarcane), Farmer's lung (mouldy hay).
- **Lung cancer** – most common cause **smoking**; types: **Non-small cell (85 %)** – adenocarcinoma (most common, non-smokers & women), squamous cell (central, smokers); **Small cell (oat cell)** – most aggressive, central, **SIADH/paraneoplastic**. S/S: **persistent cough / change in cough** (most

common), haemoptysis, weight loss, hoarseness. **Pancoast tumour** (apex) → **Horner's syndrome** (ptosis, miosis, anhidrosis).

- **Pneumonectomy** → position **supine or operated side down** (healthy lung expand kare); **no chest tube usually** (space fluid se bhar jata hai). **Lobectomy** → chest tube present; position on **unaffected side** or back as ordered.

6.9 Tracheostomy & Suctioning

- Tracheostomy – opening in trachea (**2nd–3rd tracheal ring**). Indications: prolonged ventilation, upper airway obstruction, secretion removal.
- **Obturator** & spare tube (same + one size smaller) **bedside** rakho.
- First tube change by surgeon **after 5–7 days** (tract mature).
- **Accidental decannulation** (first 72 hrs) → **call for help, use obturator / hold stoma open with tracheal dilator**, O_2; if fresh tract – bag-mask.
- Cuff pressure **20–25 mmHg (25–30 cmH_2O)** – zyada se tracheal necrosis.
- Tie change: 2 persons; **one finger** space under ties.

! Suctioning rules

- **Hyperoxygenate (100 % O_2) before & after** – 30–60 sec.
- **Sterile technique** (trach/ET); oral suction (Yankauer) – clean.
- Suction **only while withdrawing**, **intermittent with rotation**.
- Each pass **max 10–15 seconds**; max **3 passes**; rest 20–30 sec between.
- Pressure: adult **100–150 mmHg** (80–120 common); child 80–100; infant 60–80.
- Catheter size ≤ **half the internal diameter** of tube.
- Insert **without suction**; stop if bradycardia/desaturation.

? Practice MCQs – Chapter 6

1. Foreign body aspiration zyada kis side hota hai?
   (a) Both equal (b) Trachea (c) Right bronchus (d) Left bronchus
2. COPD patient ke liye best O_2 delivery device –
   (a) Simple face mask (b) Nasal cannula 6 L (c) Non-rebreather (d) Venturi mask
3. Mantoux test kitne ghante baad padhte hain?
   (a) 6–12 (b) 24 (c) 48–72 (d) 96
4. Rifampicin lene par urine ka colour –
   (a) Orange-red (b) Blue (c) Black (d) Green
5. Ethambutol ka side effect –
   (a) Gout (b) Neuropathy (c) Optic neuritis / red-green colour blindness (d) Ototoxicity
6. Tension pneumothorax me trachea shift hoti hai –
   (a) Same side (b) Opposite side (c) No shift (d) Downward
7. Chest drainage bottle me continuous bubbling in water-seal chamber –
   (a) Normal (b) Air leak (c) Lung expanded (d) Tube blocked
8. Chest tube accidentally bahar nikal jaye to –
   (a) Bottle upar uthao (b) Kuch na karo (c) Site ko sterile occlusive dressing se dhako (d) Wapas daalo

9. Thoracentesis ke liye position –
(a) Sitting, leaning forward on table (b) Lithotomy (c) Prone (d) Supine

10. Status asthmaticus me ominous sign –
(a) Silent chest (b) Cough (c) Loud wheeze (d) Tachycardia

11. Suction ka maximum time per pass –
(a) 5 sec (b) 10–15 sec (c) 30 sec (d) 1 min

12. Flail chest me characteristic sign –
(a) Barrel chest (b) Paradoxical chest movement (c) Funnel chest (d) Pigeon chest

13. INH ke saath kaunsa vitamin dete hain?
(a) C (b) B12 (c) B1 (d) B6 (pyridoxine)

14. Pursed-lip breathing ka main purpose –
(a) Sputum ↑ (b) O_2 ↑ (c) Airway collapse rokna & CO_2 nikalna (d) Cough ↑

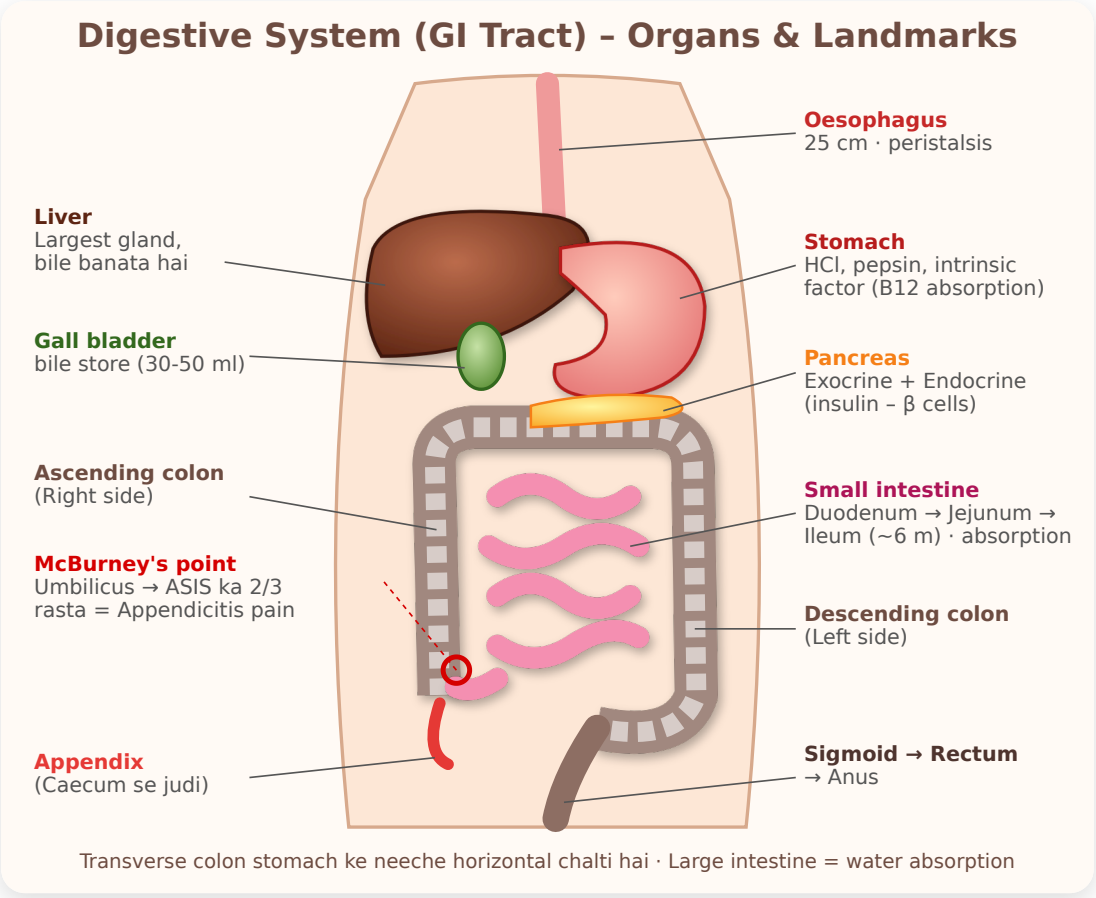
15. Pneumonectomy ke baad position –
(a) Trendelenburg (b) Prone (c) Operated side up only (d) Supine / operated side down

Answer Key – Chapter 6

1-c · 2-d · 3-c · 4-a · 5-c · 6-b · 7-b · 8-c · 9-a · 10-a · 11-b · 12-b · 13-d · 14-c · 15-d

7 Gastrointestinal System Disorders

7.1 Applied Anatomy



GI tract ke organs aur important landmarks

Digestion – quick facts

- GI tract length ~**9 m** (mouth → anus). Layers: **mucosa → submucosa → muscularis → serosa**.
- **Saliva**: 1–1.5 L/day, **ptyalin (salivary amylase)** – starch digestion starts in mouth.
- **Stomach**: HCl (parietal cells) – also **intrinsic factor** (B12 absorption); **pepsinogen** (chief cells); **gastrin** (G cells). Capacity ~1.5 L. Gastric emptying 2–6 hrs.
- **Small intestine** (~6 m): duodenum (25 cm, C-shaped) → jejunum → ileum. **Main site of digestion & absorption**. **B12 & bile salts absorbed in terminal ileum**. Iron absorbed in **duodenum**.
- **Large intestine** (~1.5 m): caecum, appendix, colon (ascending, transverse, descending, sigmoid), rectum, anus → **water & electrolyte absorption**, Vit K synthesis by bacteria.
- **Pancreas**: exocrine (trypsin, lipase, amylase) + endocrine (insulin, glucagon).

Abdominal quadrants & regions

Quadrant	Organs
RUQ	Liver, gallbladder, duodenum, head of pancreas, right kidney
LUQ	Stomach, spleen, body/tail of pancreas, left kidney

Quadrant	Organs
RLQ	**Appendix**, caecum, right ovary/tube
LLQ	Sigmoid colon, left ovary/tube

- 9 regions: Right/left hypochondriac, **epigastric**, right/left lumbar, **umbilical**, right/left iliac (inguinal), **hypogastric (suprapubic)**.
- Bowel sounds: normal **5–30/min** (5–35); absent = **no sound in 3–5 min** (listen all 4 quadrants) → paralytic ileus/peritonitis; **high-pitched tinkling** = early obstruction.

GI investigations – nursing points

- **Barium swallow / meal** (upper GI) – NPO 6–8 hrs; after: **fluids & laxative** (barium → constipation, **white chalky stool for 24–72 hrs**).
- **Barium enema** (lower GI) – clear liquid diet + bowel prep; position **left lateral/Sims**.
- **Upper GI endoscopy (OGD)** – NPO 6–8 hrs, local throat spray; after: **NPO till gag reflex returns**, watch for perforation (pain, fever, bleeding).
- **Colonoscopy** – clear liquid diet 1–2 days, **bowel prep (PEG solution)** till clear yellow output; left lateral with knees flexed; after: watch bleeding, perforation; flatus normal.
- **Stool for occult blood (guaiac)** – avoid red meat, iron, vitamin C, NSAIDs 3 days before.
- **Paracentesis** – **empty bladder first** (bladder puncture prevention), sitting upright/high Fowler's; measure abdominal girth & weight before & after; max **4–5 L** at a time (albumin for large volume).
- **Liver biopsy** – check **PT/INR & platelets**, vitamin K before; position **supine with right arm above head**, patient **holds breath after exhalation** during insertion; after: **right lateral position for 2–3 hrs** with pillow under costal margin.

7.2 Disorders of Mouth & Oesophagus

- **Stomatitis** – mouth inflammation; soft bland diet, saline rinse, avoid hot/spicy; chemo patients – soft brush.
- **Oral cancer** – most common in India (male), **tobacco chewing** (gutkha), leukoplakia (white patch – premalignant), **erythroplakia (red patch – more malignant)**.
- **Achalasia** – LES fails to relax → **dysphagia to both solids & liquids**, regurgitation; "bird's beak" on barium; Rx: pneumatic dilatation, **Heller's myotomy**.
- **Oesophageal cancer** – progressive **dysphagia (solids → liquids)**, weight loss; risk: smoking, alcohol, Barrett's oesophagus (adenocarcinoma).

GERD (Gastro-Oesophageal Reflux Disease)

- **LES weakness** → acid reflux into oesophagus. Risk: obesity, pregnancy, smoking, fatty food, caffeine, chocolate, alcohol, hiatus hernia.
- S/S: **heartburn (pyrosis)** – worse after meals & lying down, regurgitation, dysphagia, chronic cough. Complication: **Barrett's oesophagus** (premalignant – squamous → columnar).
- Rx: **PPIs (omeprazole, pantoprazole – 30 min before breakfast)**, H_2 blockers (ranitidine/famotidine), antacids, prokinetics.

GERD / Hiatus hernia – lifestyle teaching

- **Small frequent meals**; avoid eating **2–3 hrs before bedtime**
- **Elevate head of bed 15–20 cm (6–8 inches)** – blocks under head legs
- Avoid fatty food, chocolate, peppermint, caffeine, alcohol, spicy/acidic foods, smoking
- Weight loss; avoid tight clothes & bending/lifting after meals
- Remain **upright 1–2 hrs after meals**

- **Hiatus hernia** – stomach part diaphragm ke upar chest me; **sliding (most common)** vs paraesophageal (rolling). Same management as GERD; surgery – **Nissen fundoplication**.

7.3 Gastritis & Peptic Ulcer Disease

- **Gastritis** – acute (NSAIDs, alcohol, stress) / chronic (**H. pylori**, autoimmune – pernicious anaemia).
- **Peptic ulcer** – erosion of mucosa in stomach/duodenum. Most common cause: **Helicobacter pylori** (gram –ve, spiral; produces urease); 2nd: **NSAIDs**. Others: Zollinger-Ellison syndrome (gastrin-secreting tumour), stress (**Curling's ulcer – burns**; **Cushing's ulcer – head injury**).

Feature	Gastric ulcer	Duodenal ulcer (more common, ~80 %)
Age	> 50 yrs	30–60 yrs
Pain timing	**½–1 hr after meal**; food **worsens** pain	**2–3 hrs after meal & at night (midnight–3 am)**; food **relieves** pain
Acid	Normal / low	**High**
Weight	**Weight loss**	Weight gain (eats to relieve pain)
Vomiting	Common	Uncommon
Malignancy	Possible	Rare
Bleeding	Haematemesis more	**Melena** more

G vs D

Gastric = **G**ets worse with food (pain soon after eating).
Duodenal = **D**ecreases with food (pain when stomach empty / night).

- Diagnosis: **upper GI endoscopy with biopsy** (gold standard), **H. pylori tests** – **urea breath test**, stool antigen, rapid urease (CLO) test, serology.
- **H. pylori triple therapy** (14 days): **PPI + Clarithromycin + Amoxicillin** (or Metronidazole); quadruple – PPI + bismuth + metronidazole + tetracycline.
- Drugs: PPIs, H_2 blockers, **Sucralfate** (coats ulcer – **give 1 hr before meals**, separate from other drugs by 2 hrs), **Misoprostol** (NSAID-induced; **contraindicated in pregnancy**), antacids (1–3 hrs after meals & HS; **Al – constipation, Mg – diarrhoea**).
- Diet: regular meals, avoid foods that cause pain, **avoid alcohol, smoking, caffeine, NSAIDs**; bland diet & milk **no longer recommended** (milk → rebound acid).

Complications of peptic ulcer

- **Haemorrhage** (most common) – haematemesis (**coffee-ground vomitus**), melena (**black tarry stool** – needs ~60 ml blood in upper GI). Rx: 2 large bore IV, blood, **NG lavage (room temperature saline)**,

IV PPI, endoscopic therapy.

- **Perforation** (most serious) – **sudden severe "knife-like" epigastric pain spreading to shoulder (referred)**, **board-like rigid abdomen**, rebound tenderness, absent bowel sounds, shock; X-ray – **air under diaphragm**. Rx: NPO, NG suction, IV fluids, antibiotics, **emergency surgery**.
- **Pyloric (gastric outlet) obstruction** – projectile vomiting of undigested food, visible peristalsis, succussion splash.

Gastric surgery & Dumping syndrome

- Surgeries: **Billroth I** (gastroduodenostomy), **Billroth II** (gastrojejunostomy), vagotomy, pyloroplasty, total gastrectomy.
- **Dumping syndrome** – food (hypertonic chyme) rapidly enters jejunum.
  - **Early (10–30 min)**: fluid shift → dizziness, sweating, palpitation, tachycardia, cramps, diarrhoea, fainting.
  - **Late (1.5–3 hr)**: reactive **hypoglycaemia**.



Dumping syndrome diet – "Dry, Lie, Protein-fat high"

- **Small frequent meals**, **high protein, moderate fat, LOW carbohydrate** (no simple sugars)
- **No fluids with meals** – drink **30–60 min before/after** meals
- **Lie down (recumbent/semi-recumbent) 20–30 min after meals**
- Total gastrectomy → lifelong **Vitamin B12 injections** (no intrinsic factor → pernicious anaemia)

- **Gastric cancer** – risk: H. pylori, smoked/salted foods, nitrates; **Virchow's node** (left supraclavicular), **Krukenberg tumour** (ovary metastasis), Sister Mary Joseph nodule (umbilicus).

7.4 Upper vs Lower GI Bleeding

	Upper GI (above ligament of Treitz)	Lower GI
Causes	Peptic ulcer (most common), **oesophageal varices**, Mallory-Weiss tear (vomiting)	Diverticulosis, haemorrhoids, colorectal cancer, IBD
Signs	**Haematemesis, melena**	**Haematochezia** (bright red blood per rectum)

7.5 Appendicitis

- Inflammation of **vermiform appendix** (obstruction by **faecalith** most common); age 10–30 yrs; most common surgical emergency.
- S/S: **periumbilical pain → shifts to RLQ (McBurney's point)** within hours, anorexia (almost always), nausea/vomiting, low-grade fever, ↑WBC.

Signs of appendicitis

- **McBurney's point tenderness** – 1/3 distance from **ASIS to umbilicus** (from ASIS side; i.e. junction of lateral 1/3 & medial 2/3)
- **Rovsing's sign** – pressure on **LLQ → pain in RLQ**
- **Rebound tenderness (Blumberg's sign)** – peritoneal irritation
- **Psoas sign** – hip extension se pain (retrocaecal)
- **Obturator sign** – internal rotation of flexed hip se pain (pelvic appendix)
- **Sudden relief of pain = possible perforation (danger!)** → peritonitis

Suspected appendicitis me NEVER

- **No heat (hot pad)** on abdomen → rupture
- **No laxatives / enema** → perforation
- **No pain killers (strong)** before diagnosis (mask symptoms) – as per doctor
- Keep **NPO**, IV fluids, ice pack may be used; position **semi-Fowler's / right side-lying with knees flexed**

- Rx: **Appendectomy** (laparoscopic). Perforation → peritonitis → drain; position **Fowler's** (localize infection in pelvis).

7.6 Peritonitis

- Inflammation of peritoneum – most common cause: **perforated appendix / peptic ulcer**; E. coli.
- S/S: severe abdominal pain, **rigid board-like abdomen**, rebound tenderness, **absent bowel sounds**, fever, tachycardia, shock; patient lies **still with knees flexed**.
- Rx: NPO, **NG suction**, IV fluids & antibiotics, analgesics, **semi-Fowler's**, surgery.

7.7 Inflammatory Bowel Disease (IBD)

Feature	Crohn's disease	Ulcerative colitis
Area	**Mouth to anus** (any part), mostly **terminal ileum**	**Colon & rectum only** (starts in rectum, goes up)
Pattern	**Skip lesions**, transmural (full thickness)	**Continuous**, mucosal only
Stool	3–5/day, **non-bloody**, fatty (steatorrhoea)	**10–20/day, bloody diarrhoea with mucus**
Look	**Cobblestone appearance**, fistulas, strictures, abscess	Pseudopolyps, **"lead-pipe" colon**
Complications	Fistula, obstruction, malabsorption (B12)	**Toxic megacolon**, haemorrhage, **colon cancer risk ↑**
Surgery	Not curative (recurs)	**Total colectomy = curative**
Smoking	Worsens	(Protective)

- Rx: **5-ASA (sulfasalazine/mesalamine)**, corticosteroids, immunosuppressants (azathioprine), biologics (infliximab); **low-residue, high-protein, high-calorie diet** during flare; avoid milk (lactose), caffeine, raw fruits; TPN for bowel rest.

- **Irritable Bowel Syndrome (IBS)** – functional; alternating diarrhoea/constipation, pain relieved by defecation; no structural damage; stress-related; high-fibre diet.
- **Diverticulosis** – sac-like pouches (sigmoid), low-fibre diet; **Diverticulitis** – inflamed → **LLQ pain**, fever ("left-sided appendicitis"). Acute phase: NPO/clear liquids, antibiotics; later **high-fibre diet**; avoid straining.

7.8 Intestinal Obstruction

	Mechanical	Paralytic (Adynamic) ileus
Cause	**Adhesions (most common in small bowel)**, hernia (strangulated), volvulus, intussusception, tumour (**most common large bowel**)	**Post-abdominal surgery (most common)**, hypokalaemia, peritonitis, opioids
Bowel sounds	**High-pitched, hyperactive** above obstruction (early), later absent	**Absent / hypoactive**

- **Small bowel obstruction**: early **profuse vomiting (faecal vomiting later)**, colicky pain, mild distension, dehydration, **metabolic alkalosis**.
- **Large bowel obstruction**: **marked distension**, constipation/**obstipation** (no flatus), vomiting late, cramping lower abdominal pain.
- Rx: **NPO, NG tube decompression (Ryle's)**, IV fluids & electrolytes, **I/O chart**, abdominal girth measurement, surgery if strangulation.
- **Volvulus** – twisting (sigmoid – elderly); **Intussusception** – telescoping (infants – **red currant jelly stool**, sausage-shaped mass).

7.9 Hernia

- Protrusion of organ through weak muscle wall.
- **Inguinal** (most common overall) – **Indirect** (most common; through inguinal ring – young males, congenital) / Direct (through posterior wall – elderly); **Femoral** (more in females); **Umbilical**; **Incisional** (surgical scar); Hiatal.
- **Reducible** → **Irreducible (incarcerated)** → **Strangulated** (blood supply cut – **emergency**: severe pain, vomiting, tender, red mass, fever).
- Pre-op: avoid straining, **truss** (reducible only). Post-op herniorrhaphy: **avoid coughing** (deep breathing ok – **splint incision**), **scrotal support & ice bag** (scrotal swelling), avoid heavy lifting **6 weeks**, stool softeners.

7.10 Haemorrhoids, Fissure & Fistula

- **Haemorrhoids (piles)** – dilated veins of anal canal; **internal (painless bright red bleeding)** above dentate line; external (painful, thrombosed). Causes: constipation, straining, pregnancy, portal HTN. Rx: **high-fibre diet, fluids, sitz bath (warm 3–4 times/day)**, ointments, banding, **haemorrhoidectomy** (post-op: **pain severe**, first BM painful – stool softener, sitz bath, watch bleeding & urinary retention, side-lying).
- **Anal fissure** – tear, **severe pain on defecation**, streak of blood; sitz bath, high fibre.
- **Fistula-in-ano** – abnormal tract; pus discharge; fistulotomy, **seton**.
- **Pilonidal sinus** – sacrococcygeal, hairy young men ("jeep driver's disease").

7.11 Colorectal Cancer

- Most are **adenocarcinoma**; risk: age > 50, **low-fibre high-fat diet**, red meat, polyps, IBD (UC), family history, smoking, obesity.
- S/S: **change in bowel habits (most common)**, rectal bleeding, anaemia (right side – occult blood, fatigue), **ribbon/pencil-like stools** (left/rectal), tenesmus, weight loss.
- Screening: **stool occult blood yearly from 45–50 yrs**, **colonoscopy every 10 yrs**. Marker: **CEA** (monitoring recurrence).
- Surgery: **Abdominoperineal resection (APR)** → permanent colostomy; post-op perineal wound – sitz bath, side-lying.

Stoma (Colostomy / Ileostomy) Care

Stoma – nursing points

- **Healthy stoma = beefy red/pink, moist, slightly raised**. **Pale = anaemia**; **dark red/purple/black = ischaemia/necrosis → report immediately**.
- Mild oedema first 6 weeks normal; stoma functions in **2–4 days** (colostomy).
- **Ileostomy** – **liquid, continuous, enzyme-rich** output → **skin excoriation risk**, **dehydration & electrolyte loss** → fluids 2–3 L/day; no irrigation.
- **Colostomy** – sigmoid/descending → **formed stool**; **irrigation possible** (sigmoid colostomy) – 500–1000 ml **lukewarm water**, bag hung **45–50 cm (18–20 inch) above stoma**, patient sitting; cone tip; time ~ same daily.
- Transverse colostomy – semi-formed/mushy.
- Pouch **empty when 1/3 – 1/2 full**; change every 3–7 days; opening **1/16–1/8 inch (2–3 mm) larger** than stoma.
- Skin: clean with **warm water**, pat dry; avoid alcohol-based products.
- Odour/gas control: avoid **beans, cabbage, onion, eggs, fish, beer, carbonated drinks**; yogurt, buttermilk, parsley reduce odour.
- Ileostomy – avoid **high-fibre foods (nuts, corn, popcorn, raw vegetables)** → blockage; chew well.
- Body image – encourage patient to **look at & touch stoma**; **Ostomy association**.

7.12 Nasogastric (Ryle's) Tube & Enteral Feeding

- Length: **NEX** – **N**ose tip → **E**arlobe → **X**iphoid (+ few cm). Position: **high Fowler's**, head flexed (chin to chest) during insertion; patient swallows (sips of water).
- **Placement check**: **X-ray = gold standard (most accurate)**; bedside – **aspirate gastric content pH ≤ 5 (1–5)**; auscultation (whoosh test) **no longer reliable**.
- Before each feed: check placement & **residual volume** (> 250–500 ml as per policy → hold/inform); return aspirate.
- Feed by **gravity**, syringe held **15–30 cm (6–12 inch) above stomach**; **flush 30 ml water** before & after feed/medicine.
- Patient **semi-Fowler's (30–45°) during feed & 30–60 min after** → aspiration prevention.
- Feed temperature: **room temperature**; continuous feed bag change every 24 hrs (hang time 4–8 hrs for open system).

- **Diarrhoea** – most common complication of tube feeding; **aspiration** – most serious.
- Removal: patient **holds breath**, quick steady pull.

Practice MCQs – Chapter 7

1. Duodenal ulcer ka pain typically –
   (a) Sirf subah (b) Khana khate hi badhta hai (c) Dard nahi hota (d) Khane ke 2–3 hrs baad & raat me, food se relief
2. Peptic ulcer ka most common cause –
   (a) H. pylori (b) Stress (c) Alcohol (d) Spicy food
3. Rovsing's sign kis disease me?
   (a) Peritonitis (b) Pancreatitis (c) Cholecystitis (d) Appendicitis
4. Suspected appendicitis me kya contraindicated hai?
   (a) Heating pad & enema (b) Ice pack (c) NPO (d) IV fluid
5. Dumping syndrome me advice –
   (a) High carbohydrate (b) Large meals (c) Fluids with meals (d) Lie down after meals, no fluids with meals
6. NG tube placement check ka most accurate method –
   (a) Bubbles in water (b) Air insufflation sound (c) Patient ki awaaz (d) X-ray
7. Healthy stoma ka colour –
   (a) Pale white (b) Dark purple (c) Black (d) Beefy red, moist
8. Ulcerative colitis ka characteristic –
   (a) Bloody diarrhoea, continuous lesions from rectum (b) Skip lesions (c) Cobblestone (d) Fistula
9. Colostomy irrigation me bag ki height stoma se –
   (a) 15 cm (b) 45–50 cm (c) 1 m (d) 2 m
10. Red currant jelly stool kis condition me?
    (a) Crohn's (b) Volvulus (c) Intussusception (d) Hernia
11. Liver biopsy ke time patient breath –
    (a) Normal breathing (b) Exhale and hold (c) Deep inspiration hold (d) Pant fast
12. Perforated peptic ulcer ka X-ray finding –
    (a) Bird beak (b) Ground glass (c) Air under diaphragm (d) Step ladder pattern
13. Barium meal ke baad advice –
    (a) NPO 24 hrs (b) Bed rest (c) Fluids & laxative (d) Antacids
14. GERD me head of bed –
    (a) Trendelenburg (b) Flat (c) Elevated 15–20 cm (d) Legs up
15. Total gastrectomy ke baad lifelong kya dena padta hai?
    (a) Vit B12 injection (b) Iron only (c) Vit K (d) Insulin

Answer Key – Chapter 7

1-d · 2-a · 3-d · 4-a · 5-d · 6-d · 7-d · 8-a · 9-b · 10-c · 11-b · 12-c · 13-c · 14-c · 15-a

8 Liver, Biliary System & Pancreas

8.1 Liver – Applied Physiology

Liver – facts

- **Largest internal organ & largest gland** (~1.5 kg), RUQ; **dual blood supply** – hepatic artery (25 %, O_2) + **portal vein (75 %, nutrients)**. Functional unit = **lobule**; **Kupffer cells** (macrophages).
- Functions: **bile production (500–1000 ml/day)**, glycogen storage, **protein synthesis (albumin, clotting factors II, VII, IX, X – Vit K dependent)**, detoxification (drugs, **ammonia → urea**), bilirubin metabolism, storage of **Vit A, D, B12, iron**.

Liver Function Tests (LFT)

Test	Normal
Total bilirubin	**0.3 – 1.0 (up to 1.2) mg/dL**
Direct (conjugated)	0.1 – 0.3 mg/dL
SGPT / ALT (**most specific for liver**)	7 – 56 U/L
SGOT / AST	10 – 40 U/L
Alkaline phosphatase (ALP)	40 – 130 U/L (↑ in obstructive jaundice, bone disease)
Serum albumin	**3.5 – 5.0 g/dL**
Total protein	6 – 8 g/dL
PT	11 – 13.5 sec (INR 0.8–1.2)
Serum ammonia	15 – 45 µg/dL

- **Jaundice (icterus)** – clinically visible when bilirubin **> 2–3 mg/dL**; first seen in **sclera** (also under tongue, palms).

Type	Cause	Features
Pre-hepatic (haemolytic)	Excess RBC breakdown – sickle cell, malaria, transfusion reaction	**Unconjugated ↑**, urine normal colour, dark stool
Hepatic (hepatocellular)	Hepatitis, cirrhosis, drugs	Both ↑, dark urine
Post-hepatic (obstructive)	Gallstones in CBD, **carcinoma head of pancreas**	**Conjugated ↑, dark (tea-coloured) urine, clay-coloured (pale) stool, pruritus** (bile salts in skin)

- **Pruritus care**: cool baths, **baking soda / calamine**, short nails, cotton clothes, **cholestyramine**.

8.2 Viral Hepatitis

	Hep A	Hep B	Hep C	Hep D	Hep E
Virus	RNA (Picorna)	**DNA** (only DNA one)	RNA	RNA (defective – **needs HBV**)	RNA
Route	**Faeco-oral**	**Blood, sexual, perinatal**	**Blood** (IV drug, transfusion)	Blood (with B)	**Faeco-oral** (water)
Incubation	15–50 (~30) days	**30–180 (~60–90) days**	15–160 (~50) days		15–64 days
Chronicity	No	**Yes** (5–10 % adults, **90 % neonates**)	**Yes – most chronic (75–85 %)**	Yes	No
Vaccine	**Yes**	**Yes** (0, 1, 6 months)	**No**	(HBV vaccine protects)	No (available in China)
Special	Children; mild	**Carrier state**, HCC risk	Cirrhosis, HCC; curable with DAAs	Superinfection – severe	**Pregnant women – high mortality (~20 %)**

♫ Vowels hit your Bowels

Hepatitis **A** & **E** (vowels) → **faeco-oral (bowels)**. Baaki (B, C, D) → **blood/body fluids**.

Hepatitis B markers

Marker	Meaning
HBsAg	Infection present (acute or chronic) – **first marker**
Anti-HBs	**Immunity** (recovery or vaccination)
HBeAg	**High infectivity / active replication**
Anti-HBc IgM	**Acute / recent infection**
Anti-HBc IgG	Past/chronic infection

- **Phases of hepatitis**: **Pre-icteric (prodromal)** – anorexia (most common), nausea, fatigue, **aversion to smoking/food smell**, RUQ discomfort, flu-like → **Icteric** – jaundice, dark urine, clay stools → **Post-icteric (convalescent)**.
- Management: **rest**, **high-calorie, high-carbohydrate, moderate protein, low-fat diet**, small frequent meals, avoid **alcohol & hepatotoxic drugs (paracetamol)**; antivirals for chronic B (**tenofovir, entecavir**) & C (**sofosbuvir-based DAAs**).
- **Post-exposure (Hep B)**: unvaccinated → **HBIG (within 24 hrs, max 7 days) + vaccine**. **Newborn of HBsAg +ve mother → HBIG + Hep B vaccine within 12 hrs of birth** (different sites).
- Hep B vaccine: **IM deltoid** (adult), anterolateral thigh (infant); **NOT gluteal** (poor immune response).
- **National Viral Hepatitis Control Programme** launched **28 July 2018** – **28 July = World Hepatitis Day** (birthday of Baruch Blumberg, discoverer of HBV).

8.3 Liver Cirrhosis

D **Cirrhosis**

Chronic, **irreversible** liver disease – normal tissue **fibrosis & nodules** se replace ho jata hai. Types: **Alcoholic (Laennec's – most common)**, **Post-necrotic** (viral hepatitis B/C), **Biliary** (obstruction), Cardiac (RHF). NAFLD/NASH increasingly common.

Clinical features (2 mechanisms)

- **Hepatocellular failure**: jaundice, **↓ albumin → oedema & ascites**, **↓ clotting factors → bleeding**, **spider angiomas**, **palmar erythema**, **gynaecomastia** & testicular atrophy (↑ oestrogen), **asterixis**, fetor hepaticus, **hepatic encephalopathy**.
- **Portal hypertension** (> 10–12 mmHg): **oesophageal varices**, **caput medusae**, **splenomegaly** (→ thrombocytopenia), **ascites**, haemorrhoids.

Ascites management

- **Sodium restriction (≤ 2 g/day)**, fluid restriction if hyponatraemic
- **Diuretics – Spironolactone** (first choice, K-sparing) **+ Furosemide**
- **Daily weight & abdominal girth** (same place – mark the skin, same time)
- **Paracentesis** (large volume → **IV albumin**), TIPS
- Position: **semi/high Fowler's** (breathing easy)
- Monitor for SBP (spontaneous bacterial peritonitis)

Oesophageal Varices

- Dilated veins at lower oesophagus due to portal HTN → rupture → **massive painless haematemesis – medical emergency**, most common cause of death in cirrhosis.
- Rx: **ABC, 2 large-bore IV, blood transfusion**, **IV Octreotide / Vasopressin / Terlipressin** (splanchnic vasoconstriction), **Endoscopic band ligation (EVL – first choice)** / sclerotherapy, antibiotics, **Sengstaken-Blakemore tube** (balloon tamponade – temporary; **scissors at bedside** to cut tube if airway obstruction; check balloon pressure, deflate per order every 8–12 hrs), **TIPS**.
- Prevention of re-bleeding: **non-selective beta-blocker (propranolol)**; avoid straining, alcohol, NSAIDs, hot/rough foods.

Hepatic Encephalopathy

- **↑ Ammonia (NH_3)** crosses blood-brain barrier (liver ammonia ko urea nahi bana pata).
- Precipitating factors: **GI bleeding (most common)**, high-protein diet, constipation, infection, hypokalaemia, sedatives/opioids, dehydration, alcohol.
- S/S: confusion, sleep reversal, **asterixis ("liver flap" – flapping tremor of outstretched hands)**, **fetor hepaticus** (musty sweet breath), **constructional apraxia** (can't draw star/ handwriting change), → stupor → coma. Grades I–IV (West Haven).
- Rx: **Lactulose** (acidifies colon, $NH_3 \rightarrow NH_4^+$ trapped & excreted; goal **2–3 soft stools/day**), **Rifaximin**/Neomycin (kill ammonia-producing gut bacteria), **avoid sedatives**, treat precipitating cause; protein – **not restricted long-term** (moderate 1–1.5 g/kg; plant/dairy proteins better) – acute stage short restriction.

- Nursing: neuro checks, **handwriting/star drawing daily**, safety (side rails), bleeding precautions, skin care.

! Cirrhosis – bleeding precautions

Soft toothbrush, electric razor, avoid IM injections (or small needle + pressure 5 min), check stool for blood, **Vitamin K** as ordered, avoid aspirin/NSAIDs, monitor PT/INR & platelets.

- **Liver cancer (Hepatocellular carcinoma)** – risk: cirrhosis, HBV, HCV, aflatoxin; marker **α-fetoprotein (AFP)**.
- **Liver transplant** – lifelong immunosuppressants (tacrolimus, cyclosporine); watch rejection & infection.
- **Hepatic/liver abscess** – amoebic (**Entamoeba histolytica**, right lobe, "**anchovy sauce pus**") vs pyogenic.

8.4 Gallbladder Disorders

- Gallbladder stores & concentrates bile (**30–50 ml**); **CCK (cholecystokinin)** → contraction after fatty meal.
- **Cholelithiasis (gallstones)** – most are **cholesterol stones**. Risk: **"5 F" – Fat, Female, Forty, Fertile, Fair** (+ family history, rapid weight loss, OCP).
- **Cholecystitis** – inflamed gallbladder (90 % due to stones).

★ Cholecystitis – S/S

- **RUQ / epigastric pain after fatty meal**, radiating to **right shoulder/scapula** (referred – phrenic nerve)
- **Biliary colic** – severe colicky pain with nausea/vomiting, fever
- **Murphy's sign** – deep inspiration par RUQ palpation se pain → patient **saans rok leta hai**
- Flatulence, dyspepsia, fat intolerance; **CBD stone (choledocholithiasis)** → obstructive jaundice
- Diagnosis: **Ultrasound abdomen (best initial)**, HIDA scan; **ERCP** (diagnostic + stone removal) – watch for **pancreatitis** after ERCP

- Rx: low-fat diet, analgesics (**Morphine was avoided earlier → sphincter of Oddi spasm; now opioids acceptable**, meperidine historic), **ursodeoxycholic acid** (dissolves cholesterol stones), **ESWL**, **Laparoscopic cholecystectomy (gold standard)**.
- **Post-lap chole**: **shoulder pain** (CO_2 gas irritating diaphragm) → early ambulation, **Sims' / left lateral position** with knees bent, heat; discharge usually same/next day.
- **Open cholecystectomy + CBD exploration → T-tube**: bag **below gallbladder level**, **drainage 300–500 ml first 24 hrs**, decreases gradually; clamp before meals as ordered (trial); report > 1000 ml/day; stool colour returns brown = bile flowing to duodenum.
- **Low-fat diet** long-term (4–6 weeks) after surgery; fat tolerance improves.

8.5 Pancreatitis

Acute Pancreatitis

- **Auto-digestion** of pancreas by its own enzymes (trypsin).

- Causes (**I GET SMASHED**): **I**diopathic, **G**allstones (most common overall), **E**thanol (alcohol), **T**rauma, **S**teroids, **M**umps, **A**utoimmune, **S**corpion sting, **H**ypercalcaemia/**H**ypertriglyceridaemia, **E**RCP, **D**rugs.
- S/S: **severe constant epigastric / LUQ pain radiating to the back**, **worse lying supine & after eating; relieved by sitting up, leaning forward or fetal (knee-chest) position**; nausea, vomiting, fever, **abdominal guarding**, shock.
  - **Cullen's sign** – bluish discolouration **around umbilicus** (haemorrhagic)
  - **Grey Turner's sign** – bluish discolouration of **flanks**
  - **Hypocalcaemia** (Chvostek/Trousseau) – calcium binds to fat necrosis → **poor prognosis**; hyperglycaemia
- Diagnosis: **Serum lipase (more specific; stays elevated longer, 7–14 days)**, **serum amylase** (↑ within 2–12 hrs, normal in 3–5 days; normal 30–110 U/L), CT abdomen (best imaging). Severity: **Ranson's criteria**, BISAP.

Acute pancreatitis – management ("rest the pancreas")

- **NPO**, **NG suction** (gastric juice → stimulation ↓), IV fluids (**aggressive – RL preferred**)
- **Pain relief – opioids** (morphine/hydromorphone/fentanyl) – pain control is priority
- Position: **sitting up / knee-chest / side-lying with knees flexed**
- Early **enteral nutrition** (NJ tube) preferred over TPN when oral not possible; resume orally with **low-fat, high-carbohydrate** diet
- Monitor **blood glucose**, calcium, respiratory status (**ARDS**, pleural effusion – left), shock
- Antibiotics only if infected necrosis; avoid alcohol

Chronic Pancreatitis

- Most common cause **chronic alcoholism**; progressive destruction → **steatorrhoea** (fatty, foul, frothy, floating stools – exocrine loss), **diabetes mellitus** (endocrine loss), chronic pain, weight loss, calcifications on X-ray.
- Rx: **pancreatic enzyme replacement (Pancrelipase) – with meals & snacks** (don't crush/mix with protein foods; wipe lips – skin irritation), **low-fat diet**, insulin, **stop alcohol**, analgesics.

Pancreatic Cancer

- Mostly **adenocarcinoma, head of pancreas**; risk – **smoking (most important)**, chronic pancreatitis, DM. S/S: **painless progressive obstructive jaundice**, weight loss, **Courvoisier's sign** (palpable non-tender gallbladder), back pain. Marker **CA 19-9**. Surgery: **Whipple's procedure (pancreaticoduodenectomy)**. Very poor prognosis.

Practice MCQs – Chapter 8

1. Faeco-oral route se failne wale hepatitis –
   (a) B & C (b) A & E (c) B & D (d) C & D
2. Hepatitis B ki active replication / high infectivity marker –
   (a) Anti-HBc IgG (b) HBeAg (c) Anti-HBs (d) HBsAg only
3. Hepatic encephalopathy me lactulose ka purpose –
   (a) Constipation karna (b) Protein badhana (c) Bleeding rokna (d) Ammonia kam karna
4. Asterixis kis condition ka sign hai?
   (a) Hepatic encephalopathy (b) Hypoglycaemia (c) Pancreatitis (d) Hypertension

5. Cirrhosis me ascites ke liye first diuretic –
(a) Acetazolamide (b) Spironolactone (c) Furosemide alone (d) Mannitol

6. Murphy's sign positive hota hai –
(a) Appendicitis (b) Cholecystitis (c) Hepatitis (d) Pancreatitis

7. Acute pancreatitis ka most specific test –
(a) Bilirubin (b) ALT (c) Serum lipase (d) Serum amylase

8. Cullen's sign –
(a) Spider naevi (b) Jaundice (c) Flank discolouration (d) Periumbilical bluish discolouration

9. Obstructive jaundice me stool ka colour –
(a) Red (b) Green (c) Clay-coloured / pale (d) Black

10. Bleeding oesophageal varices ki first-line endoscopic therapy –
(a) Colonoscopy (b) Gastrectomy (c) Liver biopsy (d) Band ligation

11. Lap cholecystectomy ke baad shoulder pain ka kaaran –
(a) MI (b) Bleeding (c) CO_2 gas (d) Infection

12. Pancreatic enzyme replacement kab dete hain?
(a) Once a week (b) With meals & snacks (c) Bedtime (d) Empty stomach morning

13. Hep B vaccine adult me kahan dete hain?
(a) Intradermal forearm (b) Deltoid IM (c) SC abdomen (d) Gluteal

14. Gallstones ke risk factor "5 F" me kaunsa nahi hai?
(a) Fat (b) Forty (c) Fever (d) Female

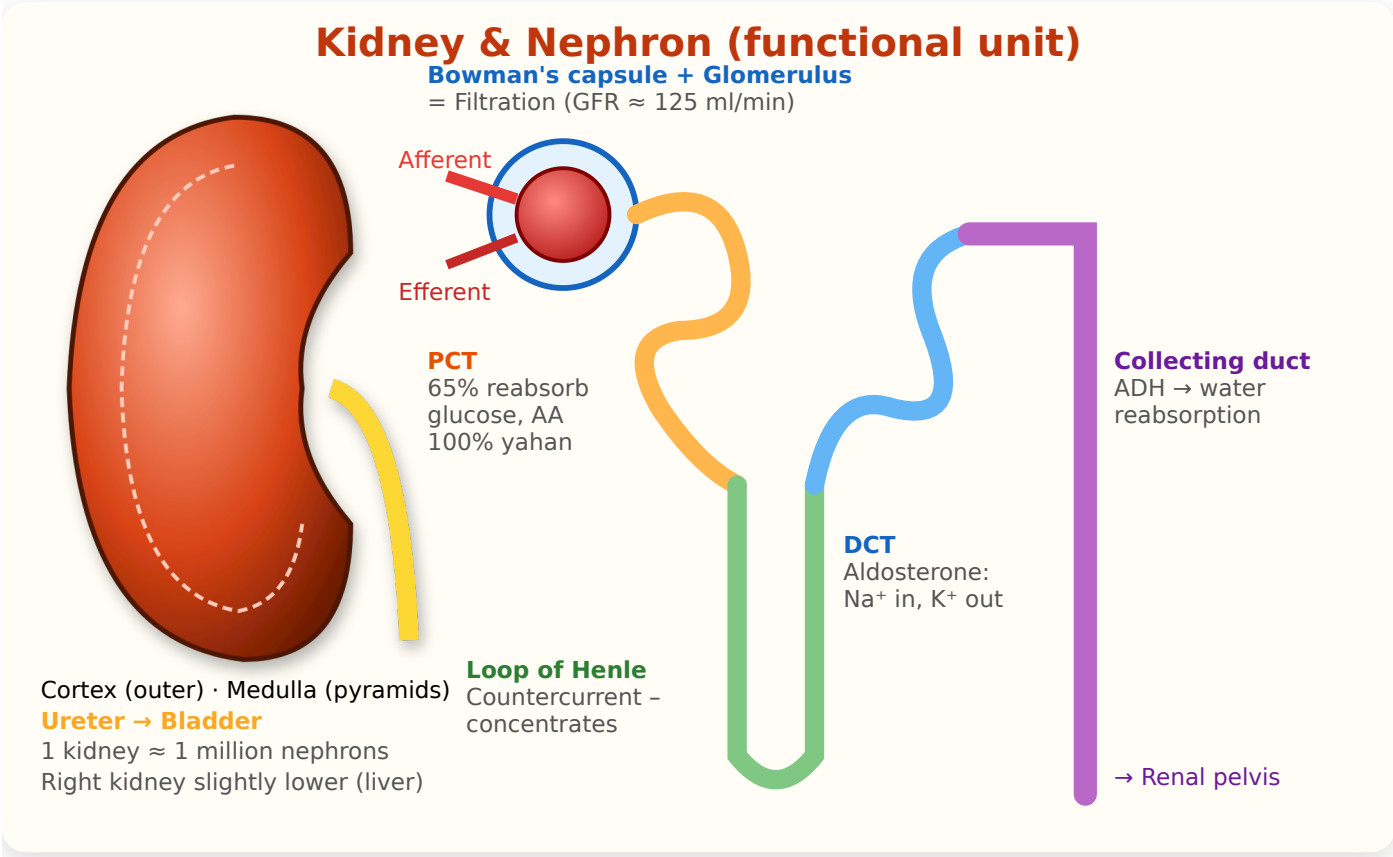
15. Most chronic hepatitis kaunsa hai?
(a) C (b) E (c) A (d) D

Answer Key – Chapter 8

1-b · 2-b · 3-d · 4-a · 5-b · 6-b · 7-c · 8-d · 9-c · 10-d · 11-c · 12-b · 13-b · 14-c · 15-a

9 Renal, Urinary & Male Reproductive Disorders

9.1 Applied Anatomy & Physiology



Kidney aur nephron ke parts – kaun kya karta hai

Kidney – facts

- **Retroperitoneal**, T12–L3; right kidney slightly lower (liver). Weight ~150 g each.
- Functional unit = **Nephron** (~1 million/kidney). Kidneys receive **20–25 % of cardiac output (~1200 ml/min)**.
- **GFR ≈ 125 ml/min (180 L/day)**; urine output ~1–1.5 L/day (99 % reabsorbed).
- Functions (**"A WET BED"**): **A**cid-base balance, **W**ater balance, **E**lectrolytes, **T**oxin removal, **B**P control (renin-angiotensin), **E**rythropoietin (RBC production), **D** – Vitamin **D** activation (calcitriol).
- **Renin** – juxtaglomerular cells → Angiotensin II (vasoconstriction) → **Aldosterone** (Na & water retain, K excrete). **ADH** – water reabsorption in collecting duct.
- Bladder capacity: **300–600 ml** (urge to void at ~150–300 ml). Female urethra **3–4 cm** (UTI more), male **18–20 cm**.

Renal Function Tests & Urine

Test	Normal
Blood urea	15 – 40 mg/dL

Test	Normal
BUN	7 – 20 mg/dL
Serum creatinine (best single indicator)	0.6 – 1.2 mg/dL (M), 0.5 – 1.1 (F)
Creatinine clearance (≈ GFR)	90 – 125 ml/min
BUN : Creatinine ratio	10 : 1 – 20 : 1
Uric acid	3.5 – 7 mg/dL
Urine specific gravity	**1.005 – 1.030** (1.010–1.025)
Urine pH	4.5 – 8 (avg 6)
Urine	Clear, pale yellow (urochrome); no glucose, protein, ketones, blood

- **24-hr urine collection**: **discard first morning void**, then collect all urine for 24 hrs **including next morning's first void**; keep on ice/preservative. Ek bhi sample miss → restart.
- **Clean-catch midstream** urine – clean meatus front to back, first part discard, midstream collect.
- From catheter – **aspirate from sampling port with sterile syringe** (never from bag).
- **IVP / CT with contrast** – check iodine/shellfish allergy, **creatinine** (contrast nephropathy), metformin hold 48 hrs, hydrate after.
- **Renal biopsy** – check coagulation; after: **prone/supine with sandbag**, bed rest 24 hrs, watch **haematuria & flank pain**, fluids.
- **Cystoscopy** – lithotomy position; after: pink-tinged urine & burning normal; bright red/clots → report; fluids.

Urinary terms

Dysuria (painful), **Frequency**, **Urgency**, **Nocturia**, **Haematuria**, **Pyuria** (pus), **Oliguria** (< 400 ml/day), **Anuria** (< 100 ml/day), **Polyuria** (> 3 L/day), **Enuresis** (bed wetting), **Retention** (can't empty), **Incontinence**, **Proteinuria**, **Glycosuria**.

9.2 Urinary Tract Infection (UTI)

- Most common organism: **E. coli** (~80 %). Females > males (short urethra). Risk: **catheter (most common cause of HAI/CAUTI)**, pregnancy, DM, BPH, stones, sexual activity, post-menopause.
- **Cystitis** (lower): **dysuria, frequency, urgency**, suprapubic pain, cloudy foul urine, haematuria. Elderly – **confusion**.
- **Pyelonephritis** (upper – kidney): **high fever with chills, flank pain, CVA (costovertebral angle) tenderness**, nausea/vomiting.
- Diagnosis: urine culture & sensitivity (**> 10^5 CFU/ml** significant), urinalysis (nitrite, leukocyte esterase).



UTI prevention teaching

- Fluids **2–3 L/day**, **void every 2–3 hrs** (don't hold), **void after intercourse**
- **Wipe front to back** (females), cotton undergarments, showers instead of tub/bubble bath
- Cranberry juice (may help), avoid irritants (caffeine, alcohol, spicy)
- Complete the **full antibiotic course**; **Phenazopyridine** – urine **orange-red** (analgesic, max 2 days)

9.3 Urolithiasis (Renal / Urinary Calculi)

- Most common stone: **Calcium oxalate** (~75–80 %); others – calcium phosphate, **uric acid** (gout), **struvite** (infection – Proteus, staghorn), **cystine**.
- Risk: dehydration, hot climate, immobility, hypercalcaemia, high animal protein/salt, gout, UTI.
- S/S: **Renal colic – excruciating, intermittent flank pain radiating to groin/testis/labia ("loin to groin")**, nausea/vomiting, **haematuria**, restlessness (patient can't lie still), frequency if near bladder.
- Diagnosis: **Non-contrast CT KUB (gold standard)**, X-ray KUB, USG, urinalysis; **strain all urine** (stone analysis).
- Rx: **analgesia (NSAIDs – ketorolac/diclofenac; opioids)** – priority; **fluids 3–4 L/day** (output ≥ 2–2.5 L/day); **tamsulosin** (α-blocker – stone passage); stones **< 5 mm usually pass**; **ESWL** (lithotripsy – shock waves; bruising on flank, haematuria few days, strain urine), **PCNL**, **ureteroscopy (URS)**, open surgery.

Stone	Diet advice
Calcium oxalate	Avoid **oxalate-rich**: **spinach, tomato (seeds), chocolate, tea, nuts, beetroot**, rhubarb; ↓ salt & animal protein; **don't restrict dietary calcium** (normal Ca intake protects)
Uric acid	**Low purine** – avoid organ meat, red meat, sardines, alcohol; **allopurinol**; alkalinize urine
Struvite	Treat infection; acidify urine
Cystine	High fluids, alkalinize urine

9.4 Acute Kidney Injury (AKI / ARF)

D AKI

Sudden (hours–days) loss of kidney function → ↑ BUN & creatinine, ↓ urine output. **Potentially reversible.**

Type	Cause
Pre-renal (most common, ~60–70 %)	**↓ renal perfusion** – hypovolaemia, haemorrhage, dehydration, shock, HF, burns
Intra-renal (intrinsic)	Kidney tissue damage – **Acute Tubular Necrosis (ATN – most common intrarenal)**, **nephrotoxic drugs (aminoglycosides – gentamicin, NSAIDs, contrast dye, amphotericin B)**, glomerulonephritis, rhabdomyolysis, transfusion reaction
Post-renal	**Obstruction** – BPH (most common in men), stones, tumours, blocked catheter

Phases of AKI

1. **Initiation (onset)** – injury se oliguria start tak (hours–days)
2. **Oliguric phase** (1–3 weeks) – **urine < 400 ml/day**, **↑ K^+ (most dangerous)**, ↑ BUN/creatinine, fluid overload, **metabolic acidosis**, ↓ Na, ↑ PO_4, ↓ Ca, uraemic symptoms
3. **Diuretic phase** (1–3 weeks) – urine **3–5 L/day** (kidneys can't concentrate) → **dehydration, hypokalaemia, hyponatraemia** risk
4. **Recovery phase** – up to 12 months; GFR improves

- Rx: treat cause, **fluid restriction = previous day's output + 500–600 ml** (insensible loss), daily weight, I/O, **manage hyperkalaemia**, diet – **high-calorie (carbohydrate), low protein (adequate), low K, low Na, low PO_4**; dialysis if needed (AEIOU).

AEIOU – Indications for urgent dialysis

Acidosis (severe metabolic) · **E**lectrolytes (**refractory hyperkalaemia**) · **I**ntoxication (lithium, aspirin, methanol) · **O**verload (fluid – pulmonary oedema) · **U**raemia (pericarditis, encephalopathy, bleeding)

9.5 Chronic Kidney Disease (CKD)

CKD

Kidney damage or **GFR < 60 ml/min/1.73 m^2 for ≥ 3 months**; progressive, **irreversible**. **ESRD/ESKD = GFR < 15** (stage 5) → dialysis/transplant.
Most common causes: **1. Diabetes mellitus (diabetic nephropathy)**, **2. Hypertension**, 3. glomerulonephritis, polycystic kidney disease.

Stage	GFR (ml/min/1.73 m^2)
1	≥ 90 (kidney damage with normal GFR)
2	60 – 89
3a / 3b	45 – 59 / 30 – 44
4	15 – 29
5 (ESRD)	**< 15**

Clinical features – every system (uraemia)

- **Fluid & electrolytes**: oedema, HTN, **hyperkalaemia**, **hyperphosphataemia + hypocalcaemia** → secondary hyperparathyroidism → **renal osteodystrophy**, metabolic acidosis
- **Haematological**: **Anaemia (↓ erythropoietin)** – normocytic normochromic; bleeding tendency
- **Cardiovascular**: HTN, HF, **uraemic pericarditis** (friction rub), CAD (most common cause of death)
- **Skin**: **pruritus**, yellow-grey (sallow) skin, **uraemic frost** (white urea crystals – late)
- **GI**: anorexia, nausea, vomiting, **uraemic fetor (ammonia/urine smell breath)**, metallic taste, stomatitis
- **Neuro**: fatigue, confusion, peripheral neuropathy, **restless legs**, seizures, coma
- **Respiratory**: Kussmaul breathing, pulmonary oedema
- Reproductive: amenorrhoea, impotence

CKD management

- Control **BP (ACE inhibitor/ARB – kidney protective)**, blood sugar
- **Erythropoietin (epoetin alfa)** SC – monitor Hb (target 10–11.5 g/dL; > 13 avoid – thrombosis, HTN); give **iron** supplements
- **Phosphate binders (calcium carbonate, sevelamer) – WITH meals**; **calcitriol (active Vit D)**
- Hyperkalaemia management, **sodium bicarbonate** for acidosis

- **Avoid**: Mg-containing antacids/laxatives (**milk of magnesia**), **Fleet (phosphate) enema**, NSAIDs, nephrotoxic drugs; adjust drug doses
- Diet: **↓ protein (0.6–0.8 g/kg – high biological value: egg, meat, milk)** before dialysis; **on dialysis – ↑ protein (1.2 g/kg HD, 1.2–1.3 g/kg PD)**; **low Na, low K (avoid banana, orange, potato, tomato, coconut water, dry fruits), low phosphorus (milk products, cola)**, fluid = urine output + 500 ml, adequate calories
- Skin care for pruritus, oral hygiene, daily weight



Potassium kam karne ki trick (leaching)

Sabziyon (aloo etc.) ko **chhote tukde me kaat ke, garam paani me 2+ ghante bhigo ke, paani phenk ke, naye paani me pakao** – potassium kafi kam ho jata hai.

9.6 Dialysis

	Haemodialysis (HD)	Peritoneal Dialysis (PD)
Membrane	Artificial (**dialyser – "artificial kidney"**)	**Peritoneum** (natural semi-permeable)
Access	**AV fistula (best – "lifeline")**, AV graft, temporary catheter (subclavian/jugular/femoral)	Tenckhoff catheter in abdomen
Schedule	**3 times/week, 3–5 hrs**	Daily – CAPD (4–5 exchanges/day) or APD at night
Anticoagulant	**Heparin** needed	No
Main complications	**Hypotension (most common)**, muscle cramps, **disequilibrium syndrome** (headache, nausea, confusion, seizures – fast urea removal → cerebral oedema; first sessions slow), bleeding, infection, air embolism	**Peritonitis (most common & serious – cloudy effluent**, abdominal pain, fever), hernia, hyperglycaemia (dextrose), protein loss, back pain, catheter exit-site infection

AV Fistula care

- **Thrill (palpate – vibration/buzz) & bruit (auscultate – whooshing sound)** = fistula **patent** → check every shift; absent → report.
- Fistula maturity: **4–6 weeks (up to 3 months)** before use; arm exercises (squeezing ball) to mature.
- **NO BP, NO venipuncture, NO IV, NO injections** in fistula arm; **no tight clothes, watch, jewellery, no sleeping/lifting heavy on that arm**.
- "**Fistula arm – No touch arm**" sign on bed.
- Bleeding after dialysis → direct pressure.

- **Before HD**: weigh, vitals, hold antihypertensives (as ordered – hypotension), check access. **After HD**: weigh (wt loss = fluid removed), vitals, bleeding check, avoid invasive procedures 4–6 hrs (heparin).
- **PD**: warm dialysate to **body temperature (dry heat – never microwave)**; strict asepsis; inflow **10–20 min**, dwell **as prescribed (4–6 hrs CAPD)**, outflow **20–30 min** by gravity; **outflow slow → change position, check kinks, constipation**; outflow should be **clear, straw-coloured**; **cloudy = peritonitis**; record balance; if respiratory distress during inflow → **elevate HOB**, slow.

Kidney Transplant

- Living donor or cadaver; graft placed in **iliac fossa** (own kidneys usually left).
- Post-op: **urine output hourly** (can be very high – diuresis), fluid replacement; lifelong **immunosuppressants** (tacrolimus, cyclosporine, mycophenolate, prednisolone) → **infection risk** – reverse isolation measures, hand hygiene.
- **Rejection**: **Hyperacute** (minutes–hours – remove graft), **Acute** (1 week–2 years; **fever, graft tenderness, ↓ urine output, ↑ creatinine, weight gain, HTN**), **Chronic** (months–years).

9.7 Glomerular Diseases

Feature	Acute Glomerulonephritis (AGN – Nephritic)	Nephrotic Syndrome
Cause	**Post-streptococcal** (group A β-haemolytic strep throat/skin, **1–3 weeks earlier**); children 2–12 yrs	Minimal change disease (children), FSGS, membranous (adults), DM, SLE
Key feature	**Haematuria ("cola / smoky / tea-coloured" urine)**, **HTN**, mild proteinuria, oliguria, **periorbital oedema**, ↑ ASO titre	**Massive proteinuria (> 3.5 g/day)**, **hypoalbuminaemia (< 3 g/dL)**, **generalized oedema (anasarca)**, **hyperlipidaemia** (lipiduria), frothy urine, thrombosis risk
BP	↑	Normal / low
Rx	Bed rest (till haematuria/HTN resolve), **penicillin**, **restrict Na, fluid & protein (if azotemia)**, diuretics, antihypertensives	**Corticosteroids (prednisolone)**, diuretics, ACE-I, statins; **↓ Na**, normal protein (**not high**), skin care, daily weight, infection prevention

- **Polycystic kidney disease** – autosomal dominant (adult); flank pain, haematuria, HTN, enlarged kidneys.
- **Wilms' tumour (nephroblastoma)** – children; abdominal mass – **don't palpate abdomen**.
- **Renal cell carcinoma** – classic triad: **haematuria, flank pain, palpable mass**; smoking risk; nephrectomy (flank incision → pain on breathing – splint).
- **Bladder cancer** – most common sign **painless haematuria**; smoking, aniline dyes; TURBT, **cystectomy with ileal conduit** (urine continuous; mucus threads normal).

9.8 Urinary Retention, Incontinence & Catheterization

- **Retention** – acute (painful, distended bladder) – causes: BPH, post-op, anaesthesia, drugs (anticholinergics, opioids). First try: privacy, running water sound, warm water over perineum, normal voiding position; then catheter. **Bladder scan** – PVR > 100–150 ml significant. Drain **not more than 750–1000 ml at once** (as per policy) – hypotension/haematuria risk.
- **Incontinence**: **Stress** (cough/sneeze/laugh – weak pelvic floor; **Kegel exercises**), **Urge** (overactive bladder – bladder training, anticholinergics like oxybutynin), **Overflow** (BPH, neurogenic – catheter), **Functional** (mobility/cognition issue – toileting schedule), **Reflex** (spinal cord injury).

> **Kegel (pelvic floor) exercise**
> Muscles jaise urine ka flow beech me rokte ho waise **tighten 10 sec → relax 10 sec**, **10–15 times, 3 times/day**. Results 6–12 weeks. Pet, thigh, buttocks ke muscles use nahi karne.

Urinary catheterization – key points

- **Sterile technique**; size adult **14–16 Fr** (female 12–14, male 14–16), child 8–10 Fr; balloon **5–10 ml sterile water** (not saline – crystallize; not air – floats).
- Female – insert **5–7.5 cm (2–3 inch)** then 2.5 cm more after urine; Male – penis held **90° (perpendicular)**, insert **17–22.5 cm (7–9 inch)** till urine + 2.5–5 cm (to "Y"); **replace foreskin** (paraphimosis prevention).
- Secure catheter – **female: inner thigh; male: lower abdomen / upper thigh**.
- Bag **always below bladder level**, never on floor, no kinks; **closed drainage system** – don't disconnect.
- **Catheter care with soap & water** daily; fluids 2–3 L.
- **CAUTI prevention** – remove as early as possible (daily review); no routine bladder irrigation.
- **Suprapubic catheter** – long-term; site care.
- **Continuous Bladder Irrigation (CBI)** after **TURP** – **3-way catheter**; normal saline; keep output **light pink/rose**; bright red/clots → ↑ irrigation flow rate; **True urine output = Total output − Irrigation fluid**.

9.9 BPH & Prostate

- **Benign Prostatic Hyperplasia** – men > 50 yrs; **enlarged prostate** compresses urethra.
- S/S (LUTS): **hesitancy**, **weak/interrupted stream**, dribbling, **incomplete emptying**, frequency, **nocturia** (often first), urgency, retention, UTI, hydronephrosis.
- Diagnosis: **DRE (digital rectal exam – smooth, firm, rubbery, symmetrically enlarged)**, **PSA** (normal < 4 ng/ml), USG/PVR, uroflowmetry.
- Drugs: **α-blockers – Tamsulosin, Prazosin** (smooth muscle relax – **orthostatic hypotension**, rise slowly); **5-α-reductase inhibitors – Finasteride, Dutasteride** (shrink prostate – **6 months** to effect; pregnant women should **not handle crushed tablets** – teratogenic).
- Avoid: **antihistamines, decongestants, anticholinergics** (retention), alcohol, caffeine at night, large fluid volume at once.
- Surgery: **TURP (gold standard)**, open prostatectomy, laser.

! Post-TURP care

- **CBI with 3-way catheter** (traction on catheter – taped to thigh – for haemostasis; **leg straight**).
- **Bladder spasms** – common (oxybutynin/B&O suppository); avoid pulling catheter.
- **TUR syndrome** – absorption of irrigation fluid (glycine/water) → **dilutional hyponatraemia** → confusion, nausea, bradycardia, HTN, seizures.
- After catheter removal – **dribbling & incontinence** temporary → **Kegel exercises**; avoid straining, heavy lifting, sitting long, intercourse **4–6 weeks**; fluids 2–3 L; **retrograde ejaculation** common.

- **Prostate cancer** – most common cancer in men (western), **adenocarcinoma, posterior lobe** – **hard, nodular** on DRE; **PSA ↑**; bone metastasis (**osteoblastic, back pain**); Rx – radical prostatectomy, radiation, **hormone therapy (orchiectomy, leuprolide)**.
- **Testicular cancer** – age **15–35**; painless lump; **monthly testicular self-exam (TSE) after warm bath**; markers **AFP, β-hCG**; cryptorchidism risk.
- **Hydrocele** (fluid – transillumination +ve), **Varicocele** ("bag of worms", left side, infertility), **Testicular torsion** (sudden severe pain – **surgical emergency within 4–6 hrs**), **Phimosis/Paraphimosis**,

Epididymitis (scrotal support, ice).

Practice MCQs – Chapter 9

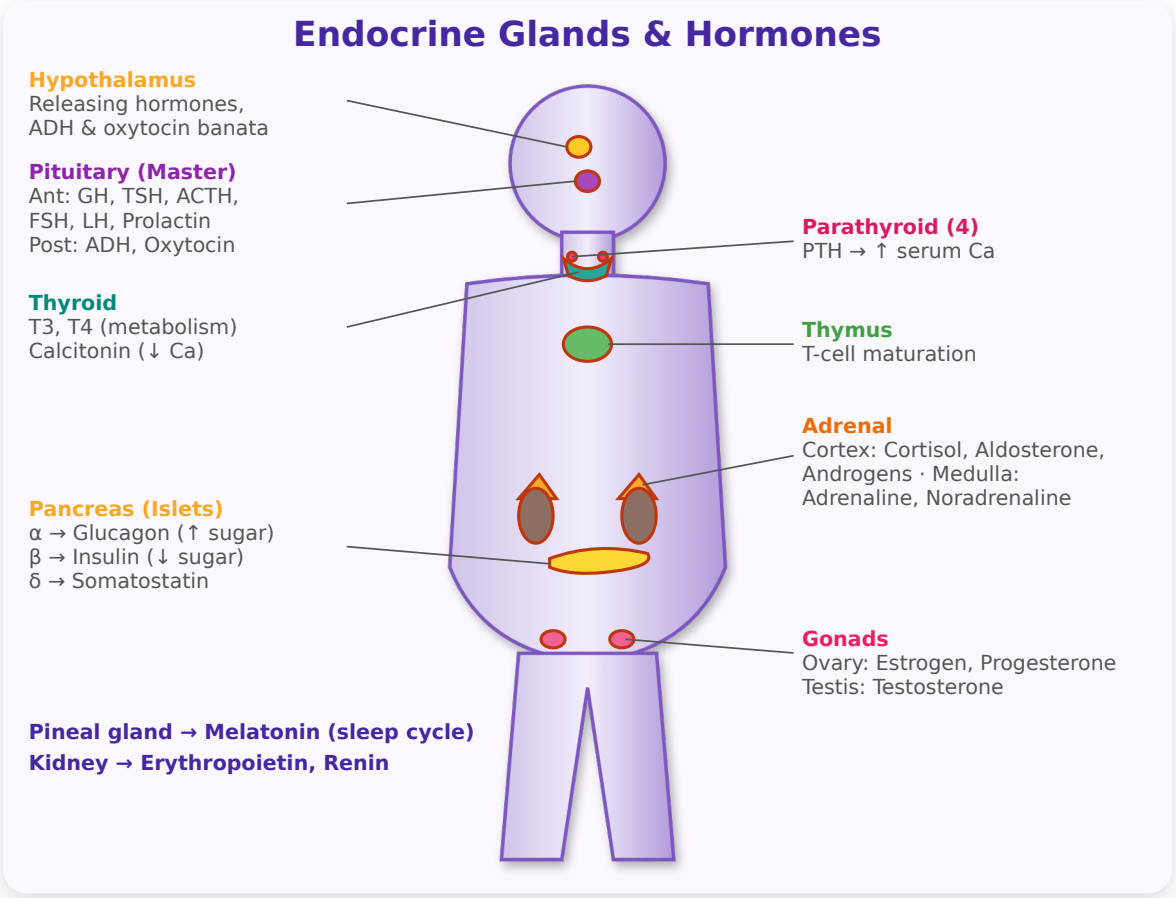
1. UTI ka most common organism –
   (a) Proteus (b) Staph (c) Klebsiella (d) E. coli
2. 24-hr urine collection me –
   (a) Random sample (b) Sirf raat ka urine (c) First morning void discard, next day ka first void include (d) First morning sample collect karte hain
3. CKD ka most common cause –
   (a) UTI (b) Stones (c) Diabetes mellitus (d) Glomerulonephritis
4. AKI ke oliguric phase me most dangerous electrolyte imbalance –
   (a) Hyponatraemia (b) Hypocalcaemia (c) Hyperkalaemia (d) Hypermagnesaemia
5. AV fistula patent hai, iska sign –
   (a) Cold arm (b) Thrill & bruit present (c) Pain (d) No pulse
6. Peritoneal dialysis me cloudy outflow ka matlab –
   (a) Peritonitis (b) Normal (c) Hyperkalaemia (d) Dehydration
7. Calcium oxalate stone me kaunsa food avoid?
   (a) Apple (b) Rice (c) Bread (d) Spinach, tomato, chocolate
8. Nephrotic syndrome ka hallmark –
   (a) Haematuria (b) Fever (c) Hypertension (d) Massive proteinuria
9. Post-streptococcal GN me urine colour –
   (a) Milky (b) Green (c) Clear (d) Cola / smoky
10. TURP ke baad CBI me true urine output –
    (a) Irrigation only (b) Total output (c) Total output − irrigation fluid (d) Total + irrigation
11. Foley balloon kis se fill karte hain?
    (a) Tap water (b) Sterile water (c) Normal saline (d) Air
12. Haemodialysis ka most common complication –
    (a) Hypertension (b) Peritonitis (c) Hypotension (d) Hernia
13. BPH me tamsulosin ka side effect –
    (a) Bradycardia (b) Orthostatic hypotension (c) Constipation (d) Hyperkalaemia
14. CKD me anaemia ka main cause –
    (a) B12 deficiency (b) ↓ Erythropoietin (c) Haemolysis (d) Bleeding
15. Stress incontinence ke liye best exercise –
    (a) Breathing (b) Push-ups (c) Running (d) Kegel

Answer Key – Chapter 9

1-d · 2-c · 3-c · 4-c · 5-b · 6-a · 7-d · 8-d · 9-d · 10-c · 11-b · 12-c · 13-b · 14-b · 15-d

10 Endocrine System Disorders

10.1 Glands & Hormones



Endocrine glands aur unke hormones

Feedback & facts

- Most hormones **negative feedback** se control hote hain (e.g. ↑ T4 → ↓ TSH).
- **Pituitary = master gland**, but controlled by **hypothalamus**. Anterior pituitary = **adenohypophysis**, posterior = **neurohypophysis** (stores ADH & oxytocin made in hypothalamus).
- **Pancreas** – both endocrine & exocrine; **Islets of Langerhans**: α (glucagon), **β (insulin – ~70 %)**, δ (somatostatin).
- **Normal fasting blood sugar 70–100 mg/dL**; post-prandial (2 hr) **< 140 mg/dL**.

10.2 Diabetes Mellitus (DM)

Diabetes Mellitus

Chronic metabolic disorder with **hyperglycaemia** due to **insulin deficiency, insulin resistance or both**.

Diagnosis (ADA)

Test	Diabetes	Pre-diabetes
Fasting plasma glucose (≥ 8 hr fast)	**≥ 126 mg/dL**	100 – 125 (IFG)
2-hr OGTT (75 g glucose)	**≥ 200**	140 – 199 (IGT)
HbA1c	**≥ 6.5 %**	5.7 – 6.4 %
Random plasma glucose + classic symptoms	**≥ 200**	–

- **HbA1c (glycosylated Hb)** = average blood sugar of **past 2–3 months** (RBC lifespan 120 days); target in diabetics **< 7 %**; **no fasting needed**.

Type 1 vs Type 2

	Type 1	Type 2 (~90–95 %)
Cause	**Autoimmune destruction of β cells** → absolute insulin deficiency	**Insulin resistance** + relative deficiency
Age	Children/young (< 30)	Adults > 40 (now younger too), **obese**
Onset	Sudden	Gradual
Body	Thin	**Obese** (central)
Ketosis	**DKA prone**	HHS prone; rarely DKA
Treatment	**Insulin – always**	Diet, exercise, **oral hypoglycaemics**, insulin later

- **Gestational DM** – first detected in pregnancy (24–28 weeks screening); risk of Type 2 later.
- Classic symptoms (**3 P's**): **Polyuria, Polydipsia, Polyphagia** + weight loss (Type 1), fatigue, blurred vision, slow wound healing, recurrent infections (candidiasis), pruritus vulvae.

Insulin

Type	Example	Onset	Peak	Duration
Rapid-acting	**Lispro, Aspart**, Glulisine	**10–15 min**	1 hr	3–4 hr
Short-acting (Regular)	**Regular (Actrapid) – clear**	**30–60 min**	2–4 hr	6–8 hr
Intermediate	**NPH (Insulatard) – cloudy**	1–2 hr	4–12 hr	12–18 hr
Long-acting (basal)	**Glargine (Lantus)**, Detemir, Degludec	1–2 hr	**No peak**	**24 hr** (Degludec 42 hr)

Insulin administration – golden rules

- **Only Regular (clear) insulin can be given IV** (DKA, hyperkalaemia).
- Mixing: "**Clear before Cloudy**" – Regular (clear) **pehle draw**, then NPH (cloudy) – **"RN"** (Regular → NPH). Air pehle NPH me daalo, phir Regular me.
- **Glargine – never mix** with other insulins.
- Route **SC**, angle **45–90°**; **abdomen = fastest absorption** (preferred) > arms > thighs > buttocks; **2 inches (5 cm) away from umbilicus**; **rotate sites** within one area (**lipodystrophy/lipohypertrophy** prevention); don't massage.
- Storage: unopened vials **refrigerator (2–8 °C)**, never freeze; in-use vial room temperature up to **28–30 days**; **roll cloudy insulin gently (don't shake)**.
- Insulin syringe **U-100** (100 units/ml); also pens.
- **Somogyi effect** – night **hypoglycaemia (2–3 am)** → rebound morning hyperglycaemia → **reduce bedtime insulin/give snack**. **Dawn phenomenon** – early morning hyperglycaemia due to GH/cortisol (no night hypo) → **increase/adjust evening insulin**. Check **3 am blood sugar** to differentiate.

Oral hypoglycaemic agents

Group	Drug	Key point
Biguanide	**Metformin** (first-line Type 2)	**Lactic acidosis**, GI upset, B12 deficiency; **no hypoglycaemia alone**; **stop 48 hrs before/after iodine contrast** & surgery
Sulfonylureas	Glibenclamide, **Glimepiride**, Gliclazide	**Hypoglycaemia** (most), weight gain; before meals
Meglitinides	Repaglinide	Take with meal; skip meal = skip dose
DPP-4 inhibitors (-gliptin)	Sitagliptin, Vildagliptin	Weight neutral
SGLT-2 inhibitors (-gliflozin)	Empagliflozin, Dapagliflozin	**Genital/urinary infections**, euglycaemic DKA, heart/kidney protective
Thiazolidinediones	Pioglitazone	Fluid retention – avoid in HF
α-glucosidase inhibitor	Acarbose	Take with **first bite**; flatulence; treat hypo with **glucose (not sucrose)**
GLP-1 agonists (injectable)	Liraglutide, Semaglutide	Weight loss, nausea

Diet & exercise

- Diet: complex carbs (**50–60 %**), protein 15–20 %, fat < 30 %, **high fibre**, **low glycaemic index** foods, avoid simple sugars, **regular meal timing** (don't skip), artificial sweeteners in moderation.
- Exercise: **150 min/week** moderate; best time **1 hr after meals**; check sugar before; **avoid if sugar > 250 mg/dL with ketones** (or < 100 → take snack); carry sugar; don't inject insulin in exercising limb.

10.3 Acute Complications of Diabetes

Hypoglycaemia (Blood sugar < 70 mg/dL)

- Causes: **too much insulin/OHA, skipped/delayed meal, excess exercise, alcohol**, vomiting.
- S/S (**adrenergic**): **sweating, tremors, palpitation, tachycardia, hunger, anxiety, pallor**; (**neuroglycopenic**): headache, confusion, blurred vision, slurred speech, irritability, **seizures, coma**. **Cold, clammy skin**. Beta-blockers **mask** symptoms. "**Cold & clammy → need some candy.**"

Rule of 15 (conscious patient)

Give **15 g fast-acting carbohydrate** (½ cup / 120 ml fruit juice, 3–4 glucose tablets, 1 tbsp sugar/honey, regular soda ½ cup) → **recheck after 15 min** → still < 70 → repeat → when normal give **complex carb + protein snack** (if next meal > 1 hr).
Unconscious: **IV 50 % dextrose (D50) 20–50 ml** or **Glucagon 1 mg IM/SC** (turn patient on side – vomiting). **Nothing by mouth** to unconscious patient.

DKA vs HHS

	DKA (Diabetic Ketoacidosis)	HHS (Hyperosmolar Hyperglycaemic State)
Type	**Type 1**	**Type 2** (elderly)
Cause	Infection (most common), missed insulin, stress	Infection, dehydration
Onset	**Rapid (< 24 hr)**	Slow (days)
Blood sugar	**> 250 mg/dL** (usually 300–800)	**> 600 mg/dL** (often > 1000)
Ketones	**Present (++)**	Absent / minimal
pH	**< 7.3 (metabolic acidosis)**, HCO_3^- < 18	> 7.3
Osmolality	Variable	**> 320 mOsm/kg** (very high)
Breathing	**Kussmaul**, **fruity (acetone) breath**	Normal
Others	Abdominal pain, nausea, vomiting, dehydration	**Severe dehydration**, neuro signs (seizures, hemiparesis, coma)
Mortality	Lower	Higher

DKA management – order matters

1. **Fluids FIRST – 0.9 % NS** (1 L in first hour) → later 0.45 % NS; **when glucose falls to ~200–250 mg/dL → add dextrose (D5 ½NS)** (hypoglycaemia & cerebral oedema prevention)
2. **IV Regular insulin infusion** (0.1 unit/kg/hr) – start **after $K^+ \geq 3.3$** checked; drop sugar **50–75 mg/dL/hr** (not faster)
3. **Potassium** – insulin pushes K into cells → **hypokalaemia** risk → K replacement (once urine output adequate)
4. Bicarbonate only if pH < 6.9
5. Monitor: hourly blood glucose, K^+, ABG, I/O, neuro status, ECG

10.4 Chronic Complications of Diabetes

- **Macrovascular**: CAD/MI (**most common cause of death**), stroke, PVD (gangrene).
- **Microvascular**: **Retinopathy** (most common cause of blindness in adults – yearly fundus exam), **Nephropathy** (most common cause of ESRD – **microalbuminuria** first sign; ACE-I), **Neuropathy** (most common complication – **glove & stocking** numbness; foot ulcers; autonomic – gastroparesis, impotence, orthostatic hypotension, silent MI).



Diabetic foot care – teaching

- **Daily inspect feet** (use mirror) – cuts, blisters, redness
- Wash with **lukewarm water** (check temp with elbow/thermometer), **dry between toes**, moisturize but **not between toes**
- **Cut nails straight across** (podiatrist best); never cut corns yourself
- **Never walk barefoot**, well-fitting closed shoes, cotton socks, check inside shoes
- **No hot water bottles / heating pads**, no tight socks; stop smoking
- Report any wound immediately (gangrene/amputation risk)

- **Sick day rules**: **never stop insulin** (stress ↑ sugar), check sugar every **3–4 hrs** & **urine ketones**, fluids **every hour** (sugar-free if high, sugar-containing if not eating), soft foods/liquids for carbs, call doctor if vomiting, sugar > 250 persistent, ketones +.

10.5 Thyroid Disorders

Feature	Hyperthyroidism (e.g. Graves' disease – most common)	Hypothyroidism (e.g. Hashimoto's – most common; iodine deficiency)
Metabolism	↑↑ (everything fast)	↓↓ (everything slow)
Lab	**↓ TSH, ↑ T3/T4** (primary)	**↑ TSH, ↓ T4** (primary)
Weight	**Loss** despite ↑ appetite	**Gain**, ↓ appetite
Heat/cold	**Heat intolerance**, sweating, warm moist skin	**Cold intolerance**, dry coarse skin
Heart	**Tachycardia**, palpitation, **AF**, ↑ SBP (wide pulse pressure)	**Bradycardia**, ↓ cardiac output, hyperlipidaemia
GI	**Diarrhoea**	**Constipation**
Neuro	**Nervous, irritable, tremors**, insomnia, hyperreflexia	**Lethargy, slow speech, depression**, memory ↓, slow reflexes
Eyes/face	**Exophthalmos** (Graves), lid lag, stare	**Periorbital puffiness**, myxoedema face, **loss of outer 1/3 eyebrow**
Hair	Fine, soft, hair loss	**Coarse, brittle**, hair loss
Menstruation	Amenorrhoea / scanty	**Menorrhagia**
Crisis	**Thyroid storm**	**Myxoedema coma**

♫ Hyper = fast, Hypo = slow

Hyper patient ko **"garam, bhookha, patla, bechain"** socho. Hypo patient ko **"thanda, mota, sust, sona chahta"** socho.
Cretinism = congenital hypothyroidism in children (mental retardation) – newborn screening TSH.

Hyperthyroidism management

- **Antithyroid drugs**: **Methimazole** (Carbimazole), **PTU (Propylthiouracil – preferred in 1st trimester of pregnancy & thyroid storm)**; side effect – **agranulocytosis** (sore throat, fever → report), hepatotoxicity (PTU).
- **Beta-blockers (Propranolol)** – symptomatic (tachycardia, tremors).
- **Iodine (Lugol's solution / SSKI)** – ↓ vascularity before surgery (**mix in juice/milk, use straw – stains teeth**), 10–14 days pre-op.
- **Radioactive iodine (I-131)** – **contraindicated in pregnancy & lactation**; avoid close contact/children & pregnant women for a few days, flush toilet 2–3 times, separate utensils; risk of **hypothyroidism later** (lifelong thyroxine).
- Diet: **high-calorie, high-protein**, frequent meals; avoid caffeine & stimulants; cool quiet environment.
- **Exophthalmos care**: **elevate HOB**, artificial tears, **tape eyelids closed at night**/eye patches, dark glasses, ↓ salt (oedema).

! Thyroid storm (Thyrotoxic crisis)

Life-threatening – precipitated by surgery, infection, stress, stopping drugs. **Very high fever (> 38.5–41 °C)**, severe tachycardia (> 130), delirium, vomiting, HF, shock.
Rx: **PTU** → (1 hr later) **iodine**, **propranolol**, **hydrocortisone**, **cooling (paracetamol – NOT aspirin**; aspirin frees T4), O_2, fluids.

Thyroidectomy – post-op care

- Position: **semi-Fowler's/Fowler's**, head & neck **supported with sandbags/pillows**, avoid neck hyperextension/flexion; support neck (hands behind head) while moving.
- Bedside: **tracheostomy set, O_2, suction, calcium gluconate IV**.
- Watch for:
  - **Haemorrhage** – check **back of neck & dressing** (blood pools behind), tight dressing, frequent swallowing.
  - **Respiratory obstruction** (oedema, haematoma, laryngeal nerve injury) – **stridor** = emergency.
  - **Recurrent laryngeal nerve damage** → **hoarseness**; check voice every 2–4 hrs (patient bole "**ee**"); minimize talking (but some hoarseness normal first days due to ET tube).
  - **Hypocalcaemia/tetany** (parathyroid removal/damage) – **tingling around mouth & fingers** (first sign), Chvostek, Trousseau → **IV calcium gluconate** (usually 1–7 days post-op).
  - **Thyroid storm**.
- Total thyroidectomy → **lifelong levothyroxine**.

Hypothyroidism management

- **Levothyroxine (T4)** – **empty stomach, morning, 30–60 min before breakfast**, same time daily; **lifelong**; start low in elderly & cardiac patients ("start low, go slow" – angina/MI risk); s/e – symptoms of hyperthyroidism (tachycardia, palpitation) if overdose; check TSH every 6–8 weeks initially.
- Low-calorie, **high-fibre** diet (constipation), warmth, skin care. Sedatives/opioids – **very sensitive** (avoid/small doses).
- **Myxoedema coma** – hypothermia, bradycardia, hypoventilation, hyponatraemia, coma; precipitated by cold, infection, sedatives → **IV levothyroxine**, hydrocortisone, ventilatory support, **passive rewarming (blankets – no heating pads)**.
- **Goitre** – enlarged thyroid; endemic (iodine deficiency) – **iodized salt (≥ 15 ppm at consumer level)**; NIDDCP. **21 October = Global Iodine Deficiency Disorders Prevention Day**.

10.6 Parathyroid Disorders

- **PTH** → ↑ serum Ca (bone resorption, kidney reabsorption, Vit D activation), ↓ phosphate.
- **Hyperparathyroidism** – adenoma; **hypercalcaemia** – **"bones, stones, groans, moans, psychic overtones"** (bone pain/fractures, renal stones, abdominal pain/constipation, depression/confusion). Rx: **fluids 3–4 L**, mobility (weight bearing), **cranberry juice**, furosemide, calcitonin, bisphosphonates, **parathyroidectomy**.
- **Hypoparathyroidism** – most common cause **accidental removal during thyroidectomy**; **hypocalcaemia + hyperphosphataemia** → **tetany**, Chvostek & Trousseau +, laryngospasm, seizures. Rx: **IV calcium gluconate (acute)**, oral calcium + **Vit D (calcitriol)**, **high-calcium, low-phosphorus diet** (avoid milk excess, cola, processed foods), phosphate binders.

10.7 Adrenal Disorders

	Addison's disease (Adrenal insufficiency – hypo)	Cushing's syndrome (↑ cortisol – hyper)
Cause	**Autoimmune** (most common), TB (India), **sudden stoppage of steroids**	**Long-term corticosteroid therapy (most common – iatrogenic)**; Cushing's **disease** = pituitary adenoma (ACTH ↑)
Hormones	**↓ Cortisol, ↓ Aldosterone**	**↑ Cortisol**
Skin	**Bronze/hyperpigmentation** (↑ ACTH/MSH) – skin creases, gums	**Purple striae** on abdomen, thin fragile skin, bruising, acne, **hirsutism**
Body	Weight loss, weakness, fatigue	**Moon face, buffalo hump, truncal (central) obesity**, thin arms & legs
BP	**Hypotension**	**Hypertension**
Sugar	**Hypoglycaemia**	**Hyperglycaemia**
Na / K	**↓ Na, ↑ K** (salt craving)	**↑ Na, ↓ K** (fluid retention)
Others	Nausea, vomiting, **salt craving**	Osteoporosis, infections, poor wound healing, mood swings, muscle wasting

	Addison's disease (Adrenal insufficiency – hypo)	Cushing's syndrome (↑ cortisol – hyper)
Rx	**Lifelong hydrocortisone + fludrocortisone**; **↑ dose during stress/illness**; medical ID bracelet; **high-Na, low-K diet**	Treat cause, taper steroids gradually, adrenalectomy/transsphenoidal surgery; **low Na, high K, high protein, low calorie diet**; infection & fall prevention

! Addisonian crisis (Acute adrenal crisis)

Severe **hypotension, shock**, hypoglycaemia, hyperkalaemia, severe abdominal/back/leg pain, vomiting, confusion → **IV hydrocortisone (100 mg bolus) + IV NS & dextrose** rapidly; treat precipitating cause (stress, infection, surgery, abrupt steroid withdrawal).
Steroid therapy rule: never stop suddenly – **taper**; take **with food** (morning – mimic body rhythm); watch infection, sugar, BP.

- **Primary hyperaldosteronism (Conn's syndrome)** – **HTN + hypokalaemia** + hypernatraemia, metabolic alkalosis; Rx – spironolactone, adrenalectomy.
- **Pheochromocytoma** – tumour of **adrenal medulla** → excess **catecholamines** → **paroxysmal severe HTN**, headache, palpitation, sweating (**triad**), anxiety, hyperglycaemia. Test: **24-hr urine VMA / metanephrines** (avoid coffee, tea, banana, chocolate, vanilla, aspirin before test). **Avoid palpation of abdomen** (crisis). Rx: **α-blocker first (phenoxybenzamine)** then β-blocker → **adrenalectomy**.

10.8 Pituitary Disorders

	Diabetes Insipidus (DI)	SIADH
Problem	**↓ ADH** (central – head injury, pituitary surgery) or kidney not responding (nephrogenic – lithium)	**↑ ADH** (excess) – **small cell lung cancer**, head injury, meningitis, drugs (carbamazepine, SSRIs)
Urine	**Polyuria – large volume (4–20 L/day), very dilute (SG < 1.005)**	**Oliguria, concentrated (SG ↑)**
Serum	**↑ Na (hypernatraemia)**, **↑ osmolality**	**↓ Na (dilutional hyponatraemia < 135)**, ↓ osmolality
Body	**Dehydration**, severe thirst (polydipsia)	**Water retention**, weight gain **without oedema**, confusion, seizures
Rx	**Desmopressin (DDAVP – nasal/oral/IV)**, vasopressin; fluids; I/O, daily weight	**Fluid restriction (500–1000 ml/day – first-line)**, 3 % NS slowly (severe), **demeclocycline**, tolvaptan, furosemide; seizure precautions

♫ DI vs SIADH

DI = "Dry Inside" → paani bahar nikal jata hai (pee a lot, dehydrated, Na high).
SIADH = "Soaked Inside" → paani andar ruk jata hai (pee little, Na low).

- **Acromegaly** – **↑ GH in adults** (after epiphyseal closure) → enlarged hands, feet (shoe/ring size ↑), jaw (prognathism), coarse facial features, thick lips/tongue, DM, HTN. **Gigantism** – ↑ GH in children (before epiphyseal closure). **Dwarfism** – ↓ GH in children.
- **Transsphenoidal hypophysectomy** – post-op: **HOB 30°**, **avoid coughing, sneezing, blowing nose, bending, straining** (CSF leak); **check nasal drainage for glucose (halo sign/ glucose > 30 mg/dL =**

CSF); mouth care without tooth-brushing (10 days); watch **DI** (urine output).

- **Hyperprolactinaemia** – galactorrhoea, amenorrhoea, infertility → bromocriptine/cabergoline.

? Practice MCQs – Chapter 10

1. Diabetes diagnosis ke liye fasting plasma glucose –
(a) ≥ 100 (b) ≥ 110 (c) ≥ 126 mg/dL (d) ≥ 200

2. Kaunsa insulin IV de sakte hain?
(a) Lente (b) Glargine (c) NPH (d) Regular

3. Insulin mixing me pehle kaun draw karte hain?
(a) Regular (clear) (b) NPH (cloudy) (c) Glargine (d) Koi bhi

4. DKA me pehla treatment –
(a) Bicarbonate (b) Potassium (c) IV 0.9 % NS (d) Insulin bolus

5. Conscious hypoglycaemic patient ko –
(a) Insulin (b) 15 g fast-acting sugar (c) NPO (d) Glucagon IV

6. Kussmaul breathing & fruity breath kis condition me?
(a) Hypoglycaemia (b) DKA (c) Myxoedema (d) HHS

7. HbA1c kitne mahine ka average batata hai?
(a) 1 week (b) 2–3 months (c) 6 months (d) 1 year

8. Graves' disease ka characteristic sign –
(a) Bronze skin (b) Moon face (c) Periorbital puffiness (d) Exophthalmos

9. Thyroidectomy ke baad bedside kya hona chahiye?
(a) Heparin (b) Insulin (c) Tracheostomy set & calcium gluconate (d) Digoxin

10. Thyroidectomy ke baad tingling around mouth – kya suspect karein?
(a) Hypoglycaemia (b) Hyperkalaemia (c) Hypocalcaemia (d) Hypercalcaemia

11. Cushing's syndrome ka feature –
(a) Bronze skin (b) Moon face, buffalo hump (c) Hypotension (d) Weight loss

12. Addison's disease me –
(a) ↓ Na, ↑ K (b) ↑ Na, ↓ K (c) Hyperglycaemia (d) Hypertension

13. Diabetes insipidus ki drug –
(a) Furosemide (b) Insulin (c) Demeclocycline (d) Desmopressin

14. SIADH ka first-line treatment –
(a) Insulin (b) Desmopressin (c) Fluid restriction (d) Fluid bolus

15. Levothyroxine kab lena chahiye?
(a) Kabhi bhi (b) Subah khali pet (c) Raat ko khane ke saath (d) Dopahar khane ke baad

16. Pheochromocytoma ka test –
(a) Urine VMA / metanephrines (b) TSH (c) HbA1c (d) Cortisol

17. Metformin ka serious side effect –
(a) Weight gain (b) Lactic acidosis (c) Hyperkalaemia (d) Hypoglycaemia

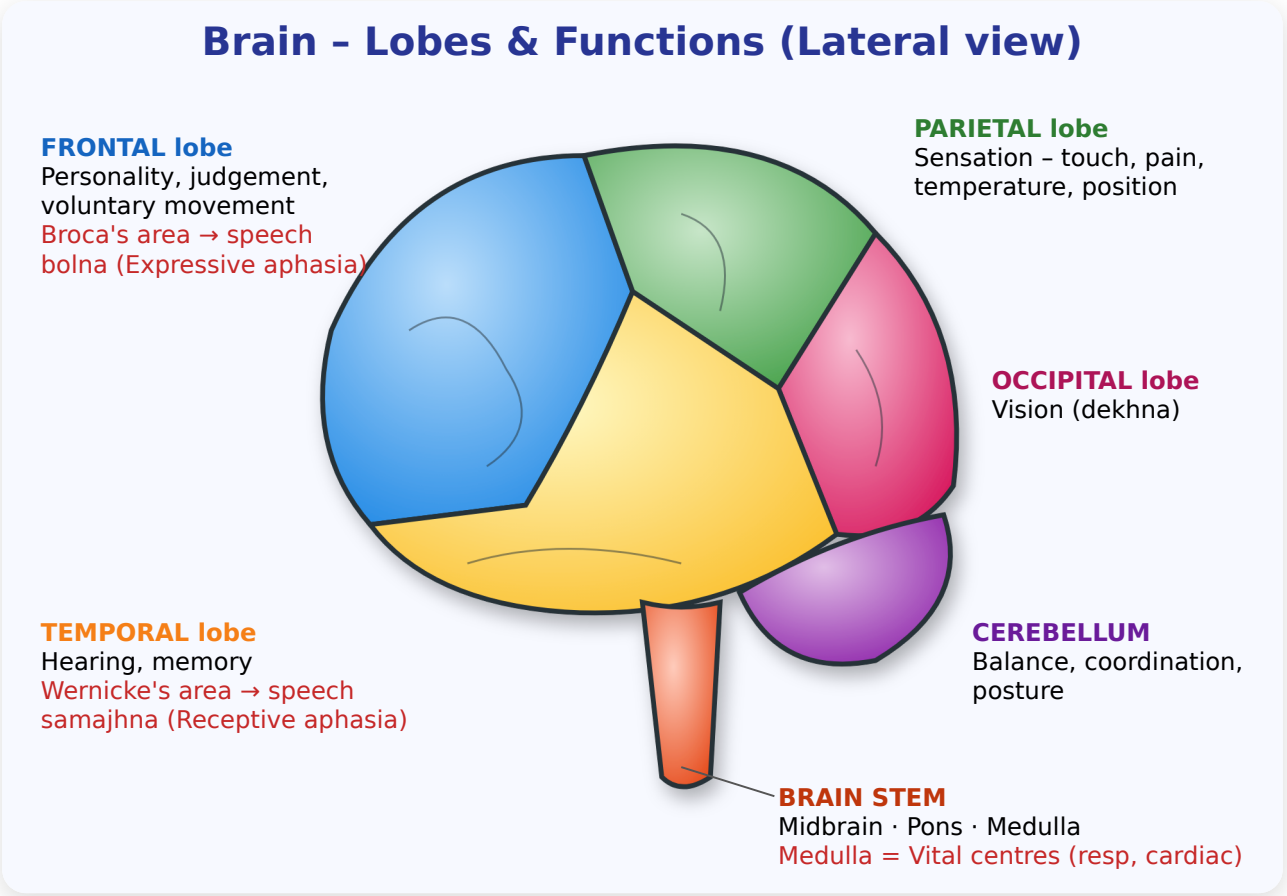
18. Somogyi effect me morning hyperglycaemia ka kaaran –
(a) Too little insulin (b) Dawn hormones (c) Infection (d) Night hypoglycaemia se rebound

Answer Key – Chapter 10

1-c · 2-d · 3-a · 4-c · 5-b · 6-b · 7-b · 8-d · 9-c · 10-c · 11-b · 12-a · 13-d · 14-c · 15-b · 16-a · 17-b · 18-d

11 Neurological System Disorders

11.1 Applied Anatomy



Brain ke lobes aur unke kaam

Nervous system – facts

- **CNS** = brain + spinal cord; **PNS** = 12 cranial + **31 pairs spinal nerves**.
- Brain weight ~**1.4 kg**; receives **15–20 % cardiac output**, **20 % of body O_2**; glucose = main fuel.
- Meninges: **Dura mater** (outer) → **Arachnoid** → **Pia mater** (inner). **CSF in subarachnoid space**.
- **CSF**: produced by **choroid plexus** (~500 ml/day), total volume **~150 ml**, normal pressure **60–180 mmH_2O (5–15 mmHg)**, **clear, colourless**; protein 15–45 mg/dL; glucose 50–80 mg/dL (~2/3 of blood glucose).
- Spinal cord ends at **L1–L2** (adult) → **lumbar puncture at L3–L4 / L4–L5** (safe).
- **Circle of Willis** – arterial ring at brain base; **Middle cerebral artery** – most common stroke site.
- Dominant hemisphere (speech) = **left** in most people.

12 Cranial Nerves

No.	Nerve	Type	Function / Test
I	**Olfactory**	S	Smell – coffee/soap with eyes closed
II	**Optic**	S	Vision – Snellen chart, visual fields

No.	Nerve	Type	Function / Test
III	**Oculomotor**	M	Eye movement, **pupil constriction**, eyelid raise (**ptosis if damaged**)
IV	**Trochlear**	M	Eye down & in (superior oblique)
V	**Trigeminal**	B	**Face sensation, mastication**, **corneal reflex** (sensory) – **trigeminal neuralgia**
VI	**Abducens**	M	Lateral eye movement (lateral rectus)
VII	**Facial**	B	**Facial expression**, taste anterior 2/3 tongue, tears/saliva – **Bell's palsy**
VIII	**Vestibulocochlear (Auditory)**	S	**Hearing & balance** – Rinne/Weber
IX	**Glossopharyngeal**	B	Taste posterior 1/3, **gag reflex (sensory)**, swallowing
X	**Vagus**	B	**Gag reflex (motor)**, swallowing, voice, parasympathetic to heart/GI
XI	**Accessory (Spinal)**	M	**Shoulder shrug** (trapezius), head turn (SCM)
XII	**Hypoglossal**	M	**Tongue movement**

♫ **Names & type**

Names: "**O**h **O**h **O**h **T**o **T**ouch **A**nd **F**eel **V**ery **G**ood **V**elvet, **A**h **H**eaven"
Type (S/M/B): "**S**ome **S**ay **M**arry **M**oney, **B**ut **M**y **B**rother **S**ays **B**ig **B**rains **M**atter **M**ore"

11.2 Neurological Assessment

Glasgow Coma Scale (GCS) – Teasdale & Jennett (1974)

Eye opening (E)		Verbal response (V)		Motor response (M)	
Spontaneous	**4**	Oriented	**5**	Obeys commands	**6**
To voice/speech	3	Confused	4	Localizes pain	5
To pain	2	Inappropriate words	3	Withdraws from pain	4
None	1	Incomprehensible sounds	2	**Abnormal flexion (decorticate)**	3
		None	1	**Extension (decerebrate)**	2
				None	1

✎ **GCS – must remember**

- Total **3 – 15**. **15 = normal**; **≤ 8 = severe injury / coma → "8 = intubate"**; 9–12 moderate; 13–14 mild.
- Lowest score **3** (not 0). Intubated patient – verbal "T" (e.g. 10T).
- **Decorticate** (flexion – arms flexed to chest, "**to the core**") = cerebral hemisphere damage. **Decerebrate** (extension – arms straight, wrists pronated) = **brainstem damage – worse prognosis**.
- **Most sensitive & earliest indicator of neuro deterioration = change in level of consciousness (LOC)**.

- **Pupils** – **PERRLA** (Pupils Equal, Round, Reactive to Light & Accommodation); normal 2–6 mm. **Unilateral fixed dilated pupil** = **uncal herniation (CN III compression – same side)**; pinpoint = pons lesion/opioids; bilateral fixed dilated = severe hypoxia/brain death.
- **Babinski sign** – sole stroke → **toes fanning & big toe dorsiflexion = positive (abnormal in adults)**, normal in infants < 1–2 yrs → UMN lesion.
- **Romberg test** – feet together, eyes closed → sway = positive (proprioception/cerebellar).
- **Kernig's & Brudzinski's** signs → meningeal irritation.
- **Reflex grading**: 0 absent, 1+ diminished, **2+ normal**, 3+ brisk, 4+ hyperactive with clonus.

Diagnostic procedures

- **Lumbar puncture**: empty bladder; position – **lateral recumbent, fetal position (knees to chest, chin to chest)** or sitting leaning forward; **L3–L4/L4–L5**; **contraindicated in ↑ICP** (herniation) – check fundus/CT first. After: **lie flat (supine) 4–12 hrs** (post-LP headache prevention), ↑ fluids, check site & legs sensation.
- **CT scan** (first choice in acute stroke/head injury – bleeding rule out), **MRI** (remove metal; pacemaker contraindicated; claustrophobia), **EEG** (seizures – hair wash, **no caffeine** 24–48 hr before, sleep deprivation may be ordered; continue/hold antiepileptics as ordered).
- **Cerebral angiography** – iodine allergy check; after: puncture site & distal pulses, leg straight.

11.3 Increased Intracranial Pressure (↑ICP)

- Normal ICP **5–15 mmHg** (0–15); **> 20 mmHg sustained = treat**.
- **Monro-Kellie doctrine**: skull is rigid box with **brain (80 %), blood (10 %), CSF (10 %)** – one ↑ → others must ↓, otherwise ICP ↑.
- **CPP (Cerebral perfusion pressure) = MAP − ICP**; normal **60–100 mmHg** (keep **≥ 60–70**).
- Causes: head injury, haemorrhage, tumour, cerebral oedema, hydrocephalus, meningitis.

Signs of ↑ICP

- **Earliest sign: change in level of consciousness** (restlessness, confusion, drowsiness)
- **Headache** (worse in morning), **projectile vomiting** (without nausea), **papilloedema**
- Pupillary changes (ipsilateral dilated, sluggish), motor weakness, seizures
- **Late sign – Cushing's triad**: **↑ Systolic BP (widened pulse pressure) + Bradycardia + Irregular respiration (Cheyne-Stokes/ bradypnoea)**
- Infants: **bulging fontanelle**, ↑ head circumference, high-pitched cry, **sunset eyes**

↑ICP – nursing management

- **HOB 30°**, head in **neutral midline** (no neck flexion/rotation – jugular drainage)
- Avoid **hip flexion > 90°**, **Valsalva (straining, coughing, sneezing)**, **isometric exercise** – stool softeners
- Avoid clustering of nursing activities; **suction < 10 sec** (only when needed), hyperoxygenate
- Quiet dark environment, limit visitors
- **Osmotic diuretic – Mannitol 20 %** (use **filter**; crystals – warm; monitor serum osmolality < 320, I/O), hypertonic saline 3 %
- Maintain **$PaCO_2$ 35–40** (hyperventilation briefly only in herniation), normothermia (treat fever – ↑ metabolism), seizure prophylaxis
- **Fluid** – **isotonic NS**; avoid hypotonic (D5W) fluids
- Corticosteroids (**dexamethasone**) – for **tumour oedema** (not head injury)
- **No opioids/sedatives that mask LOC** (unless ventilated), avoid restraints (struggling ↑ICP)

11.4 Head Injury

- **Scalp** laceration – bleeds profusely. **Skull fracture**: linear, depressed, basilar.
- **Basilar skull fracture**: **Battle's sign** (bruising **behind ear – mastoid**), **Raccoon eyes** (periorbital bruising), **CSF rhinorrhoea/otorrhoea** – **halo/ring sign** on gauze (blood centre, yellow ring), **glucose positive** (dextrostix). **Do NOT** put NG tube (use orogastric), no nasal suction, no packing nose/ear, don't let patient blow nose.
- **Concussion** – temporary loss of consciousness, no structural damage, amnesia; observe 24 hrs.
- **Contusion** – bruise of brain tissue.

	Epidural (Extradural) haematoma	Subdural haematoma
Bleed	**Arterial – middle meningeal artery** (temporal bone fracture)	**Venous – bridging veins**
Course	**Brief loss of consciousness → "Lucid interval" → rapid deterioration**	Acute (24–48 hr), subacute, **chronic (weeks – elderly, alcoholics, anticoagulants)**
CT	**Biconvex (lens-shaped)**	**Crescent-shaped**
Rx	**Emergency craniotomy / burr hole**	Evacuation

- **Subarachnoid haemorrhage** – ruptured **berry aneurysm** → "**worst headache of my life**" (thunderclap), neck stiffness, photophobia; nimodipine (vasospasm prevention).
- Head injury first priority: **ABC with cervical spine immobilization** (assume C-spine injury in every head injury).

11.5 Stroke (CVA – Cerebrovascular Accident / Brain Attack)

D Stroke

Sudden loss of brain function due to **disruption of blood supply**. **Ischaemic (~85 %)** – thrombotic (most common; during sleep/rest, atherosclerosis) or embolic (**AF** – most common source; sudden, during activity); **Haemorrhagic (~15 %)** – intracerebral (**hypertension** – most common cause) or subarachnoid.

TIA (Transient Ischaemic Attack) – symptoms resolve **within 24 hrs (usually < 1 hr)** without infarction – **warning sign of stroke**.

♫ BE FAST – stroke recognition

Balance loss · **E**yes (vision loss) · **F**ace drooping · **A**rm weakness · **S**peech difficulty · **T**ime to call **108** – note the **time of onset / last known well**

Clinical features (depends on site) – left vs right hemisphere

Left hemisphere stroke	Right hemisphere stroke
Right-sided hemiplegia	**Left-sided hemiplegia**
Aphasia (speech problems)	**Spatial-perceptual deficits**, **left neglect**
Slow, cautious behaviour	**Impulsive**, poor judgement, denies deficits
Depression, anxiety	Short attention span; **safety risk ↑**

- **Expressive (Broca's) aphasia** – frontal lobe; patient **understands but can't speak** (frustrated). **Receptive (Wernicke's)** – temporal; **fluent but meaningless speech**, can't understand. **Global** – both.
- **Dysphagia** (aspiration risk), **dysarthria** (slurred – motor), **homonymous hemianopia** (loss of same half of visual field both eyes → teach to **turn head (scan) toward affected side**, place items on **unaffected side**), **agnosia** (can't recognize objects), **apraxia** (can't perform learned movements).
- Hemiplegia → **flaccid initially → spastic** later.

Management

- **CT scan first (non-contrast)** – rule out haemorrhage **before thrombolytics**.
- **Thrombolysis (tPA – Alteplase 0.9 mg/kg, max 90 mg; 10 % bolus)** – within **3–4.5 hrs** of symptom onset; BP must be **< 185/110**; no anticoagulants/IM injections/ invasive lines for 24 hrs; watch **bleeding & neuro status**. **Mechanical thrombectomy** – up to 24 hrs (large vessel).
- Ischaemic: aspirin (after 24 hrs if tPA given), antiplatelets, statins; BP permissive (don't lower aggressively unless > 220/120). Haemorrhagic: BP control, reverse anticoagulants, surgery.
- **HOB 30°** (haemorrhagic/↑ICP), ischaemic – flat or ≤ 30° as ordered.

Stroke – nursing care

- **Swallow screen (bedside) before any oral intake** – NPO until cleared; **thickened liquids**, upright **90° for meals**, chin tuck, food on **unaffected side** of mouth, check pocketing.
- **Positioning**: prevent contractures – turn every 2 hrs, **hand roll / splint**, **trochanter roll** (external rotation of hip ↓), **footboard/ high-top sneakers** (footdrop), arm on pillow (shoulder subluxation – **don't pull affected arm**).
- **ROM exercises** within 24–48 hrs, early mobilization.
- Transfer/ambulation – **stand on/ assist from affected side**; cane in **unaffected (strong) hand**.
- Dressing: **put clothes on affected side first, remove from unaffected side first**.
- Communication: simple short sentences, picture boards, give time, one question at a time.
- Bladder training, bowel program, skin care, emotional lability support.

11.6 Seizures & Epilepsy

- **Seizure** = abnormal, excessive, sudden discharge of neurons. **Epilepsy** = **≥ 2 unprovoked seizures > 24 hrs apart**.
- **Generalized tonic-clonic (Grand mal)**: aura (in focal onset) → **cry**, loss of consciousness → **tonic phase (stiff, 10–20 sec)** → **clonic phase (jerking, 30–60 sec)** → **post-ictal** (confusion, sleep, headache), **incontinence, tongue bite**.
- **Absence (Petit mal)** – children, **brief staring (5–10 sec)**, eye blinking; many times/day.
- **Myoclonic** – brief jerks; **Atonic (drop attacks)** – loss of muscle tone → fall; **Focal (partial)** – simple (conscious) / complex (impaired awareness, automatisms – lip smacking).
- **Status epilepticus** – seizure **> 5 min** or repeated seizures without regaining consciousness → **medical emergency** → **IV Lorazepam (drug of choice)** / Diazepam (IV/rectal), Midazolam (IM/buccal) → then **Phenytoin/Fosphenytoin**, levetiracetam, valproate.

During a seizure – Do's & Don'ts

DO: stay with patient, **note time & characteristics**, protect head (pillow), **loosen tight clothing**, move dangerous objects, **turn to side (lateral) after/ during** to drain saliva, maintain airway, O_2, suction ready, side rails padded.
DON'T: **restrain**, **put anything in mouth** (no tongue blade/spoon/finger), give oral fluids, leave patient.
After: recovery position, reorient, check injuries, document.

Antiepileptic drugs

Drug	Key side effects / points
Phenytoin	**Gingival hyperplasia** (oral hygiene, soft brush), **hirsutism**, ataxia, nystagmus, **teratogenic**; therapeutic level **10–20 µg/ml**; IV only with **NS** (precipitates in dextrose), **≤ 50 mg/min** (hypotension, arrhythmia); purple glove syndrome
Carbamazepine	Trigeminal neuralgia drug of choice, **agranulocytosis/aplastic anaemia**, SJS, hyponatraemia
Valproate (Sodium valproate)	**Hepatotoxicity**, **pancreatitis**, weight gain, hair loss, **neural tube defects** (avoid in pregnancy)

Drug	Key side effects / points
Phenobarbital	Sedation, neonatal seizures drug of choice
Levetiracetam	Behaviour changes, safer
Ethosuximide	**Absence seizures** drug of choice

- Teaching: **never stop suddenly** (status epilepticus), regular dose, avoid alcohol & sleep deprivation, flashing lights, **no driving / swimming alone / heights** till seizure-free (as per law), medical ID, contraception interactions.

11.7 Infections of CNS

Meningitis

- Inflammation of meninges. **Bacterial** (most serious): **N. meningitidis** (meningococcal – **petechial/purpuric rash**, epidemics in crowded places), **S. pneumoniae** (adults – most common), **H. influenzae** (children – vaccine). Viral (aseptic) – milder. TB meningitis.
- S/S: **severe headache, fever, nuchal (neck) rigidity**, **photophobia**, vomiting, altered consciousness, seizures; infants – bulging fontanelle, poor feeding, high-pitched cry.
- **Kernig's sign** – hip & knee flexed 90° → **knee extension painful/resisted**. **Brudzinski's sign** – **neck flexion → involuntary hip & knee flexion**.
- Diagnosis: **Lumbar puncture – CSF**:

CSF	Bacterial	Viral	TB
Appearance	**Turbid/cloudy**	Clear	Clear/cobweb
Cells	**↑ Neutrophils**	↑ Lymphocytes	↑ Lymphocytes
Protein	**↑↑**	Normal/slight ↑	**↑↑**
Glucose	**↓↓**	**Normal**	↓

- Rx: **droplet isolation for first 24 hrs of antibiotics** (meningococcal), **IV antibiotics ASAP** (after cultures – don't delay), dexamethasone, quiet **dark room**, HOB 30°, seizure & ↑ICP precautions, fluids; **close contacts prophylaxis – Rifampicin / Ciprofloxacin**.
- **Encephalitis** – brain tissue inflammation; viral (HSV – **acyclovir**; **Japanese encephalitis** – Culex mosquito, vaccine in UIP).
- **Brain abscess**, **Tetanus** (see communicable chapter).

11.8 Spinal Cord Injury (SCI)

- Common cause: RTA, falls, diving; most common sites **C5–C7, T12–L1**; young males.
- **Complete** (no function below) vs **incomplete** – central cord, Brown-Séquard, anterior cord syndromes.
- **Level**: **C1–C4 → quadriplegia + ventilator dependent** (C3-4-5 diaphragm); **C5–C8 → tetraplegia** with some arm function; **T1 & below → paraplegia**.
- **Spinal shock** – temporary loss of all reflexes below injury (**flaccid paralysis, areflexia**), lasts days–weeks; return of reflexes = end.

- **Neurogenic shock** – injury above **T6** → **hypotension + bradycardia + warm dry skin** (loss of sympathetic tone) → fluids, vasopressors, atropine.

> **! Autonomic Dysreflexia (Hyperreflexia) – EMERGENCY**
> Injury **at/above T6**; triggered by noxious stimulus below injury – **full bladder (most common)**, constipation/faecal impaction, tight clothes, pressure ulcer.
> S/S: **Sudden severe hypertension (up to 300 mmHg)**, **pounding headache**, **bradycardia**, flushing & sweating **above** injury, cold pale skin below, nasal stuffiness, blurred vision.
> Rx: **Sit patient upright immediately (HOB 90°, legs down)** → loosen clothes → **find & remove cause – check bladder (catheter kinked? catheterize)** → bowel (with anaesthetic gel) → skin; **antihypertensive (nifedipine/nitrate)** if BP stays high. Risk: stroke, seizure.

- First aid & care: **immobilize spine** (cervical collar, spine board), **log rolling** (3–4 persons), **jaw thrust** for airway; **methylprednisolone – no longer routinely recommended**.
- **Traction/stabilization**: Crutchfield tongs, **Halo vest** (keep **wrench taped to vest** for emergency CPR; pin site care; never lift by halo frame).
- Long-term: **intermittent self-catheterization**, bowel program (suppository, digital stimulation), **skin care – turn every 2 hrs**, pressure relief, DVT prevention, contracture prevention, respiratory care, sexuality counselling, rehab.

11.9 Degenerative & Neuromuscular Disorders

Disease	Key pathology	Key features	Treatment / Nursing
Parkinson's disease	**↓ Dopamine** in substantia nigra (basal ganglia)	**TRAP**: **T**remor (**resting, "pill-rolling"** – first sign), **R**igidity (cogwheel), **A**kinesia/**Bradykinesia**, **P**ostural instability; **shuffling/festinating gait**, **mask-like face**, micrographia, drooling, soft monotone voice	**Levodopa-Carbidopa** (avoid **Vit B6/pyridoxine & high-protein meals** with dose; s/e – dyskinesia, orthostatic hypotension, dark urine/sweat), dopamine agonists, MAO-B inhibitors, anticholinergics (tremor); **fall prevention**, raised toilet seat, walk with wide base, swing arms, march to music, thickened liquids
Myasthenia gravis	**Autoimmune – antibodies against acetylcholine receptors** at neuromuscular junction; thymus abnormality	**Muscle weakness that ↑ with activity & ↓ with rest**; **ptosis & diplopia** (first signs), dysphagia, weak voice, respiratory weakness	**Anticholinesterases – Pyridostigmine (Mestinon)** – give **on time, 30–60 min before meals**; **Tensilon (edrophonium) test**; thymectomy, plasmapheresis, IVIG, steroids; **plan activities in peak energy (morning)**

Disease	Key pathology	Key features	Treatment / Nursing
Guillain-Barré syndrome (GBS)	**Autoimmune demyelination of peripheral nerves** – after viral/**Campylobacter** infection, vaccine	**Ascending symmetrical flaccid paralysis** (legs → upward), areflexia, paresthesia, **respiratory failure** (major risk), autonomic instability; **CSF – ↑ protein, normal cells (albuminocytological dissociation)**	**Monitor respiration – vital capacity** (< 15–20 ml/kg → ventilate), **plasmapheresis / IVIG**, DVT prevention, pain mx; most recover
Multiple sclerosis	**Autoimmune demyelination of CNS** (young women 20–40)	**Relapsing-remitting**; visual problems (**optic neuritis, diplopia**), fatigue, numbness, spasticity, **Charcot's triad (nystagmus, intention tremor, scanning speech)**, bladder problems; **Uhthoff's** (heat worsens)	Steroids for relapse, interferon-β, disease-modifying drugs; **avoid heat (hot baths)**, stress, infection; energy conservation
Alzheimer's disease	Neurofibrillary tangles, **amyloid plaques**, ↓ acetylcholine	Progressive memory loss (**recent memory first**), confusion, wandering, sundowning, aphasia	**Donepezil**, rivastigmine (cholinesterase inhibitors), memantine; safe environment, routine, reality orientation (early), validation therapy (late), **ID bracelet**
Amyotrophic lateral sclerosis (ALS)	Degeneration of **upper & lower motor neurons**	Progressive weakness, fasciculations, dysphagia, respiratory failure; **cognition intact**	Riluzole, edaravone; supportive
Huntington's disease	Autosomal dominant, chorea	Choreiform movements, dementia	Supportive, genetic counselling

✎ Myasthenic vs Cholinergic crisis

	Myasthenic crisis	Cholinergic crisis
Cause	**Too little** medication / infection / stress	**Too much** anticholinesterase
Tensilon (edrophonium) test	**Improves (strength ↑)**	**Worsens**
Features	Weakness, respiratory failure, ↑ HR, ↑ BP	**SLUDGE** (Salivation, Lacrimation, Urination, Diarrhoea, GI cramps, Emesis), bradycardia, miosis, fasciculations
Rx	More pyridostigmine, ventilate, IVIG/plasmapheresis	**Atropine** (antidote), stop drug, ventilate

- **Bell's palsy** – CN VII, sudden unilateral facial paralysis (can't close eye, drooling) → **protect eye (patch, artificial tears)**, steroids, facial exercises; most recover in weeks–months.
- **Trigeminal neuralgia (Tic douloureux)** – CN V, **severe paroxysmal stabbing facial pain** triggered by touch, chewing, cold air, brushing → **carbamazepine**; room temperature food, chew on unaffected

side, avoid triggers.

- **Brain tumours** – most common primary **glioma (glioblastoma multiforme – most malignant)**, meningioma (benign); S/S of ↑ICP, seizures, focal deficits.
- **Craniotomy post-op**: **Supratentorial → HOB 30–45°**; **Infratentorial (posterior fossa) → flat, side-lying** (avoid neck flexion), NPO initially (gag reflex); watch ↑ICP, CSF leak, DI/SIADH, seizures.

? Practice MCQs – Chapter 11

1. GCS score ≤ 8 ka matlab –
(a) Moderate (b) Severe – intubate (c) Mild injury (d) Normal

2. ↑ICP ka earliest sign –
(a) Fixed pupils (b) Cushing's triad (c) Change in level of consciousness (d) Papilloedema

3. Cushing's triad me hota hai –
(a) ↑SBP/wide pulse pressure, bradycardia, irregular respiration (b) Hypoglycaemia (c) Fever, rash, stiff neck (d) Hypotension, tachycardia, tachypnoea

4. ↑ICP me head of bed –
(a) 90° (b) 30° with neck neutral (c) Trendelenburg (d) Flat

5. Epidural haematoma ka classic feature –
(a) Slow onset weeks (b) Crescent shape (c) Venous bleed (d) Lucid interval

6. Battle's sign kahan dikhta hai?
(a) Behind ear (mastoid) (b) Flank (c) Around eyes (d) Umbilicus

7. Stroke me tPA ka window –
(a) 12 hrs (b) Within 3–4.5 hrs (c) 1 week (d) 24 hrs

8. Hemiplegic patient ko kapde pehnate waqt –
(a) Dono ek saath (b) Koi bhi (c) Pehle affected side (d) Pehle unaffected side

9. Seizure ke dauran kya nahi karna chahiye?
(a) Time note karna (b) Mouth me spoon dalna (c) Side turn (d) Head protect

10. Status epilepticus ka drug of choice –
(a) Carbamazepine (b) Phenytoin oral (c) IV Lorazepam (d) Paracetamol

11. Phenytoin ka common side effect –
(a) Hypoglycaemia (b) Ototoxicity (c) Orange urine (d) Gingival hyperplasia

12. Kernig's & Brudzinski's sign kis disease me?
(a) MG (b) Parkinson's (c) Meningitis (d) Stroke

13. Autonomic dysreflexia me first action –
(a) Give fluids (b) Flat lay (c) Trendelenburg (d) Sit upright & check bladder

14. Parkinson's disease me kis neurotransmitter ki kami?
(a) GABA (b) Serotonin (c) Dopamine (d) Acetylcholine

15. Myasthenia gravis me weakness –
(a) Activity se badhti, rest se ghatti (b) Only at night (c) Rest se badhti, activity se ghatti (d) Constant

16. GBS ki main complication –
(a) Jaundice (b) Hypertension (c) Diarrhoea (d) Respiratory failure

17. Lumbar puncture ke baad position –
(a) Prone with legs up (b) Trendelenburg (c) Flat supine (d) Fowler's

18. Cholinergic crisis ka antidote –

(a) Neostigmine (b) Atropine (c) Naloxone (d) Pyridostigmine

Answer Key – Chapter 11

1-b · 2-c · 3-a · 4-b · 5-d · 6-a · 7-b · 8-c · 9-b · 10-c · 11-d · 12-c · 13-d · 14-c · 15-a · 16-d · 17-c · 18-b

12 Musculoskeletal System Disorders

12.1 Applied Anatomy

Bones & joints – facts

- Adult skeleton = **206 bones**; newborn ~270–300. **Largest bone – Femur**; **smallest – Stapes**; longest = femur.
- Bone cells: **Osteoblasts** (bone **b**uild), **Osteoclasts** (bone **c**rush/resorb), Osteocytes (mature).
- Bone growth – epiphyseal plate; bone remodelling regulated by **PTH, calcitonin, Vit D, oestrogen**.
- Joints: **Synarthrosis** (immovable – skull sutures), **Amphiarthrosis** (slightly – pubic symphysis), **Diarthrosis/synovial** (freely – knee, hip, shoulder). **Ball & socket** – hip, shoulder; **Hinge** – elbow, knee; **Pivot** – atlas-axis.
- Muscle types: skeletal (voluntary), cardiac, smooth.
- **Ligament** = bone to bone; **Tendon** = muscle to bone.

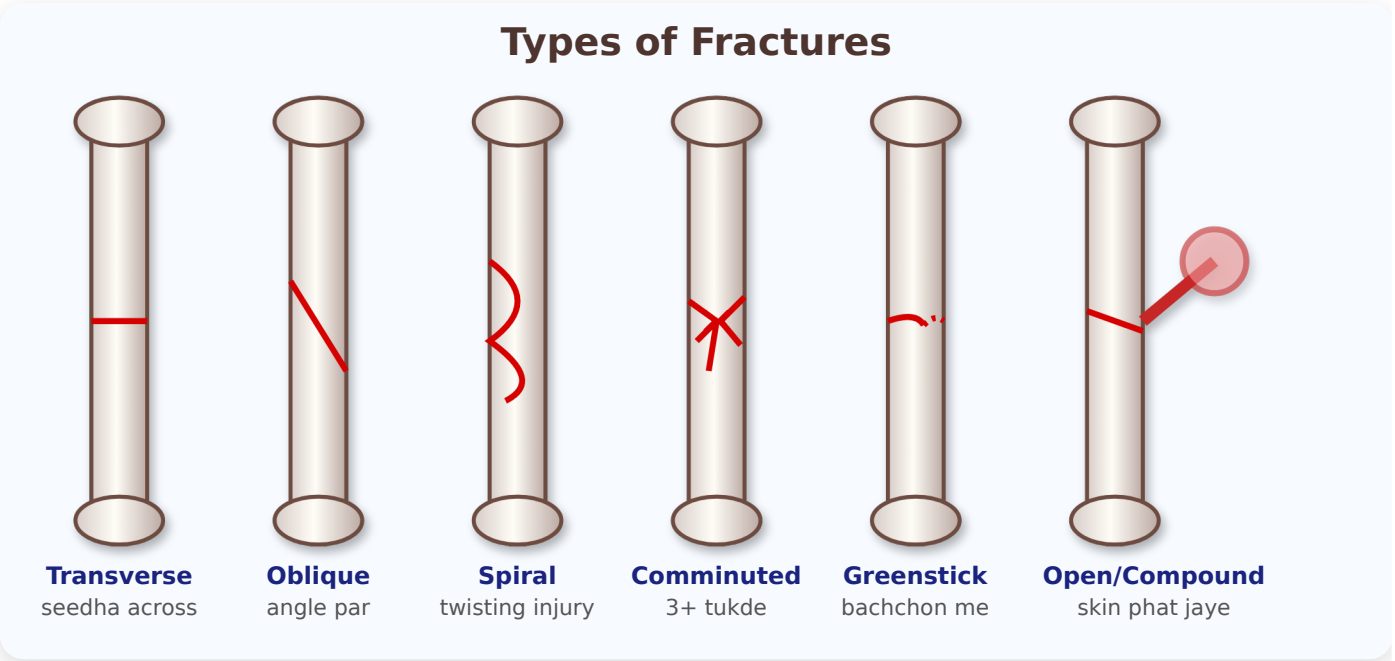
Common terms

Sprain (ligament injury), **Strain** (muscle/tendon – "pulled muscle"), **Dislocation** (complete displacement of joint surfaces), **Subluxation** (partial), **Contracture**, **Ankylosis** (joint fusion), **Crepitus** (grating), **Kyphosis** (hunchback – thoracic), **Lordosis** (sway back – lumbar), **Scoliosis** (lateral curve).

RICE – soft tissue injury (first 24–48 hrs)

Rest · **I**ce (20 min at a time, cloth wrapped; first 24–48 hrs) · **C**ompression (elastic bandage) · **E**levation (above heart level). After 48 hrs → **heat**.

12.2 Fractures



Fractures ke common types

Type	Meaning
Closed (simple)	Skin intact
Open (compound)	Bone exposed through skin → **infection (osteomyelitis) risk**
Complete / Incomplete	Full break / partial
Greenstick	Incomplete – one side breaks, other bends (**children** – soft bones)
Comminuted	**3 or more fragments** (crushed)
Transverse / Oblique / Spiral	Straight across / angle / twisting injury (child abuse suspicion in infants)
Impacted	One fragment driven into another
Compression	Vertebra crushed (osteoporosis)
Pathological	Weak bone se minimal trauma – **osteoporosis, cancer metastasis**, osteomyelitis
Avulsion	Ligament/tendon pulls a bone fragment
Depressed	Skull – fragment pushed inward
Stress / fatigue	Repeated stress – athletes, soldiers (march fracture – metatarsal)
Colles' fracture	**Distal radius**, dorsal displacement – **"dinner fork" deformity**; fall on outstretched hand (elderly women)
Smith's	Distal radius volar displacement (reverse Colles')
Pott's fracture	Ankle (bimalleolar)

- **S/S**: **pain & tenderness**, **deformity**, swelling, ecchymosis, **loss of function**, **crepitus**, abnormal movement, **shortening** (hip – also **external rotation**).
- **Bone healing stages**: Haematoma formation (1–3 days) → Fibrocartilage/granulation (3 days–2 weeks) → **Callus formation** (2–6 weeks) → Ossification (3 weeks–6 months) → Remodelling (up to 1

yr).
- Healing time approx: **Upper limb 6–8 weeks; Lower limb 12–16 weeks**; children faster.

First aid for fracture

- ABC, control bleeding (open fracture – **cover with sterile dressing; don't push bone back**)
- **Splint "as it lies"** – include **joint above & below** the fracture
- Elevate, ice, check **neurovascular status distal** before & after splinting
- Don't straighten / move unnecessarily

Treatment

- **Reduction**: **Closed** (manipulation) or **Open reduction internal fixation (ORIF)** – plates, screws, nails, rods.
- **Immobilization**: cast, splint, traction, external fixator.
- **Rehabilitation**: exercises, weight-bearing as ordered.

5 P's – Neurovascular assessment (also "6 P's")

Pain (out of proportion – earliest), **P**allor, **P**ulselessness (late), **P**aresthesia, **P**aralysis (+ **P**oikilothermia – cool).
Check distal to injury/cast **every 1 hr for first 24 hrs**, then every 4 hrs. Compare with other limb. **Capillary refill < 2–3 sec** normal.

Cast care

- **Plaster of Paris (POP)** – dries in **24–72 hrs** (feels cool & damp while wet; dry = white, shiny, resonant, odourless); **Fibreglass** – dries in 30 min, lightweight, water-resistant.
- Wet cast – handle with **palms (not fingers)** → finger dents → pressure points; place on **pillow**, keep uncovered to dry; turn every 2 hrs; no heater/hair-dryer on hot setting.
- **Elevate above heart level first 24–48 hrs**; ice packs.
- **Never insert anything inside cast to scratch** (pen, stick) → skin injury → infection; use cool air from hair-dryer.
- Petal the edges; keep dry; check **"hot spots"** (warm area on cast = infection), **foul smell, drainage** → report.
- Isometric exercises of muscles under cast; ROM of joints not immobilized.
- **Cast syndrome** (body/hip spica cast) – **superior mesenteric artery compression** → **nausea, vomiting, abdominal distension, pain** → report; small frequent meals.
- Cast removal – **cast cutter vibrates, doesn't cut skin**; skin washed gently, oil/lotion, don't scrub flaky skin.

Traction

D **Traction**

Pulling force applied to a body part to **reduce & immobilize fracture**, decrease muscle spasm, prevent deformity. **Counter-traction** (usually patient's body weight; bed elevation) is needed.

Type	Example	Use
Skin traction (max 2.5–4.5 kg / 5–10 lb; intermittent)	**Buck's extension** (straight pull – **hip fracture pre-op**, knee), **Russell's** (knee sling – femoral), **Bryant's** (children < 2 yrs/ < 14 kg – both legs vertical 90°, **buttocks slightly off bed**), Cervical halter, pelvic belt	Short-term
Skeletal traction (pin/wire through bone; 7–14 kg / 15–30 lb, continuous)	**Balanced suspension with Thomas splint + Pearson attachment** (femur), Crutchfield/Gardner-Wells tongs (cervical), **Halo**	Long-term; stronger pull

! Traction – golden rules

- **Weights hang freely** – never on floor/bed; **never remove weights** (skeletal) without order; never lift weights when moving patient.
- Ropes in pulley grooves, knots away from pulleys; **body in alignment, centre of bed**; feet away from footboard.
- **Pin site care** – sterile, as per protocol (chlorhexidine / NS) – watch infection (redness, drainage, pin loosening).
- Neurovascular checks, skin care (pressure ulcers), DVT/constipation/pneumonia prevention, **trapeze bar** to move.

Complications of fractures

- **Compartment syndrome** – ↑ pressure in a closed muscle compartment (tight cast/bandage, swelling) → ischaemia. **Earliest sign: severe pain unrelieved by opioids, increased on passive stretch** → paresthesia → pallor → **pulselessness (late)**. Rx: **Loosen/bivalve cast, remove tight bandage**, **keep limb at heart level (NOT elevated – ↓ perfusion)**, no ice, **fasciotomy within 4–6 hrs**. Complication → **Volkmann's ischaemic contracture** (forearm – supracondylar fracture).
- **Fat embolism syndrome** – **long bone (femur) / pelvic fractures**, within **24–72 hrs** → **dyspnoea, hypoxaemia (first), confusion/restlessness, petechiae (chest, axilla, conjunctiva)**, fever, tachycardia. Prevention: **immobilize fracture early, minimal movement**. Rx: **High Fowler's, O_2 high-flow**, ventilation, fluids, steroids.
- **DVT/PE**, **shock** (haemorrhage – pelvis/femur large blood loss), **infection/osteomyelitis**, **delayed union, non-union, malunion**, **avascular necrosis** (head of femur), **Complex regional pain syndrome**.

Hip fracture

- Elderly women, osteoporosis, fall. **Affected leg – shortened, externally rotated, adducted**, pain in hip/groin.
- **Intracapsular** (femoral neck – **AVN risk**; hemiarthroplasty) vs **Extracapsular** (intertrochanteric – ORIF).
- Pre-op: **Buck's traction** (spasm ↓).
- **Total hip replacement (THR/THA) – prevent dislocation**:
  - **Abduction pillow (wedge) between legs** – keep legs apart
  - **No hip flexion > 90°** (no low chairs, raised toilet seat, no bending to tie shoes – use long-handled reacher/ shoe horn)
  - **No adduction (crossing legs)**, **no internal rotation** (toes pointing in)
  - Sit in high firm chair with arms; turn to **unaffected side** with pillow between legs (as per surgeon)

- Signs of dislocation: **sudden pain, shortening of leg, rotation** → report

- **Total knee replacement (TKR)** – **CPM (continuous passive motion) machine**; goal **90° flexion**; no pillow under knee (flexion contracture).

12.3 Bone & Joint Disorders

Osteoporosis

- **↓ Bone mass/density** → porous fragile bones → **fractures (vertebral compression – most common, hip, wrist)**. "**Silent disease**".
- Risk: **post-menopausal women (↓ oestrogen) – most common**, age, **small thin frame, Caucasian/Asian**, family history, **low Ca/Vit D**, sedentary, **smoking, alcohol, caffeine**, **long-term corticosteroids**, heparin, hyperthyroidism.
- S/S: **loss of height**, **kyphosis (dowager's hump)**, back pain, fractures.
- Diagnosis: **DEXA scan (bone mineral density) – T-score ≤ −2.5 = osteoporosis**; −1 to −2.5 = osteopenia; > −1 normal.
- Rx: **Calcium (1000–1200 mg/day) + Vit D (800–1000 IU)**, **weight-bearing exercise (walking – best)**, stop smoking/alcohol, fall prevention, **Bisphosphonates – Alendronate**, raloxifene, calcitonin, denosumab, teriparatide.

> **! Alendronate (bisphosphonate) – how to take**
>
> **Early morning, empty stomach**, with **full glass (200–240 ml) of plain water**, **remain upright (sitting/standing) for 30 min**, no food/drink/other meds for **30 min** → oesophagitis/ulcer prevention.

- **Osteomalacia** (adults) / **Rickets** (children) – **↓ Vit D** → soft bones, bowing.
- **Paget's disease** – abnormal bone remodelling; enlarged skull (hat size ↑), bowing; ↑ ALP.
- **Osteomyelitis** – bone infection; Staph aureus most common; S/S – pain, fever, swelling, tenderness; **bone biopsy (definitive)**, **long-term IV antibiotics (4–6 weeks or more)**, surgical debridement (sequestrectomy – dead bone = **sequestrum**; new bone = **involucrum**); immobilize; high protein, Vit C.
- **Bone tumours** – **Osteosarcoma** (most common primary malignant, **10–25 yrs, distal femur / around knee**, sunburst/Codman's triangle), Ewing's sarcoma (onion peel), Multiple myeloma (elderly – punched-out lesions, Bence-Jones protein), **Osteochondroma** (most common benign). **Metastasis to bone** – breast, lung, prostate, kidney, thyroid ("**BLT with a Kosher Pickle**").

Arthritis – OA vs RA

Feature	Osteoarthritis (OA) – degenerative	Rheumatoid Arthritis (RA) – autoimmune
Nature	**"Wear & tear"** – cartilage degeneration; most common arthritis	**Systemic autoimmune**, synovium inflamed (pannus)
Age/Sex	> 50; obese; women slightly more	**20–50 yrs**, **women 3 : 1**
Joints	**Weight-bearing – knees, hips**, spine, DIP	**Small joints – hands (MCP, PIP), wrists, feet**; **symmetrical, bilateral**
Stiffness	**< 30 min** morning; **pain ↑ with activity, ↓ with rest**	**> 1 hr morning stiffness**; ↑ after rest

Feature	Osteoarthritis (OA) – degenerative	Rheumatoid Arthritis (RA) – autoimmune
Nodes	**Heberden's (DIP), Bouchard's (PIP)**	**Rheumatoid nodules**, **swan-neck, boutonnière deformity, ulnar deviation**
Systemic	**No**	**Yes** – fatigue, fever, weight loss, anaemia, Sjögren's, pericarditis
Lab	Normal ESR	**RA factor, Anti-CCP (most specific), ↑ ESR/CRP**
Rx	Weight loss, exercise, **paracetamol**, NSAIDs, intra-articular steroids, **joint replacement**	**DMARDs – Methotrexate (first-line**; folic acid, liver/CBC monitoring, teratogenic), hydroxychloroquine (**eye exam**), sulfasalazine, biologics, NSAIDs, steroids



Arthritis – nursing care

- **Heat** for stiffness (warm shower in morning), **cold** for acute inflammation (RA flare)
- Balance rest & exercise; **ROM exercises**; joint protection – **use large joints**, avoid prolonged same position
- Assistive devices – raised toilet seat, grab bars, built-up handles, cane on **unaffected side**
- Weight reduction (OA); energy conservation (RA)

Gout

- **↑ Uric acid (hyperuricaemia)** → **monosodium urate crystals** in joints → acute inflammatory arthritis; men > 40; **big toe (first MTP joint) – "Podagra"** most common; **tophi** (chronic deposits – ear, fingers).
- Triggers: **high-purine foods – organ meats (liver, kidney), red meat, seafood (sardines, anchovies), alcohol (beer)**, fructose drinks, thiazide diuretics, aspirin (low dose), dehydration, fasting.
- Diagnosis: joint aspiration – **needle-shaped negatively birefringent crystals**; serum uric acid.
- Acute attack: **Colchicine** (s/e – diarrhoea, GI), **NSAIDs (indomethacin)**, steroids; rest, elevate, **ice**, bed cradle (avoid sheet pressure).
- Long-term (prevention): **Allopurinol** (↓ uric acid production – **fluids 2–3 L/day**, rash → stop; **not started during acute attack**), Febuxostat, Probenecid (↑ excretion); **low-purine diet, avoid alcohol, fluids 2–3 L**.

Other disorders

- **Ankylosing spondylitis** – young men, **HLA-B27**, spine fusion ("**bamboo spine**"), low back pain & stiffness; posture exercises, firm mattress, **no pillow or thin pillow**.
- **SLE** – see immunology chapter.
- **Carpal tunnel syndrome** – **median nerve** compression; numbness thumb–index–middle fingers; **Phalen's test** (wrist flexion 60 sec) & **Tinel's sign** (tap); splint, surgery.
- **Low back pain / Herniated disc (PIVD)** – **L4–L5 / L5–S1** most common; sciatica (radiating to leg), **straight leg raising test positive**; rest (short), NSAIDs, physiotherapy, **proper body mechanics – lift with legs, back straight, object close to body**; laminectomy/discectomy – **log-roll**, check sensation & movement of legs, CSF leak.
- **Dupuytren's contracture** – palmar fascia (ring/little finger).

12.4 Amputation

- Causes: **PVD/diabetes (most common)**, trauma, tumours, infection (gangrene).
- **Post-op care**:
  - Watch **haemorrhage** – keep **tourniquet at bedside**
  - **Elevate stump on pillow only first 24 hrs** (oedema ↓) – **after 24–48 hrs, do NOT keep on pillow (hip flexion contracture**)
  - **Prone position 20–30 min, 3–4 times/day** (above-knee amputation) → hip flexion contracture prevention
  - **No pillow between thighs** (abduction contracture); don't let stump hang over bed edge
  - **Stump bandaging – figure-of-8 elastic bandage**, **distal to proximal** (cone shape for prosthesis), re-wrap every 4–6 hrs; shrinker sock
  - **Stump toughening** – push stump against soft then harder surface
  - **Phantom limb pain/sensation** – **real**; acknowledge; analgesics, gabapentin, mirror therapy, TENS
  - Body image, rehab, prosthesis fitting
- Exercise – strengthening arm muscles (**triceps**) for crutches; quadriceps setting.

12.5 Mobility Aids

Crutches, Cane & Walker – key points

Crutches:

- Length: **2–3 finger widths (~5 cm / 1.5–2 inch) below axilla**; handgrip so **elbow flexed 20–30°**; crutch tip **15 cm (6 inch) in front & to the side** of foot.
- **Weight on HANDS – not axilla** (brachial plexus injury – "crutch palsy").
- Gaits: **4-point** (most stable; partial weight both legs – R crutch, L foot, L crutch, R foot), **3-point** (non-weight bearing one leg – both crutches + affected leg together, then good leg), **2-point** (faster 4-point), **Swing-to/Swing-through** (paraplegics).
- **Stairs**: **UP – "Up with the good" (good leg first)**; **DOWN – "Down with the bad" (crutches + bad leg first)**. "**Good goes to heaven, bad goes to hell.**"
- Sitting: back up to chair, both crutches in hand on **affected side**, lower down.

Cane: held in **hand OPPOSITE (unaffected/strong side) to affected leg**; top at **greater trochanter** level, elbow 15–30° flexed; move **cane + affected leg together**, then unaffected leg.

Walker: lift & move walker **15–20 cm (6–8 inch)** ahead → step **affected leg** into walker → then unaffected leg; elbows flexed 15–30°; **never use walker on stairs**.

Practice MCQs – Chapter 12

1. Greenstick fracture kisme zyada hota hai?
   (a) Athletes (b) Elderly (c) Women (d) Children
2. Compartment syndrome ka earliest sign –
   (a) Pulselessness (b) Pallor (c) Pain unrelieved by analgesics / on passive stretch (d) Paralysis
3. Compartment syndrome me limb position –
   (a) Elevated above heart (b) Ice packed (c) At heart level (d) Dependent

4. Fat embolism sabse zyada kis fracture ke baad?
(a) Rib (b) Skull (c) Finger (d) Long bone – femur

5. Wet POP cast ko kaise handle karte hain?
(a) Fingertips se (b) Hook se (c) Nahi chhoote (d) Palms se

6. Total hip replacement ke baad kya avoid karein?
(a) High chair (b) Ambulation (c) Hip flexion > 90° & crossing legs (d) Abduction pillow

7. Crutch walking me body weight kahan?
(a) Hands (b) Elbows (c) Axilla (d) Shoulders

8. Crutches se stairs chadhte waqt pehle –
(a) Crutches (b) Bad leg (c) Dono saath (d) Good leg

9. Cane kis haath me pakadte hain?
(a) Unaffected (strong) side (b) Dominant hand always (c) Koi bhi (d) Affected side

10. RA ka most specific test –
(a) RA factor (b) Uric acid (c) ESR (d) Anti-CCP

11. Gout me commonly affected joint –
(a) Shoulder (b) Spine (c) Knee (d) Big toe (1st MTP)

12. Allopurinol lene wale patient ko –
(a) Fluids restrict (b) Alcohol allowed (c) High purine diet (d) Fluids 2–3 L/day

13. Above-knee amputation ke 24 hrs baad hip flexion contracture rokne ke liye –
(a) Pillow between thighs (b) Stump on pillow always (c) Sitting all day (d) Prone position several times a day

14. Alendronate ke baad patient ko kitni der upright rehna chahiye?
(a) Not needed (b) 30 min (c) 5 min (d) 2 hrs

15. Osteoporosis ki diagnosis ka best test –
(a) X-ray (b) Serum calcium (c) DEXA scan (d) CT

16. Bryant's traction kis age group ke liye?
(a) Adults (b) Children < 2 yrs (c) Adolescents (d) Elderly

Answer Key – Chapter 12

1-d · 2-c · 3-c · 4-d · 5-d · 6-c · 7-a · 8-d · 9-a · 10-d · 11-d · 12-d · 13-d · 14-b · 15-c · 16-b

13 Integumentary System & Burns

13.1 Skin – Applied Anatomy

Skin facts

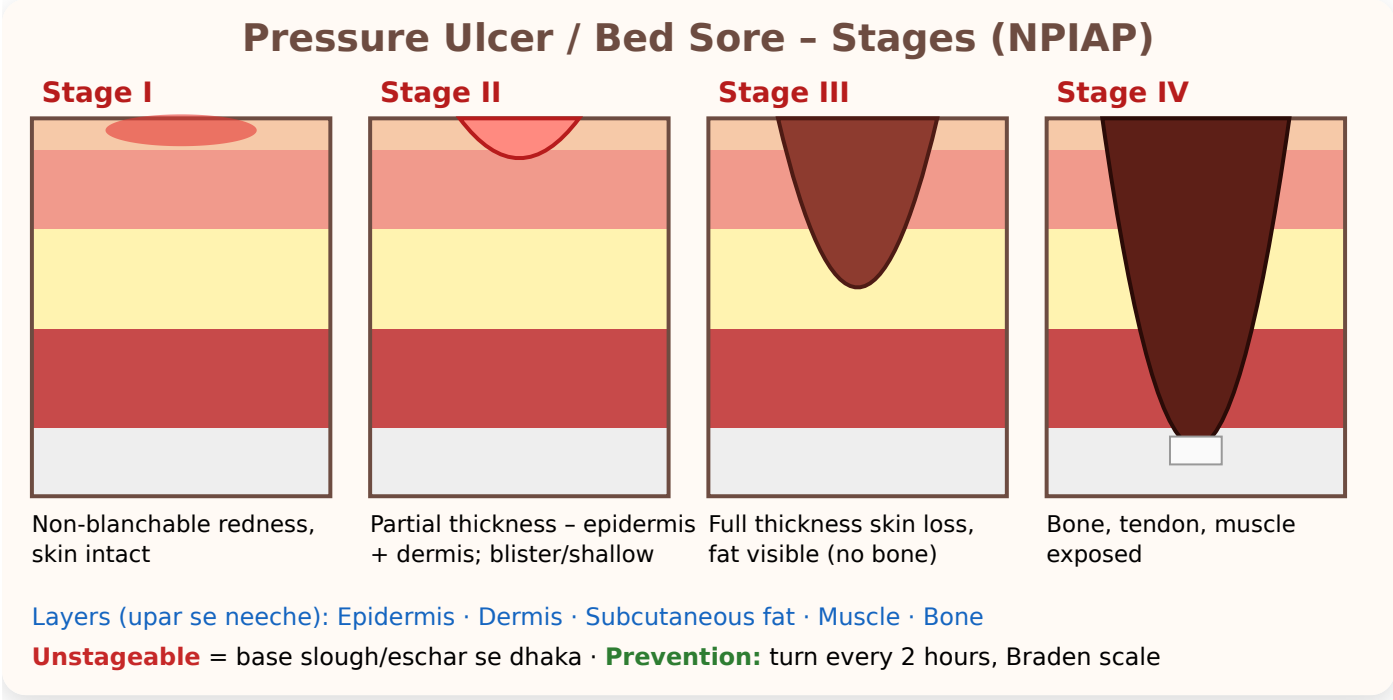
- **Largest organ** of body (~16 % body weight, ~1.5–2 m^2).
- Layers: **Epidermis** (avascular; **stratum basale** – new cells, **melanocytes**; stratum corneum – outer dead keratin) → **Dermis** (blood vessels, nerves, hair follicles, **sweat & sebaceous glands**, collagen) → **Subcutaneous tissue (hypodermis)** – fat.
- Functions: protection, **temperature regulation**, sensation, **Vitamin D synthesis**, excretion, fluid balance.
- Epidermis renews every **~28 days**.

Skin lesions

Primary lesion	Meaning	Example
Macule	Flat, colour change, **< 1 cm**	Freckle, measles rash
Patch	Flat, > 1 cm	Vitiligo
Papule	Raised solid, < 1 cm	Wart, mole
Plaque	Raised flat-topped, > 1 cm	**Psoriasis**
Nodule	Raised solid, 1–2 cm, deeper	Lipoma
Vesicle	Fluid-filled, < 1 cm	**Chickenpox, herpes**
Bulla	Fluid-filled, > 1 cm	**2nd degree burn blister**, pemphigus
Pustule	Pus-filled	Acne, impetigo
Wheal	Transient oedematous raised	**Urticaria**, insect bite, **TB test**
Secondary	**Crust, scale, erosion, ulcer, fissure, scar, keloid, lichenification**	

- **Petechiae** (pinpoint < 3 mm – platelet ↓), **Purpura** (3 mm–1 cm), **Ecchymosis** (bruise > 1 cm), **Spider angioma** (liver), **Cyanosis**, **Jaundice**, **Erythema**.

13.2 Pressure Ulcer (Pressure Injury / Bed Sore / Decubitus Ulcer)



Pressure ulcer ke 4 stages – depth ke hisaab se

- Localized damage over **bony prominences** due to **pressure** (± shear & friction). Most common site: **Sacrum / coccyx** (supine), then **heels**, ischial tuberosity (sitting), greater trochanter (side-lying), occiput, elbows, scapula.
- Risk: **immobility (most important)**, malnutrition (↓ protein, albumin < 3.5), incontinence (moisture), ↓ sensation, elderly, anaemia, ↓ LOC, **shear** (sliding down in bed when HOB > 30°), friction.
- **Assessment tool: Braden scale** – 6 parameters (**Sensory perception, Moisture, Activity, Mobility, Nutrition, Friction & shear**); score **6–23**; **≤ 18 = at risk**, **≤ 9 = very high risk**. (Lower score = higher risk.) Norton scale – 5–20; ≤ 14 at risk.
- **Deep tissue injury** – purple/maroon intact skin; **Unstageable** – base covered by slough/eschar (stage can't be determined till debrided).

Prevention & care

- **Reposition every 2 hrs** (chair – every 1 hr; shift weight every 15 min); turning schedule (clock)
- **30° lateral tilt**; **HOB ≤ 30°** (shear ↓); **lift sheet** to move – don't drag
- **Pressure-relieving mattress** (alternating air, water bed), **heels off bed** (pillow under calves)
- **Don't massage reddened bony prominences**; **no donut/ring cushions** (↓ circulation)
- Keep skin **clean & dry**, barrier cream for incontinence
- Nutrition: **high protein, high calorie, Vit C, zinc**, fluids
- Wound: **moist wound healing**; NS cleansing; debridement of necrotic tissue (**except stable dry heel eschar**); appropriate dressings (hydrocolloid, foam, alginate for exudate)

13.3 Common Skin Conditions

Condition	Cause / Key feature	Treatment / Nursing
Impetigo	Staph/Strep; **honey-coloured crusts** around mouth/nose; children; contagious	Topical mupirocin, wash crusts, separate towels
Cellulitis	Deep skin & subcutaneous infection (Strep/Staph) – red, warm, swollen, tender, spreading	Antibiotics, **elevate**, warm compress, mark borders
Herpes zoster (Shingles)	**Reactivation of varicella-zoster** in dorsal root ganglion; **unilateral, dermatomal painful vesicles** (don't cross midline); elderly/immunocompromised	**Acyclovir** (within 72 hrs), analgesics; **post-herpetic neuralgia**; contagious to non-immune (chickenpox) – contact/airborne if disseminated
Herpes simplex	HSV-1 (cold sores, lips), HSV-2 (genital)	Acyclovir
Scabies	Sarcoptes scabiei mite; **intense itching worse at night**, burrows in **finger webs, wrists, axilla, genitals**	**Permethrin 5 % cream – neck to toes**, leave 8–14 hrs, repeat after 1 week; **treat all family members simultaneously**; wash clothes/linen in hot water; contact precautions
Pediculosis (lice)	Head/body/pubic lice; nits on hair shaft	Permethrin 1 %, fine-tooth comb; wash linen hot
Tinea (Ringworm)	**Fungal (dermatophyte)**; ring-shaped with central clearing; capitis, corporis, cruris (groin), **pedis (athlete's foot)**	Antifungals (clotrimazole, terbinafine, griseofulvin for scalp); keep dry
Candidiasis	Candida albicans; white patches (oral thrush – **wipes off leaving red base**), intertrigo; diabetics, antibiotics, steroids	Nystatin, fluconazole
Psoriasis	**Chronic autoimmune**, **rapid epidermal turnover (3–4 days)**; **silvery-white scaly plaques** on **elbows, knees, scalp**; **Auspitz sign** (pinpoint bleeding on removing scale), **Koebner phenomenon** (new lesions at trauma site); nail pitting; not contagious	Topical steroids, **coal tar**, salicylic acid, vitamin D analogues (calcipotriol), **PUVA / UVB phototherapy** (eye protection), methotrexate, biologics; stress ↓
Atopic dermatitis / Eczema	Allergic; itchy, dry; flexures	Emollients, steroids, avoid triggers, short nails
Contact dermatitis	Allergen/irritant (nickel, latex)	Avoid cause, steroids
Acne vulgaris	Sebaceous glands; adolescents	Benzoyl peroxide, retinoids (**isotretinoin – teratogenic**), antibiotics
Urticaria	Wheals, itching – allergy	Antihistamines
Stevens-Johnson syndrome / TEN	Severe drug reaction (sulfa, anticonvulsants, allopurinol, nevirapine) – skin & mucosa sloughing	Stop drug, **burns-unit-like care**

Condition	Cause / Key feature	Treatment / Nursing
Pemphigus vulgaris	Autoimmune; **flaccid bullae**, **Nikolsky sign +** (skin peels with rubbing)	Steroids, burn-like care
Vitiligo	Autoimmune loss of melanocytes – white patches	Steroids, PUVA, sun protection
Leprosy	**M. leprae**; **hypopigmented anaesthetic patches**, thickened nerves	MDT (see communicable chapter)

Skin cancer

- **Basal cell carcinoma** – most common, **pearly papule with rolled edges**, rarely metastasizes; sun-exposed areas.
- **Squamous cell carcinoma** – scaly, can metastasize.
- **Malignant melanoma** – most dangerous; **ABCDE**: **A**symmetry, **B**order irregular, **C**olour variation, **D**iameter > 6 mm, **E**volving.
- Prevention: avoid sun 10 am–4 pm, **sunscreen SPF ≥ 30**, protective clothing.

13.4 Burns

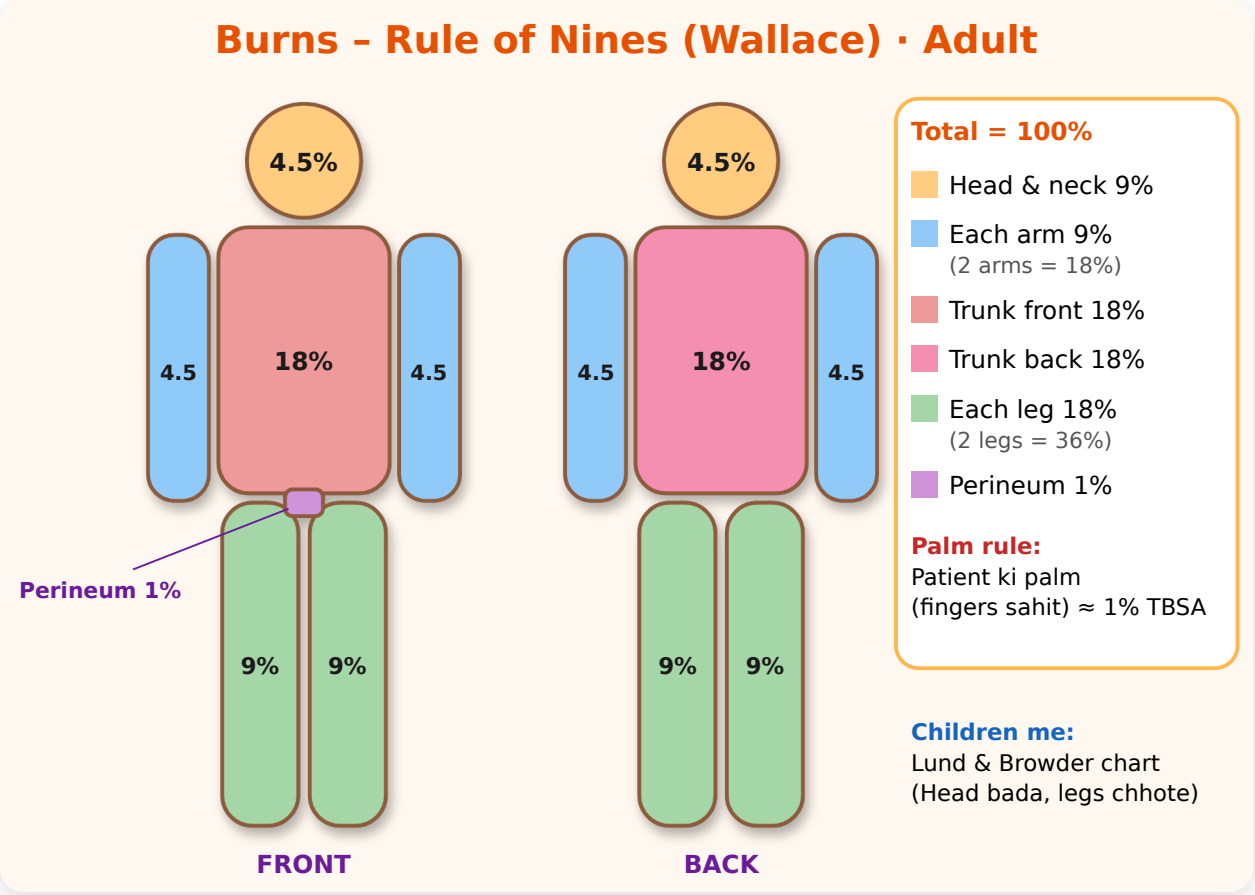
D Burns

Tissue injury caused by **thermal (most common), chemical, electrical, radiation** sources. In India, burns common in women (kitchen, kerosene stoves), children (scalds).

Depth of burns

Depth	Old name	Layers	Appearance	Pain	Healing
Superficial	1st degree	**Epidermis only**	**Red, dry, no blisters**, blanches (sunburn)	**Painful**	3–6 days, no scar
Superficial partial thickness	2nd degree	Epidermis + upper dermis	**Moist, red, blisters (bullae)**, blanches	**Very painful**	10–21 days, minimal scar
Deep partial thickness	2nd degree (deep)	Epidermis + deep dermis	Red/white, mottled, less moist, slow blanch	Painful (pressure)	3–6 weeks, scar; may need graft
Full thickness	**3rd degree**	**Epidermis + dermis + subcutaneous**	**White, leathery, charred, dry, eschar**, no blanching	**Painless (nerve endings destroyed)**	**Skin graft required**, contractures
Deep full thickness	**4th degree**	**Muscle, tendon, bone**	Black, charred	Painless	Amputation/ grafting

Extent – TBSA (Total Body Surface Area)



Rule of Nines (Wallace) – adult TBSA calculation

- **Rule of Nines** (adults): Head & neck 9 %, each arm 9 %, front trunk 18 %, back trunk 18 %, each leg 18 %, perineum 1 %.
- **Lund & Browder chart** – **most accurate** (age-adjusted; children – head larger).
- **Palm method** – patient's palm (with fingers) ≈ **1 %** – scattered small burns.
- **Superficial (1st degree) burns are NOT counted** in TBSA for fluid calculation.
- **Major burns**: > 20–25 % TBSA (adult), > 10 % (children/elderly), any full thickness > 10 %, **burns of face, hands, feet, perineum, major joints**, **inhalation**, electrical, chemical burns → **burn centre**.

Phases of burn management

Phase	Duration	Key problems
1. Emergent / Resuscitative	**Injury → first 24–72 hrs** (till fluid mobilization & diuresis)	**Fluid shift (plasma → interstitial) → hypovolaemic "burn shock"**, oedema, **hyperkalaemia** (cell destruction), hyponatraemia, **↑ Hct (haemoconcentration)**, metabolic acidosis, ↓ urine, airway oedema
2. Acute	48–72 hrs → wound closure	**Fluid remobilization (diuresis)** – **hypokalaemia**, fluid overload risk; **infection (sepsis – most common cause of death after 72 hrs)**, wound care, grafting, nutrition, pain, **Curling's ulcer**
3. Rehabilitation	Wound closure → return to function	**Contractures, hypertrophic scars, keloids**, body image, pressure garments (worn **23 hrs/day for 12–24 months**)

Emergent phase – management

First aid (at scene)

- **STOP the burning** – "**Stop, Drop & Roll**" (clothes on fire); remove from source
- **Cool the burn with cool running water (tap water ~15–25 °C) for 20 min** – **NO ice** (vasoconstriction, hypothermia), **no butter, toothpaste, oil, ghee**
- **Remove jewellery, watches, belts & non-adherent clothing** (before swelling); don't pull stuck clothing
- **Cover with clean dry sheet / cling film**; keep patient **warm** (hypothermia)
- **Don't break blisters**
- **Chemical burns**: brush off dry powder first, then **flush with copious water 20–30 min** (eyes – 15–30 min irrigation)
- **Electrical burns**: **switch off power source first** (don't touch victim); look for **entry & exit wounds**; **ECG monitoring (arrhythmias – VF)**; **myoglobinuria** (rhabdomyolysis → AKI) – urine output 75–100 ml/hr; cervical spine immobilization (fall)

- **Airway – first priority**. **Inhalation injury** suspected if: **facial burns, singed nasal hair/eyebrows, soot in mouth/sputum (carbonaceous)**, **hoarseness, stridor**, burns in enclosed space, cough → **early intubation** (before oedema), **100 % O_2** (CO poisoning – **SpO_2 falsely normal**; check **carboxyhaemoglobin**; cherry-red skin).
- **2 large-bore IV lines** (through unburned skin if possible), **Foley catheter** (hourly urine output), NG tube (paralytic ileus common if > 20 % TBSA), tetanus prophylaxis, IV opioids for pain (**not IM** – poor absorption).

Fluid resuscitation – Parkland (Baxter) Formula

Parkland formula – must know

Total fluid in first 24 hrs = 4 ml × body weight (kg) × % TBSA burned (**Ringer's Lactate**)

- **½ (50 %) in first 8 hrs** – counted **from the time of burn** (not from arrival)
- **¼ (25 %) in next 8 hrs**, **¼ (25 %) in next 8 hrs**
- **Example**: 60 kg, 40 % TBSA → 4 × 60 × 40 = **9600 ml** → first 8 hrs **4800 ml (600 ml/hr)**, next 16 hrs 4800 ml (300 ml/hr).
- **Best guide of adequate resuscitation = urine output**: **adult 0.5 ml/kg/hr (30–50 ml/hr)**; **child 1 ml/kg/hr**; electrical burns 1–1.5 ml/kg/hr (75–100 ml/hr).
- Colloids (albumin) – usually **after 24 hrs** (capillaries leaky in first 24 hrs). **(Modified Brooke formula: 2 ml × kg × % TBSA.)**

Acute phase – wound care

- **Strict asepsis**, **protective (reverse) isolation**, no plants/flowers, sterile linen; **bed cradle**; room **warm (~ 30 °C)** – heat loss.
- **Escharotomy** – circumferential full-thickness burns (limbs/chest) → constriction → compartment syndrome/breathing restriction.
- Topical antimicrobials:

- **Silver sulfadiazine 1 % (SSD)** – most common; s/e **transient leukopenia**; avoid in sulfa allergy, pregnancy near term, neonates; **not on face**.
- **Mafenide acetate** – penetrates eschar (**ear cartilage**), **painful**, **metabolic acidosis** (carbonic anhydrase inhibitor).
- **Silver nitrate 0.5 %** – **stains black**, electrolyte loss (hyponatraemia, hypokalaemia).
- Silver-impregnated dressings, mupirocin, bacitracin (face).

- **Hydrotherapy**/wound cleansing – **premedicate with analgesic 30 min before** dressing.
- **Debridement**: natural, mechanical, **enzymatic (collagenase)**, **surgical (tangential excision)**.
- **Skin grafts**: **Autograft (own skin – permanent; best)**, Allograft/homograft (cadaver – temporary), Xenograft/heterograft (pig – temporary), synthetic (Biobrane, Integra). **Split-thickness** (epidermis + part dermis) – most common; full-thickness (face, hands). **Graft site – immobilize 3–5 days**, elevate, no pressure; **donor site – more painful** (keep dressing undisturbed, moist/dry as per protocol).
- **Nutrition**: **hypermetabolic state** → **high calorie (up to 3000–5000 kcal), high protein (1.5–2 g/kg)**, Vit C, zinc, **early enteral feeding (within 24 hrs)**; **daily weight**.

Rehabilitation – contracture prevention

- **Position of comfort = position of contracture** → avoid. **Neck – extension (no pillow / small roll under shoulders)**, **shoulders – abducted 90°**, elbows extended, hands – functional/ splinted, hips – neutral extended, knees – extended, **ankles – dorsiflexed 90°** (footboard/splint).
- Early **ROM exercises**, splints, ambulation; **pressure garments** (scar ↓), massage with lotion after healing, sun protection 1 year, body image & psychosocial support.

Practice MCQs – Chapter 13

1. Pressure ulcer risk assessment ke liye scale –
   (a) Morse (b) Braden scale (c) GCS (d) APGAR
2. Braden score kitna ho to "at risk"?
   (a) 25 (b) ≥ 20 (c) ≤ 18 (d) 23
3. Bone ya muscle exposed – pressure ulcer stage?
   (a) II (b) III (c) IV (d) I
4. Full-thickness (3rd degree) burn ka feature –
   (a) Red, dry, painful (b) Blanches easily (c) Very painful with blisters (d) Painless, white leathery
5. Rule of nines me ek poori leg (adult) –
   (a) 9 % (b) 18 % (c) 4.5 % (d) 36 %
6. Parkland formula –
   (a) 2 ml × kg × %TBSA (b) 4 ml × kg × %TBSA (c) 1 ml × kg × %TBSA (d) 8 ml × kg × %TBSA
7. Parkland me first half fluid kitne ghante me?
   (a) 4 hrs (b) 8 hrs from time of burn (c) 12 hrs (d) 24 hrs
8. Burn resuscitation adequacy ka best indicator –
   (a) Thirst (b) BP (c) Urine output (d) Pulse
9. Burn first aid me –
   (a) Ghee lagao (b) Cool running water 20 min (c) Blister phodo (d) Ice lagao
10. Scabies treatment me permethrin lagate hain –
    (a) Only face (b) Only patient (c) Only on lesions (d) Neck to toes, all family members

11. Silver nitrate dressing ka side effect –
(a) Black staining & hyponatraemia (b) Hyperkalaemia (c) Metabolic acidosis (d) Leukopenia

12. Inhalation injury ka sign –
(a) Singed nasal hair & soot in sputum (b) Leg burns (c) Back burns (d) Hand blisters

13. Psoriasis ka characteristic lesion –
(a) Vesicles in dermatome (b) Silvery-white scaly plaque (c) Burrows (d) Honey crust

14. Neck burn me contracture prevent karne ke liye –
(a) Big pillow under head (b) No pillow / neck extension (c) Prone with neck flexed (d) Chin on chest

15. Emergent phase me kaunsa electrolyte badhta hai?
(a) Magnesium (b) Calcium (c) Potassium (d) Sodium

Answer Key – Chapter 13

1-b · 2-c · 3-c · 4-d · 5-b · 6-b · 7-b · 8-c · 9-b · 10-d · 11-a · 12-a · 13-b · 14-b · 15-c

14 Blood & Haematological Disorders

14.1 Blood – Applied Physiology

Blood facts

- Blood volume ≈ **5–6 L** (**7–8 % body weight**, ~70 ml/kg); pH **7.35–7.45**.
- **Plasma 55 %** (90 % water; albumin, globulins, fibrinogen) + **Formed elements 45 %** (RBC, WBC, platelets).
- Haematopoiesis: adult – **red bone marrow** (flat bones, vertebrae, sternum, pelvis); fetus – liver, spleen.
- **RBC lifespan 120 days**; **platelets 7–10 days**; neutrophils ~6 hrs in blood. Old RBCs destroyed in **spleen** ("graveyard of RBCs").
- **Erythropoietin** (kidney) → RBC production; needs **iron, B12, folic acid**, protein.
- **Universal donor = O negative**; **Universal recipient = AB positive**. Plasma: AB = universal plasma donor.

Normal values (adult)

Parameter	Normal
Haemoglobin	**Male 13–17 g/dL; Female 12–15 g/dL**
RBC count	M 4.5–5.5; F 4.0–5.0 million/mm³
Haematocrit (PCV)	M 40–50 %; F 36–46 % (≈ 3 × Hb)
WBC (TLC)	**4,000 – 11,000 /mm³**
Differential	**Neutrophils 40–75 %**, **Lymphocytes 20–40 %**, Monocytes 2–8 %, **Eosinophils 1–6 %** (allergy, worms), Basophils 0–1 %
Platelets	**1.5 – 4.5 lakh /mm³ (150,000–450,000)**
MCV	80 – 100 fL (micro < 80, macro > 100)
Reticulocytes	0.5 – 2.5 %
ESR (Westergren)	M 0–15; F 0–20 mm/hr
Bleeding time	2 – 7 min (Ivy)
Clotting time	4 – 10 min
PT	11 – 13.5 sec; **INR 0.8 – 1.2**
aPTT	25 – 40 sec
Serum iron	60 – 170 µg/dL; **Ferritin** 20–250 (M) / 10–120 (F) ng/ml
Fibrinogen	200 – 400 mg/dL

- **Neutropenia** = ANC **< 1500** (severe < 500) → **neutropenic precautions**.
- **Thrombocytopenia** < 1.5 lakh; **spontaneous bleeding risk < 20,000**; < 50,000 → bleeding precautions.

WBC order (decreasing)

"**N**ever **L**et **M**onkeys **E**at **B**ananas" → **N**eutrophils > **L**ymphocytes > **M**onocytes > **E**osinophils > **B**asophils

14.2 Anaemia

Anaemia

Haemoglobin below normal (**WHO: < 13 g/dL men, < 12 g/dL non-pregnant women, < 11 g/dL pregnant women & children 6–59 months**). **Severe anaemia (WHO)**: Hb **< 7 g/dL** in pregnant women & children; **< 8 g/dL** in non-pregnant adults.
General S/S: **fatigue (most common)**, weakness, **pallor** (conjunctiva, palms, nail beds), dyspnoea on exertion, tachycardia, palpitation, dizziness, headache, cold intolerance; severe → HF.

Type	Cause / Mechanism	Key features	Treatment
Iron deficiency (most common worldwide & India)	**Blood loss** (menstruation, GI – hookworm, ulcer, cancer), poor intake, pregnancy, malabsorption	**Microcytic hypochromic**, ↓ ferritin (best test), ↑ TIBC; **koilonychia (spoon nails)**, **pica** (eating mud/ice), angular stomatitis, glossitis, brittle hair	**Oral iron (ferrous sulphate)**, IV iron (iron sucrose, ferric carboxymaltose) / IM (**Z-track**), iron-rich diet
Megaloblastic – Vit B12 deficiency	Strict vegetarian, **lack of intrinsic factor (pernicious anaemia – autoimmune)**, gastrectomy, ileal resection	**Macrocytic**; **smooth beefy red sore tongue (glossitis)**, **neurological signs (paresthesia, ataxia, subacute combined degeneration)**, mild jaundice; **Schilling test**	**Vit B12 IM (cyanocobalamin) – lifelong in pernicious anaemia**
Megaloblastic – Folic acid deficiency	Poor diet, **alcoholism**, pregnancy, drugs (methotrexate, phenytoin)	Macrocytic, **no neurological signs**	**Folic acid** oral; green leafy vegetables
Aplastic anaemia	**Bone marrow failure** – drugs (**chloramphenicol**, chemo), radiation, benzene, viral, idiopathic	**Pancytopenia** (↓ RBC + WBC + platelets) → anaemia + infection + bleeding	Remove cause, **bone marrow transplant**, immunosuppression, transfusion; neutropenic & bleeding precautions
Haemolytic	RBC destruction – sickle cell, thalassaemia, G6PD deficiency, autoimmune, malaria, transfusion reaction	**Jaundice (unconjugated)**, splenomegaly, ↑ reticulocytes, dark urine	Treat cause, folic acid, splenectomy
Anaemia of chronic disease / CKD	Chronic infection, inflammation, CKD (↓ EPO)	Normocytic	Treat cause, **erythropoietin**

! Iron therapy – teaching

- **Oral iron**: take on **empty stomach (1 hr before meals)** for best absorption – if GI upset, take with food; **with Vitamin C / orange juice** (↑ absorption); **avoid with milk, tea/coffee, antacids, calcium, eggs** (↓ absorption); **liquid iron through straw** (teeth staining); **black/dark green stools (normal)**; constipation (fibre, fluids); continue **2–3 months (up to 6)** after Hb normal to replenish stores.
- **IM iron dextran**: **Z-track technique**, deep IM upper outer quadrant gluteal, **separate needle** for drawing & injecting, **don't massage**; test dose (anaphylaxis).
- Iron poisoning antidote: **Deferoxamine (Desferrioxamine)**.
- **Anaemia Mukt Bharat (2018)** – **6×6×6 strategy**; IFA supplementation: pregnant/lactating women **60 mg elemental iron + 500 µg folic acid daily for 180 days** (red tablet).

Sickle Cell Anaemia

- **Autosomal recessive**; abnormal **HbS** (valine replaces glutamic acid at position 6 of β chain); in India – tribal populations (Gujarat, MP, Maharashtra, Chhattisgarh, Odisha). **National Sickle Cell Anaemia Elimination Mission – launched 1 July 2023** (Shahdol, MP) – target **2047**.
- RBCs **sickle under hypoxia, dehydration, cold, infection, high altitude, stress, acidosis** → **vaso-occlusive crisis** (severe pain – bones, joints, abdomen, chest), **acute chest syndrome**, stroke, **splenic sequestration**, aplastic crisis, priapism, leg ulcers, **auto-splenectomy** (infections – pneumococcal), **hand-foot syndrome (dactylitis)** – infants.
- Diagnosis: **sickling test**, **Hb electrophoresis** (confirmatory), HPLC.
- **Crisis management**: **H**ydration (IV fluids), **O**xygen, **P**ain relief (opioids – morphine; no meperidine), rest, warmth (**no cold compress** – vasoconstriction), antibiotics, transfusion; **Hydroxyurea** (↑ HbF – long-term), folic acid, vaccines (pneumococcal, Hib, meningococcal), penicillin prophylaxis in children.
- Avoid: dehydration, high altitude, unpressurized flights, strenuous exercise, cold, smoking, tight clothing.

Thalassaemia

- **Autosomal recessive** – ↓ synthesis of globin chains (**β-thalassaemia major – Cooley's anaemia**, most common severe in India – Sindhi, Punjabi, Gujarati, Bengali communities).
- Starts at **6 months** age (HbF → HbA switch); **severe microcytic anaemia**, **hepatosplenomegaly**, **chipmunk face (frontal bossing, maxillary prominence)**, growth retardation, "hair-on-end" skull X-ray; **↑ HbF** on electrophoresis.
- Rx: **regular blood transfusion (every 2–4 weeks, keep Hb > 9–10)** → **iron overload (haemosiderosis)** → **iron chelation – Deferoxamine (SC/IV), Deferasirox, Deferiprone (oral)**; folic acid; splenectomy; **bone marrow (stem cell) transplant – only cure**; genetic counselling & **premarital/antenatal screening (HbA2 > 3.5 % = trait)**.
- **8 May = World Thalassaemia Day**.

Polycythaemia vera

- Myeloproliferative – **↑ RBC, WBC, platelets**; ↑ Hct, hyperviscosity → thrombosis; **ruddy (plethoric) face**, **pruritus after hot bath**, headache, splenomegaly. Rx: **phlebotomy** (Hct < 45 %), hydroxyurea, aspirin; fluids; avoid iron. Secondary polycythaemia – COPD, high altitude, smoking.

14.3 White Blood Cell Disorders

Leukaemia

- Malignant proliferation of WBCs in bone marrow → **↓ normal cells** → **anaemia (fatigue), infection (fever – most common cause of death), bleeding (thrombocytopenia)** + bone pain, hepatosplenomegaly, lymphadenopathy, weight loss.
- **ALL** – most common **childhood** cancer (2–5 yrs), best prognosis (CNS involvement – intrathecal chemo).
- **AML** – adults; **Auer rods**.
- **CLL** – most common in **elderly** (> 60) in western; slow; often asymptomatic.
- **CML** – adults 40–60; **Philadelphia chromosome t(9;22)** – BCR-ABL → **Imatinib** (targeted therapy).
- Diagnosis: **bone marrow aspiration & biopsy** (definitive) – site: **posterior superior iliac crest** (adults; sternum alternative), **tibia in infants**; after: **pressure 5–10 min**, lie on site/supine 30–60 min, watch bleeding.
- Rx: chemotherapy (induction → consolidation → maintenance), **bone marrow / stem cell transplant**, targeted therapy, supportive.

Neutropenic (protective/reverse) precautions

- **Private room** (positive pressure/HEPA if available), **strict hand hygiene** (most important)
- **No fresh flowers/plants**, no standing water; **no fresh fruits/vegetables (raw), undercooked meat, unpasteurized dairy** – **low-microbial diet**
- No visitors with infection, no children recently vaccinated with live vaccines
- Avoid **rectal temperature, suppositories, enemas, IM injections** (as possible)
- Monitor **temperature (fever ≥ 38 °C = emergency – febrile neutropenia)** – signs of infection may be absent (no pus)
- Mask for patient outside room; dedicated equipment; daily bath; oral care with soft brush
- **G-CSF (filgrastim)** – ↑ neutrophil production (bone pain s/e)

Bleeding (thrombocytopenia) precautions

- **Soft toothbrush/ toothette**, **electric razor**; no dental floss
- **Avoid IM/SC injections**, rectal temperature, enemas, suppositories; apply **pressure 5–10 min** after venipuncture
- **No aspirin / NSAIDs**, alcohol; stool softeners (no straining)
- Avoid **blowing nose forcefully**, contact sports; padded side rails
- Check urine, stool, emesis for blood; petechiae, gum bleeding; neuro checks (intracranial bleed)
- **Platelet transfusion** if < 10,000–20,000 or bleeding

Lymphoma

	Hodgkin's lymphoma	Non-Hodgkin's lymphoma (NHL)
Hallmark	**Reed-Sternberg cells**	No RS cells; B-cell mostly
Age	**Bimodal – 15–35 & > 55 yrs**	Older; more common overall

	Hodgkin's lymphoma	Non-Hodgkin's lymphoma (NHL)
Spread	**Contiguous (orderly) – cervical lymph node first** (painless, rubbery)	Non-contiguous, extranodal
Features	**"B symptoms"** – fever, **night sweats**, **weight loss > 10 %**; **pruritus**; **pain in nodes after alcohol**	Painless lymphadenopathy
Prognosis	**Good (curable – ABVD chemo)**	Variable

- Staging: **Ann Arbor (I–IV)**; EBV associated.
- **Multiple myeloma** – plasma cell malignancy, elderly; **CRAB – hyperCalcaemia, Renal failure, Anaemia, Bone lesions (punched-out)**; **Bence-Jones protein** in urine; M-spike; fluids 3–4 L/day, fracture precautions.

14.4 Bleeding Disorders

Disorder	Defect	Features	Treatment
Haemophilia A (80 %)	**Factor VIII deficiency**; **X-linked recessive** – males affected, females carriers	**Haemarthrosis (bleeding into joints – knees) – most common**, prolonged bleeding after injury/circumcision/dental, deep muscle haematoma, **↑ aPTT**, normal PT & platelets & BT	**Factor VIII concentrate**, **desmopressin (DDAVP)** – mild; **RICE for joint bleed**, no IM, no aspirin; soft toothbrush; non-contact sports (swimming)
Haemophilia B (Christmas disease)	**Factor IX deficiency**, X-linked	Same	Factor IX concentrate
Von Willebrand disease	vWF deficiency (most common inherited bleeding disorder; autosomal dominant)	Mucosal bleeding, menorrhagia, ↑ BT	Desmopressin, vWF concentrate
ITP (Immune thrombocytopenic purpura)	**Autoimmune destruction of platelets** (children – after viral infection, acute; adults – chronic, women)	Petechiae, purpura, epistaxis, gum bleeding; ↓ platelets, normal PT/aPTT	**Corticosteroids**, IVIG, **splenectomy**, platelet transfusion (life-threatening bleed only)
DIC (Disseminated Intravascular Coagulation)	**Secondary** – sepsis (most common), trauma, obstetric (abruptio placentae, amniotic embolism, IUD), cancer, burns; **simultaneous widespread clotting & bleeding** (factors consumed)	**Bleeding from all sites – IV sites, wounds, mucosa, oozing**, petechiae, organ ischaemia; **↓ platelets, ↓ fibrinogen, ↑ PT/aPTT, ↑ D-dimer/FDP** (most specific)	**Treat underlying cause (most important)**, FFP, cryoprecipitate (fibrinogen), platelets, heparin (selected)
Vitamin K deficiency	↓ factors II, VII, IX, X	Newborn (haemorrhagic disease of newborn), malabsorption, warfarin	**Vitamin K** (newborn – 1 mg IM at birth)

14.5 Blood Transfusion

Blood components

Component	Storage	Shelf-life	Use / Points
Whole blood	**2–6 °C**	**35 days (CPDA-1)**; 21 days (CPD/ACD)	Acute massive blood loss
Packed RBCs (PRBC)	2–6 °C	35–42 days (SAGM 42)	**Anaemia**; 1 unit ↑ **Hb by ~1 g/dL** (Hct by 3 %)
Platelet concentrate	**20–24 °C (room temp) with continuous gentle agitation**	**5 days**	Thrombocytopenia; 1 unit ↑ ~5,000–10,000; **don't refrigerate**; infuse fast (~30 min)
Fresh frozen plasma (FFP)	**≤ −18 °C / −30 °C**	**1 year**	Clotting factors – DIC, warfarin reversal, liver disease; **thaw at 30–37 °C**, use within 6 hrs (24 hr if refrigerated); ABO compatible
Cryoprecipitate	≤ −18 °C	1 year	**Fibrinogen, Factor VIII, vWF, XIII** – DIC, haemophilia A, vWD
Albumin	Room temp	3–5 years	Hypovolaemia, hypoalbuminaemia, burns

Transfusion – nursing procedure

- **Informed consent**; verify **doctor's order**; check **blood group, cross-match, Rh**, unit number, expiry, patient ID – **2 nurses at bedside** (double check)
- **Baseline vitals** before transfusion; start within **30 min of receiving from blood bank**; **complete one unit within 4 hrs** (max)
- **Needle/cannula 18–20 G** (adult; 22–24 child)
- **Blood administration set with filter (170–260 micron)**, prime with **0.9 % NS only**
- **No medications** in the blood line; **don't warm** in microwave/hot water (blood warmer only for massive/rapid transfusion)
- **Start slowly – 2 ml/min (~15 drops/min) for first 15 min, stay with patient** (most reactions occur in first 15 min / 50 ml) → then increase rate
- **Vitals: before, after 15 min, then every 30 min–1 hr, and after completion**
- Don't refrigerate blood in ward fridge; return unused within 30 min

Transfusion reactions

Reaction	Cause / Timing	Signs	Action
Acute haemolytic (most dangerous)	**ABO incompatibility** – **clerical error (most common)**; within **minutes (first 15 min)**	**Fever, chills, flank / low back pain**, chest pain, dyspnoea, hypotension, tachycardia, **haemoglobinuria (dark/red urine)**, burning along vein, DIC, AKI, shock	**STOP transfusion immediately**
Febrile non-haemolytic (most common)	Antibodies against donor WBC	Fever, chills	Stop, antipyretic; **leukocyte-reduced** blood next time

Reaction	Cause / Timing	Signs	Action
Allergic (mild)	Plasma proteins	**Urticaria, itching**, flushing	Stop/slow, **antihistamine**, may restart if mild (as ordered)
Anaphylactic	IgA deficient recipient	Bronchospasm, hypotension, shock	Stop, **adrenaline**, airway
Circulatory overload (TACO)	Too fast/too much – elderly, HF	Dyspnoea, **crackles**, cough, ↑ BP, **JVD**	Slow/stop, **upright position**, diuretic, O_2
TRALI	Antibodies in donor plasma; within 6 hrs	Acute respiratory distress, hypoxaemia, **non-cardiogenic pulmonary oedema**, hypotension	Stop, respiratory support
Bacterial (septic)	Contaminated unit (platelets – room temp)	High fever, rigors, hypotension, shock	Stop, cultures, antibiotics
Massive transfusion complications		**Hypocalcaemia (citrate)**, hyperkalaemia, hypothermia, dilutional coagulopathy	Calcium gluconate, blood warmer
Delayed haemolytic	2–14 days	Fever, jaundice, ↓ Hb	

! Transfusion reaction – nursing steps

1. **STOP the transfusion immediately**
2. **Keep IV line open with 0.9 % NS** using **new tubing** (don't flush remaining blood into patient)
3. **Check vitals**, stay with patient, notify doctor & blood bank
4. **Recheck identification** (patient & blood bag)
5. **Send blood bag + tubing + fresh blood sample (EDTA & clotted) + first voided urine** to blood bank/lab
6. Treat symptoms (antihistamine, antipyretic, O_2, adrenaline, diuretic), monitor urine output
7. **Document** & incident report

- **Rh factor**: Rh –ve mother with Rh +ve fetus → **Anti-D immunoglobulin (RhoGAM) 300 µg IM within 72 hrs** of delivery/abortion & at **28 weeks** (antenatal).
- Blood donation: age **18–65 yrs**, weight **≥ 45 kg (350 ml) / ≥ 55 kg (450 ml)**, Hb **≥ 12.5 g/dL**, interval **3 months (men) / 4 months (women)**. **14 June = World Blood Donor Day** (Karl Landsteiner's birthday – discovered ABO groups, 1901). **1 October = National Voluntary Blood Donation Day**.

? Practice MCQs – Chapter 14

1. India me most common anaemia –
   (a) Aplastic (b) B12 deficiency (c) Sickle cell (d) Iron deficiency
2. Koilonychia kis anaemia ka sign hai?
   (a) Aplastic (b) Pernicious (c) Haemolytic (d) Iron deficiency
3. Vit B12 deficiency me unique feature –
   (a) Neurological signs / paresthesia (b) Pica (c) Spoon nails (d) Jaundice only

4. Oral iron ke saath kya dena chahiye absorption badhane ke liye?
(a) Antacid (b) Tea (c) Milk (d) Orange juice / Vitamin C

5. IM iron injection ki technique –
(a) Air-lock only (b) Intradermal (c) Z-track (d) SC

6. Sickle cell crisis me first management –
(a) Exercise (b) Hydration, O_2, pain relief (c) Cold compress (d) Iron

7. Haemophilia A me kis factor ki kami?
(a) VIII (b) X (c) VII (d) IX

8. Haemophilia me most common bleeding site –
(a) Brain (b) Kidney (c) Skin (d) Joints (haemarthrosis)

9. Platelets kis temperature par store hote hain?
(a) 20–24 °C with agitation (b) 2–6 °C (c) −30 °C (d) 37 °C

10. Ek unit blood maximum kitne ghante me complete karna chahiye?
(a) 1 hr (b) 4 hrs (c) 8 hrs (d) 12 hrs

11. Transfusion reaction ka first nursing action –
(a) Check temperature (b) Stop transfusion (c) Inform doctor (d) Give antihistamine

12. Acute haemolytic reaction ka common cause –
(a) ABO incompatibility / clerical error (b) Bacterial (c) Allergy (d) Fast rate

13. Blood transfusion ke saath compatible fluid –
(a) D10 (b) D5W (c) RL (d) 0.9 % NS

14. DIC ka most specific lab finding –
(a) ↑ D-dimer/FDP (b) ↑ Platelets (c) ↑ Hb (d) Normal PT

15. Thalassaemia me iron overload ka treatment –
(a) Deferoxamine (b) Iron tablets (c) Heparin (d) Vit K

16. Reed-Sternberg cells kis disease me?
(a) NHL (b) Myeloma (c) Hodgkin's lymphoma (d) ALL

17. Neutropenic patient ke liye kya allowed nahi hai?
(a) Fresh flowers & raw salad (b) Private room (c) Hand hygiene (d) Cooked food

18. Universal donor blood group –
(a) O –ve (b) AB +ve (c) A +ve (d) B –ve

Answer Key – Chapter 14

1-d · 2-d · 3-a · 4-d · 5-c · 6-b · 7-a · 8-d · 9-a · 10-b · 11-b · 12-a · 13-d · 14-a · 15-a · 16-c · 17-a · 18-a

15 Immunology, HIV/AIDS & Communicable Diseases

15.1 Immunity – Basics

Type	Meaning	Example
Innate (natural)	Born with; non-specific	Skin, mucosa, phagocytes, fever
Active natural	Disease hone ke baad	Chickenpox ke baad immunity
Active artificial	**Vaccine**	BCG, OPV, MR – long-lasting
Passive natural	Mother se	**Placenta (IgG)**, breast milk (**IgA**)
Passive artificial	Ready-made antibodies	**Immunoglobulins – ATS/TIG, anti-rabies Ig, HBIG, anti-D, antivenom** – immediate but short

- **Humoral immunity** – **B-lymphocytes** → plasma cells → antibodies. **Cell-mediated** – **T-lymphocytes** (CD4 helper, CD8 cytotoxic).
- Immunoglobulins: **IgG** (most abundant, **crosses placenta**, secondary response), **IgM** (largest, **first** antibody in primary response – acute infection), **IgA** (secretions, **colostrum**), **IgE** (**allergy, parasites**), IgD.

Hypersensitivity reactions (Gell & Coombs)

Type	Mechanism	Example
I – Immediate / Anaphylactic	**IgE**, mast cells, histamine	**Anaphylaxis, asthma, hay fever, urticaria, penicillin allergy, bee sting**
II – Cytotoxic	IgG/IgM against cells	**ABO transfusion reaction, Rh incompatibility**, Goodpasture, myasthenia gravis
III – Immune complex	Ag-Ab complexes deposit	**SLE, post-streptococcal GN, serum sickness**, RA, Arthus reaction
IV – Delayed (cell-mediated)	T cells; 24–72 hrs	**Mantoux (tuberculin) test**, contact dermatitis (poison ivy, nickel), **transplant rejection**

♫ **ACID**

Type **I – A**naphylactic · Type **II – C**ytotoxic · Type **III – I**mmune complex · Type **IV – D**elayed

Anaphylaxis

- Severe life-threatening type I reaction – **within minutes**; triggers: drugs (**penicillin – most common drug**, NSAIDs, contrast), foods (peanuts, shellfish), insect stings, **latex**.

- S/S: urticaria, flushing, **angioedema**, **stridor/wheeze**, dyspnoea, **hypotension**, tachycardia, nausea, feeling of impending doom.
- Rx: **Stop the trigger** → **Adrenaline 1:1000 (1 mg/ml) 0.5 mg IM (0.01 mg/kg; child max 0.3 mg) – anterolateral thigh**, repeat every 5–15 min → airway & **high-flow O_2** → supine with legs raised (breathing difficulty → sit up) → IV fluids → **antihistamines (chlorpheniramine), hydrocortisone**, nebulized salbutamol.
- Prevention: **test dose before penicillin/ATS/antivenom** (as per policy), allergy band, **EpiPen** for known allergies, **wait 15–30 min after injections/vaccines**.
- **Latex allergy** – banana, avocado, kiwi, chestnut cross-sensitivity.

SLE (Systemic Lupus Erythematosus)

- Chronic **autoimmune** multisystem disease; **young women (15–45), 9 : 1**.
- S/S: **butterfly (malar) rash** across cheeks & nose (sun-sensitive – **photosensitivity**), **joint pain (arthralgia)**, fatigue, fever, **lupus nephritis** (most serious), pleuritis, pericarditis, oral ulcers, alopecia, anaemia, Raynaud's.
- Tests: **ANA (screening, most sensitive)**, **anti-dsDNA & anti-Smith (most specific)**, ↓ complement.
- Rx: **hydroxychloroquine** (eye exams), NSAIDs, **corticosteroids**, immunosuppressants; **avoid sun (sunscreen SPF 30+)**, rest, infection prevention, report fever.
- **Drug-induced lupus** – hydralazine, procainamide, isoniazid.

15.2 HIV / AIDS

D HIV/AIDS

HIV = Human Immunodeficiency Virus – **retrovirus** (RNA, **reverse transcriptase** enzyme), attacks **CD4+ T-helper cells**. **AIDS** = advanced stage – **CD4 < 200 cells/mm^3** OR presence of **AIDS-defining opportunistic infection/cancer**. Normal CD4 = **500–1500 cells/mm^3**.
First case in India – **1986, Chennai** (by Dr. Suniti Solomon). **1 December = World AIDS Day**.

- **Transmission**: **sexual (most common in India ~85 %)**, blood & blood products, sharing needles, **mother to child (vertical – pregnancy, delivery, breastfeeding)**, needle-stick (risk ~0.3 %).
- **NOT transmitted** by: hugging, shaking hands, sharing utensils/toilets, mosquitoes, saliva/tears/sweat (casual contact).
- Body fluids with high virus: **blood, semen, vaginal fluid, breast milk**, CSF, amniotic fluid.
- **Window period** – infection to detectable antibodies (**~3–12 weeks**, up to 3 months) – test negative but can transmit.

Stages (WHO)

1. **Acute HIV infection (seroconversion)** – 2–4 weeks; flu-like / mononucleosis-like (fever, sore throat, rash, lymphadenopathy); **very high viral load**.
2. **Clinical latency (asymptomatic)** – years (avg 8–10); virus replicating.
3. **Symptomatic** – fever, night sweats, weight loss, chronic diarrhoea, oral thrush, herpes zoster.
4. **AIDS** – opportunistic infections.

Opportunistic infections

Infection	Note
Tuberculosis	**Most common OI in India & most common cause of death** in PLHIV
Pneumocystis jirovecii pneumonia (PCP)	Most common OI (west); dry cough, dyspnoea; **Cotrimoxazole** prophylaxis (CD4 < 200/350)
Oral/oesophageal candidiasis	White patches; often first sign
Kaposi's sarcoma	**Purple-red skin lesions** – HHV-8; most common AIDS cancer
Toxoplasmosis (brain), **Cryptococcal meningitis**, **CMV retinitis** (blindness), **MAC**, cryptosporidiosis (diarrhoea), herpes zoster, **HIV wasting syndrome**, **AIDS dementia complex**	

Diagnosis

- **ELISA / Rapid tests** – screening; India uses **3 different tests (antigen/principle) – Strategy III** for adults. **Western blot** – confirmatory (older).
- **Infants < 18 months**: **HIV DNA PCR** (maternal antibodies interfere with ELISA) – at 6 weeks.
- **CD4 count** – immune status; **Viral load (PCR)** – treatment response (goal: **undetectable < 50 copies/ml** → "**U = U**: Undetectable = Untransmittable").
- **Pre-test & post-test counselling**, **informed consent & confidentiality** mandatory (HIV/AIDS Act 2017).

ART (Antiretroviral Therapy)

- **"Test and Treat" – start ART in all HIV-positive regardless of CD4** (India since 2017).
- **Preferred first-line (NACO): TLD = Tenofovir (TDF) 300 mg + Lamivudine (3TC) 300 mg + Dolutegravir (DTG) 50 mg** – 1 tablet daily.
- Drug classes: NRTIs (zidovudine – **anaemia**; tenofovir – **renal toxicity**), NNRTIs (efavirenz – **vivid dreams, CNS**; nevirapine – **rash/SJS, hepatotoxicity**), PIs (lipodystrophy), Integrase inhibitors (dolutegravir).
- **Adherence > 95 %** essential (resistance); lifelong; same time daily.
- **PEP (Post-Exposure Prophylaxis)** – start **as early as possible, within 2 hrs ideal, max 72 hrs**; **28 days** of ART (TLD); follow-up testing at 6 weeks, 3 & 6 months.
- **PPTCT/PMTCT** – ART to all pregnant positive women (lifelong); infant **Nevirapine syrup for 6 weeks** (12 weeks if high-risk – + AZT); **exclusive breastfeeding** recommended with maternal ART (no mixed feeding); safe institutional delivery.
- **NACP (National AIDS Control Programme)** – **phase V (2021–2026)**; **NACO** est. 1992. **95-95-95 target** (by 2025/2030): 95 % know status, 95 % on ART, 95 % virally suppressed.
- Red ribbon = AIDS awareness symbol. Helpline **1097**.

Nursing care of PLHIV

- **Standard precautions** for all patients (not special isolation); protective isolation if severe neutropenia
- Oral care (thrush – nystatin), nutrition (**high calorie, high protein**, small frequent meals; food safety)
- Skin care, diarrhoea management (fluids, electrolytes), fever management
- Psychological support, **confidentiality**, **non-judgmental attitude**, address stigma
- Teach **safe sex (condoms)**, no sharing needles/razors/toothbrushes, adherence, recognise OI signs

15.3 Important Communicable Diseases

Disease	Agent / Vector / Route	Key features	Treatment / Nursing / Prevention
Dengue	**Dengue virus (4 serotypes, Flavivirus)**; **Aedes aegypti** (**day biter**, breeds in **clean stagnant water** – coolers, tyres, pots); "**Tiger mosquito**"	**High fever, severe headache, retro-orbital pain, myalgia/arthralgia ("breakbone fever")**, rash; **warning signs** – abdominal pain, persistent vomiting, bleeding, lethargy; **Dengue haemorrhagic fever / shock syndrome** – **↓ platelets, ↑ Hct (> 20 % = plasma leak)**; **Tourniquet test** (≥ 10 petechiae/sq inch)	**NS1 antigen (day 1–5)**, IgM ELISA (after day 5); **paracetamol only – NO aspirin/NSAIDs/steroids**; **oral/IV fluids**, monitor platelets & Hct; mosquito nets; **source reduction**; **16 May = National Dengue Day**
Chikungunya	Alphavirus; **Aedes**	Fever + **severe, prolonged joint pain** ("bent over")	Symptomatic
Malaria	**Plasmodium** – **P. vivax** (most common India), **P. falciparum** (most dangerous – **cerebral malaria**), P. malariae, P. ovale, P. knowlesi; **female Anopheles** (**night biter**, dusk–dawn; clean water)	**Paroxysm: Cold stage (chills, rigors) → Hot stage (high fever) → Sweating stage**; fever every **48 hrs (tertian – vivax, falciparum)**, 72 hrs (quartan – malariae); anaemia, splenomegaly; falciparum – cerebral malaria, blackwater fever, ARDS, hypoglycaemia	**Peripheral blood smear (thick & thin – gold standard)**, **RDT**; **Vivax: Chloroquine 3 days + Primaquine 14 days (0.25 mg/kg)** (check **G6PD** – haemolysis; avoid in pregnancy & infants < 6 months); **Falciparum: ACT – Artesunate + Sulfadoxine-Pyrimethamine (AS+SP)** (NE states: AL – Artemether-Lumefantrine) + Primaquine single dose; **Severe: IV/IM Artesunate**; **Chloroquine – retinopathy**; **Quinine – cinchonism, hypoglycaemia**; LLIN/ITN bed nets, IRS; **25 April = World Malaria Day**; NVBDCP → **NCVBDC**

Disease	Agent / Vector / Route	Key features	Treatment / Nursing / Prevention
Typhoid (Enteric fever)	**Salmonella Typhi**; **faeco-oral** (contaminated food/water); carriers (gallbladder – "Typhoid Mary")	**Step-ladder fever**, **relative bradycardia** (Faget's sign), **rose spots** (trunk), **coated tongue**, abdominal pain, constipation/diarrhoea, splenomegaly; **complication – intestinal perforation & haemorrhage (3rd week)**	**Blood culture (1st week – best)**, **Widal (after 1st week)**, stool culture (3rd week), **bone marrow culture (most sensitive)**; **Ceftriaxone, azithromycin**; fluids, soft high-calorie diet, **enteric precautions, hand hygiene**; **Typhoid conjugate vaccine**
Cholera	**Vibrio cholerae**; faeco-oral (water)	**Painless profuse "rice-water stools"**, vomiting, **severe dehydration**, muscle cramps, shock	**ORS (WHO low-osmolarity: 245 mOsm/L)**, IV RL, **doxycycline** (single dose) / azithromycin; zinc (children)
Tetanus	**Clostridium tetani** (spores in soil, dust, animal faeces; anaerobic); wounds, umbilical stump (**neonatal tetanus**), unclean delivery/abortion	**Trismus (lockjaw) – first sign**, **risus sardonicus** (sardonic smile), **opisthotonus** (arched back), muscle rigidity & **painful spasms triggered by light, noise, touch**, dysphagia; consciousness intact	**Quiet dark room**, minimal handling, **TIG (Human tetanus immunoglobulin) 3000–6000 IU IM** / ATS (test dose), **metronidazole**, **diazepam** (spasms), wound debridement, airway (tracheostomy), **Td vaccine**; NG feeding; **India eliminated maternal & neonatal tetanus (2015)**
Rabies (Hydrophobia)	**Rabies virus (Rhabdovirus – bullet-shaped)**; bite/scratch/lick on broken skin of **rabid dog (~95 %)**, cat, monkey, bat; incubation 1–3 months (shorter for face/neck bites)	Prodrome – **pain/paresthesia at bite site**, fever → **hydrophobia (fear of water – laryngeal spasm)**, aerophobia, hyperexcitability, hypersalivation, delirium → paralysis, coma, death (**~100 % fatal once symptoms appear**)	**Wound: wash immediately with soap & running water for 15 min** + povidone-iodine/alcohol; **don't suture** (if needed – after RIG, loose); **Category III (bites/scratches with bleeding, licks on broken skin/mucosa) → RIG (20 IU/kg human RIG, infiltrated into wound) + ARV**; **ARV IM (Essen): days 0, 3, 7, 14, 28** (deltoid; **never gluteal**) / **ID (Updated Thai Red Cross): days 0, 3, 7, 28** (2 sites); observe dog 10 days; patient care – **quiet dark room**, avoid water sound, contact precautions, staff PrEP; **28 September = World Rabies Day** (Louis Pasteur)

Disease	Agent / Vector / Route	Key features	Treatment / Nursing / Prevention
Leprosy (Hansen's disease)	**Mycobacterium leprae** (acid-fast; **slowest growing**; affects **skin & peripheral nerves**); **droplet/nasal secretions**, prolonged close contact; incubation 3–5 years	**Hypopigmented/ reddish skin patches with definite loss of sensation**, **thickened tender peripheral nerves** (ulnar, common peroneal), deformities (claw hand, foot drop, lagophthalmos); **Paucibacillary (≤ 5 patches)** vs **Multibacillary (≥ 6 patches/ nerve involvement / smear +ve)**	**MDT (WHO, 2018 uniform): Rifampicin (monthly supervised) + Clofazimine + Dapsone – PB: 6 months, MB: 12 months** (all 3 drugs both); **Clofazimine → reddish-brown skin discoloration**; self-care of anaesthetic hands/feet (soak, oil, protective footwear, check injuries); **30 January = Anti-Leprosy Day (Gandhiji)**; NLEP; **Sparsh** awareness campaign
Measles	Paramyxovirus; **airborne**	**3 C's – Cough, Coryza, Conjunctivitis** + fever → **Koplik's spots** (buccal mucosa – pathognomonic, before rash) → maculopapular rash (behind ears → face → body)	**Vitamin A** (2 doses), supportive; **MR vaccine** (9–12 & 16–24 months)
Chickenpox (Varicella)	VZV; airborne & contact	**Pleomorphic rash (macule→papule→vesicle→crust all at once)**, centripetal (trunk), itching; infectious till all lesions crusted	**Acyclovir**, calamine, **no aspirin (Reye's syndrome)**, trim nails
Mumps	Paramyxovirus; droplet	**Parotitis (painful swelling)**, complications – **orchitis**, meningitis, pancreatitis	Symptomatic, soft diet, avoid sour foods
Diphtheria	**Corynebacterium diphtheriae**; droplet	**Greyish-white pseudomembrane** in throat (bleeds on removal), **bull neck**, myocarditis	**ADS (antitoxin)**, penicillin/erythromycin; **DPT/Pentavalent**
COVID-19	**SARS-CoV-2**; droplet, aerosol, contact	Fever, cough, loss of taste/smell, dyspnoea, ARDS; risk – elderly, comorbid	**RT-PCR (gold standard)**, RAT; O_2, **awake proning**, steroids (hypoxic), anticoagulation; masks, distancing, vaccines (Covishield, Covaxin)
Leptospirosis	**Leptospira**; **rat urine** contaminated water (floods, paddy fields)	Fever, **calf muscle pain**, conjunctival suffusion; **Weil's disease** (jaundice + renal failure + bleeding)	**Doxycycline / penicillin**
Japanese encephalitis	Flavivirus; **Culex** (paddy fields); **pigs = amplifier host**	Encephalitis in children	JE vaccine (UIP)
Filariasis	**Wuchereria bancrofti**; **Culex** (dirty water)	Lymphoedema, **elephantiasis**, hydrocele	**DEC (diethylcarbamazine) + Albendazole** (MDA) ± Ivermectin
Kala-azar	**Leishmania donovani**; **sandfly**	Fever, **massive splenomegaly**, skin darkening	Liposomal amphotericin B (single dose)
Hepatitis	See Chapter 8		

Mosquito vectors

- **Anopheles** → **Malaria** (female; night)
- **Aedes** → **Dengue, Chikungunya, Zika, Yellow fever** (day biter; clean water)
- **Culex** → **Filariasis, Japanese encephalitis, West Nile** (night; dirty water)
- **Mansonia** → Brugian filariasis

Practice MCQs – Chapter 15

1. Mantoux test kis type ki hypersensitivity hai?
(a) Type IV (b) Type I (c) Type III (d) Type II

2. Anaphylaxis ka drug of choice –
(a) Adrenaline 1:1000 IM (b) Antihistamine (c) Salbutamol (d) Hydrocortisone IV

3. Placenta cross karne wali immunoglobulin –
(a) IgM (b) IgG (c) IgA (d) IgE

4. HIV kis cell ko attack karta hai?
(a) CD4 T-helper cells (b) RBC (c) Platelets (d) B cells

5. AIDS diagnosis CD4 count –
(a) < 500 (b) < 350 (c) < 200 (d) < 1000

6. PLHIV me India me most common opportunistic infection –
(a) PCP (b) Tuberculosis (c) CMV (d) Kaposi

7. HIV PEP kitne din tak lete hain?
(a) 7 (b) 14 (c) 28 (d) 90

8. Dengue me kaunsi dawa avoid karni chahiye?
(a) Paracetamol (b) ORS (c) IV fluids (d) Aspirin / NSAIDs

9. Dengue failane wala mosquito –
(a) Mansonia (b) Aedes aegypti (c) Anopheles (d) Culex

10. Primaquine dene se pehle kaunsa test?
(a) LFT (b) TSH (c) RFT (d) G6PD

11. Tetanus ka first sign –
(a) Rash (b) Opisthotonus (c) Fever (d) Trismus (lockjaw)

12. Dog bite ke baad wound ko –
(a) Bandage tight (b) Kuch na karo (c) Turant suture (d) Soap & running water se 15 min wash

13. Rabies ARV (IM) schedule –
(a) 0, 7, 21 (b) 0, 1, 6 months (c) 0, 3, 7, 14, 28 (d) Single dose

14. Leprosy me MB case ka MDT duration –
(a) 6 months (b) 12 months (c) 2 years (d) 3 months

15. Koplik's spots kis disease me?
(a) Chickenpox (b) Measles (c) Mumps (d) Rubella

16. Rice-water stools –
(a) Typhoid (b) Amoebiasis (c) Dengue (d) Cholera

17. SLE ka characteristic rash –
(a) Honey crust (b) Koplik (c) Butterfly malar rash (d) Rose spots

18. Typhoid ke 1st week me best test –

(a) Blood culture (b) Widal (c) Stool culture (d) Urine culture

Answer Key – Chapter 15

1-a · 2-a · 3-b · 4-a · 5-c · 6-b · 7-c · 8-d · 9-b · 10-d · 11-d · 12-d · 13-c · 14-b · 15-b · 16-d · 17-c · 18-a

16 Oncology Nursing

16.1 Cancer – Basics

D Cancer (Neoplasm)

Uncontrolled, abnormal **cell growth** that invades surrounding tissue and can **spread (metastasize)** to distant sites. Study = **Oncology**.

Benign tumour	Malignant tumour
Slow growth, **encapsulated**	Rapid, **not encapsulated**, invasive
Well-differentiated (cells look normal)	**Poorly differentiated / anaplastic**
No metastasis	**Metastasis** (blood, lymph, seeding)
Rarely recurs	Often recurs
Usually not fatal	Fatal if untreated

- Naming: **-oma** (benign, e.g. lipoma, fibroma); **Carcinoma** – epithelial (most common); **Sarcoma** – connective tissue (bone, muscle); **Leukaemia**, **Lymphoma**; **Carcinoma in situ** – not invaded basement membrane.
- **Carcinogens**: **Tobacco (most important preventable cause)**, alcohol, **viruses** (HPV – cervical/oral; HBV/HCV – liver; EBV – Burkitt's lymphoma, nasopharyngeal; HTLV-1; H. pylori – stomach), radiation (UV – skin), chemicals (asbestos – mesothelioma; benzene – leukaemia; **aniline dyes – bladder**; vinyl chloride – liver), **aflatoxin – liver**, diet (low fibre, high fat, smoked/salted food), obesity, genetics (**BRCA1/2 – breast & ovary**).
- Common cancers in India: **Men – oral/lip & lung**; **Women – breast (most common), then cervical**. Overall most common cause of cancer death worldwide – **lung cancer**.

♫ CAUTION – 7 warning signs (American Cancer Society)

C – Change in bowel or bladder habits
A – A sore that does not heal
U – Unusual bleeding or discharge
T – Thickening or lump in breast or elsewhere
I – Indigestion or difficulty in swallowing
O – Obvious change in wart or mole
N – Nagging cough or hoarseness
(+ **U**nexplained anaemia, **S**udden weight loss → "CAUTION US")

Grading & Staging

- **Grading** – degree of differentiation (cells kitne normal dikhte hain): **Grade I (well) → IV (undifferentiated/anaplastic)**.
- **Staging – TNM**: **T** = Tumour size (T0–T4, Tis = in situ), **N** = Node involvement (N0–N3), **M** = Metastasis (M0/M1).
- Clinical stage 0 (in situ) → I, II (localized) → III (regional spread) → **IV (distant metastasis)**.

Tumour markers

Marker	Cancer
CEA	Colorectal (also lung, breast)
AFP	**Liver (HCC)**, germ cell / testicular
CA-125	**Ovarian**
CA 19-9	**Pancreatic**
CA 15-3	**Breast**
PSA	**Prostate**
β-hCG	Choriocarcinoma, testicular, molar pregnancy
Calcitonin	Medullary thyroid

- **Biopsy = definitive diagnosis** (excisional, incisional, needle – FNAC, core).

Prevention & Screening

Cancer	Screening
Breast	**BSE monthly from age 20** (**5–7 days after menses**; post-menopausal – fixed date each month); **Clinical breast exam (CBE)**; **Mammography yearly/ 2-yearly from 40–50**
Cervix	**Pap smear** (every 3 yrs from 21 / 25), **VIA (Visual Inspection with Acetic acid – white patch)** – India's low-cost screening (30–65 yrs, every 5 yrs); HPV DNA test; **HPV vaccine (9–14 yrs girls)** – India's indigenous **Cervavac**
Oral	**OVE (Oral visual examination)** – 30+ yrs (NPCDCS/NP-NCD)
Colorectal	FOBT yearly, colonoscopy every 10 yrs (from 45–50)
Prostate	PSA, DRE
Testicular	Monthly TSE

- **National Cancer Awareness Day – 7 November** (Marie Curie's birthday); **World Cancer Day – 4 February**. India – population-based screening for **oral, breast, cervical** cancers (≥ 30 yrs) under **NP-NCD**.

16.2 Cancer Treatment

Surgery

Prophylactic, diagnostic (biopsy), **curative**, palliative (relieve symptoms), reconstructive.

Chemotherapy

- Kills **rapidly dividing cells** – cancer + normal fast-dividing cells (**bone marrow, GI mucosa, hair follicles, gonads**) → hence side effects.
- **Cell-cycle specific** (antimetabolites – methotrexate, 5-FU; vinca alkaloids – vincristine) vs **non-specific** (alkylating – cyclophosphamide, cisplatin; antitumour antibiotics – doxorubicin).

Side effect	Nursing management
Bone marrow suppression (most serious; dose-limiting) – nadir 7–14 days	**Neutropenia** → infection precautions, report **temp ≥ 38 °C**; **thrombocytopenia** → bleeding precautions; anaemia → rest, transfusion, EPO; check CBC before each cycle
Nausea & vomiting (most common)	**Antiemetics 30–60 min BEFORE chemo** (**ondansetron**, dexamethasone, aprepitant); small frequent bland meals, cold foods, avoid strong odours; ginger
Stomatitis / mucositis	Soft brush, **saline / sodium bicarbonate mouthwash**, avoid alcohol mouthwash, hot spicy foods; ice chips during 5-FU
Alopecia	Begins **2–3 weeks**; **temporary – regrowth after treatment** (may differ colour/texture); scarf/wig, cut hair short before, mild shampoo, sun protection
Diarrhoea/constipation (vincristine)	Fluids, diet
Fatigue	Energy conservation, rest, light exercise
Specific toxicities	**Doxorubicin – cardiotoxicity**, red urine; **Cisplatin – nephrotoxicity, ototoxicity**; **Cyclophosphamide – haemorrhagic cystitis** (fluids 3 L, mesna); **Vincristine – peripheral neuropathy**, constipation; **Bleomycin – pulmonary fibrosis**; **Methotrexate – leucovorin (folinic acid) rescue**; Paclitaxel – hypersensitivity
Extravasation (vesicants – doxorubicin, vincristine)	**Stop infusion immediately**, **leave needle & aspirate**, notify, antidote, cold (most) / warm (vinca alkaloids) compress, elevate
Tumour lysis syndrome	Rapid cell death → **hyperkalaemia, hyperuricaemia, hyperphosphataemia, hypocalcaemia** → AKI; **hydration + allopurinol/rasburicase**
Reproductive	Infertility – **sperm banking/egg freezing before**; contraception (teratogenic)

! Safe handling of chemotherapy (cytotoxic drugs)

- Prepare in **Biological Safety Cabinet (Class II, vertical laminar flow)**
- **PPE: double chemo-tested gloves, long-sleeved impermeable gown, eye protection, N95/mask**
- **Pregnant/ breastfeeding staff should not handle**
- Luer-lock syringes, closed-system transfer devices; **no eating/drinking** in the area
- Disposal in **labelled cytotoxic waste container** (yellow – incineration); spill kit
- Patient's **excreta are contaminated for 48 hrs** → wear gloves, **flush toilet twice with lid closed**; linen separate

Radiation therapy

- **External beam (teletherapy)** – e.g. Cobalt-60, linear accelerator; patient **NOT radioactive**.
- **Internal – Brachytherapy** (sealed source placed in/near tumour – cervix, prostate) & **unsealed** (I-131 oral/IV) – **patient IS radioactive** (emits radiation).

Radiation safety – Time, Distance, Shielding (ALARA)

- **Time** – minimize time near patient (organize care; **max ~30 min per shift**); limit visitors to **30 min/day**
- **Distance** – keep max distance; visitors **≥ 6 feet (2 m)**; **inverse square law** (2× distance → ¼ exposure)
- **Shielding** – **lead apron/ shield**; private room with lead lining
- **Dosimeter (film badge/ TLD) – worn by staff**
- **No pregnant nurses/visitors, no children < 18 yrs**
- **Dislodged implant** → **pick up with long-handled forceps** (never hands) → place in **lead container** kept in room; notify radiation safety officer
- Cervical implant: **strict bed rest, flat/ HOB ≤ 10–20°**, Foley catheter, low-residue diet (no BM), log-roll only
- Linen & dressings **kept in room** until checked by radiation officer

- **Skin care in external radiation** – **don't wash off ink markings**; wash with **lukewarm water & mild soap, pat dry**; **no lotions, powders, deodorants, perfumes on treated area** (unless prescribed); **avoid sun, heat/ice packs, tight clothes**; loose cotton clothing; electric razor.
- Side effects depend on site: skin reactions (erythema → dry/moist desquamation), **fatigue (most common)**, mucositis, xerostomia (head & neck – **dental check before**), diarrhoea (abdomen/pelvis), cystitis, bone marrow suppression, alopecia (only in treated area – may be permanent).

Other therapies

Hormone therapy (Tamoxifen – breast, ER+ ; s/e hot flashes, **endometrial cancer, DVT**; letrozole), **Immunotherapy** (checkpoint inhibitors), **Targeted therapy** (Imatinib – CML; Trastuzumab – HER2 breast – cardiotoxic), **Bone marrow / stem cell transplant**, **Biological response modifiers** (interferon).

16.3 Important Cancers – Nursing Points

Breast cancer

- Most common cancer in Indian women; **upper outer quadrant (most common site)** – tail of Spence; **painless, hard, irregular, fixed lump**; **peau d'orange** (orange-peel skin – lymphatic block), **nipple retraction**, bloody discharge, axillary nodes.
- Risk: **female, age > 50**, BRCA, family history, **early menarche (< 12), late menopause (> 55), nulliparity/first child after 30**, no breastfeeding, HRT, obesity, alcohol.
- Surgery: **Lumpectomy (breast conserving)** + radiation, **Modified radical mastectomy (MRM)**, sentinel lymph node biopsy.

Post-mastectomy care

- Position: **semi-Fowler's**, **affected arm elevated on pillow** (above heart level) – lymphoedema prevention
- **Drains (Hemovac/JP)** – empty & measure, 1–2 weeks
- Exercises: **start fingers/hand/elbow (squeeze ball) day 1**; shoulder exercises gradually (**wall climbing, rope turning, pendulum swing**) after drain removal / as per surgeon
- **Lymphoedema prevention (lifelong)** on affected side: **NO BP, injections, IV, blood draws**; avoid cuts, burns, insect bites; **gloves for gardening**, oven mitts; **no tight jewellery/sleeves**; electric razor; don't carry heavy bags; report swelling
- Body image – breast prosthesis (after healing ~6 weeks), reconstruction; support groups

Cervical cancer

- **HPV 16 & 18** (most important – ~70 %); risk – early sexual activity, multiple partners, **multiparity**, smoking, HIV, low socioeconomic, OCPs long-term.
- **Post-coital bleeding (most common early symptom)**, intermenstrual bleeding, foul watery discharge; late – pain, fistula.
- Screening: **Pap smear / VIA / HPV DNA**; CIN (pre-cancer) → cryotherapy/ LEEP; Rx – Wertheim's (radical) hysterectomy, radiation (brachytherapy), chemo.
- **HPV vaccine** – India: **Cervavac (Serum Institute)**; girls 9–14 yrs.

Oral cancer

- **Most common cancer in Indian men**; **tobacco chewing (gutkha, khaini)**, smoking, betel quid, alcohol, poor oral hygiene, HPV.
- Premalignant: **leukoplakia**, **erythroplakia (highest malignant potential)**, **oral submucous fibrosis** (areca nut – restricted mouth opening).
- Most common site in India – **buccal mucosa / gingivo-buccal sulcus** (where tobacco is kept); tongue (lateral border).
- Rx: surgery (commando operation), radiation, chemo; **post-op – airway (tracheostomy), suction, communication board, nutrition (NG/ PEG)**.
- **Laryngeal cancer** – **persistent hoarseness (> 2 weeks) – earliest sign**; **total laryngectomy → permanent tracheostomy (stoma), loss of voice** – speech rehab (oesophageal speech, electrolarynx, TEP); protect stoma (no swimming, cover during shower), humidification.

16.4 Oncological Emergencies

- **Superior vena cava syndrome** – lung cancer/lymphoma; **facial/neck/arm oedema**, distended chest veins, dyspnoea → **semi-Fowler's**, O_2, radiation, steroids; avoid IV/BP in arms.
- **Spinal cord compression** – back pain (first), weakness, bladder/bowel dysfunction → **dexamethasone**, radiation.
- **Hypercalcaemia** (bone metastasis) – fluids, bisphosphonates.
- **Tumour lysis syndrome**, **SIADH** (small cell lung), **cardiac tamponade**, **neutropenic sepsis**, **DIC**.

16.5 Palliative & End-of-Life Care

- **Palliative care** (WHO) – improves **quality of life** of patients & families facing life-threatening illness – **relief of pain & suffering** (physical, psychological, social, spiritual); **neither hastens nor postpones death**; begins early, along with curative care. **Hospice** – end-of-life (last ~6 months).
- **Kubler-Ross – 5 stages of grief/dying (1969)**: **D**enial → **A**nger → **B**argaining → **D**epression → **A**cceptance ("**DABDA**"). Not always in order.
- Common symptoms: pain (WHO ladder – Chapter 3), dyspnoea (opioids, fan, O_2, position), nausea, constipation (opioids – laxatives prophylactically), anorexia/cachexia, mouth dryness, delirium, **"death rattle"** (secretions – **side position, glycopyrrolate/hyoscine**; deep suction not useful).
- **Signs of approaching death**: ↓ consciousness, **Cheyne-Stokes breathing**, mottled cold extremities (knees/feet first), ↓ urine output, weak rapid pulse, ↓ BP, **hearing is the last sense to be lost** (keep talking to patient).
- **Signs of death**: no pulse, no respiration, fixed dilated pupils; **Algor mortis** (cooling ~1 °C/hr), **Rigor mortis** (stiffness – starts **2–4 hrs** after death, jaw first, full ~12 hrs, disappears 24–48+ hrs), **Livor mortis** (purple discolouration in dependent parts – 20 min–2 hrs).
- **Last offices (after-death care)**: confirm death by doctor, respect religious rituals, **place body supine, pillow under head (livor in face ↓), close eyes & mouth (rolled towel under chin), dentures in place**, remove tubes (unless **medico-legal case – don't remove, inform police**), clean & dress, ID tags (wrist, ankle, shroud), valuables to family with documentation, family time.

? Practice MCQs – Chapter 16

1. Cancer ka most important preventable cause –
(a) Radiation (b) Alcohol (c) Diet (d) Tobacco

2. Breast self-examination kab karna chahiye?
(a) Before menses (b) Any day (c) During menses (d) 5–7 days after menses

3. Breast cancer ka most common site –
(a) Lower outer (b) Nipple (c) Lower inner (d) Upper outer quadrant

4. CA-125 kis cancer ka marker hai?
(a) Breast (b) Liver (c) Ovarian (d) Colon

5. Chemotherapy ke nadir ka time –
(a) 1–2 days (b) 7–14 days (c) 1 month (d) 3 months

6. Antiemetic kab dena chahiye?
(a) Only if asked (b) After vomiting (c) Next day (d) 30–60 min before chemo

7. Radiation safety ke 3 principles –
(a) Speed, light, heat (b) Time, distance, shielding (c) Rest, ice, elevation (d) Gloves, mask, gown

8. Dislodged radioactive implant ko –
(a) Long forceps se lead container me (b) Haath se uthao (c) Flush karo (d) Dustbin me

9. External radiation me skin markings ko –
(a) Scrub (b) Cream lagao (c) Wash off (d) Wash nahi karna

10. Post-mastectomy patient ke affected arm me –
(a) No BP, injection, IV (b) IV line (c) Tight bangles (d) BP lena

11. Cervical cancer ka main cause –
(a) HIV (b) HSV (c) HPV 16 & 18 (d) EBV

12. Kubler-Ross ka first stage –
(a) Denial (b) Depression (c) Acceptance (d) Anger

13. Dying patient me last sense jo khatam hota hai –
(a) Taste (b) Vision (c) Hearing (d) Touch

14. Doxorubicin ki major toxicity –
(a) Cystitis (b) Ototoxicity (c) Cardiotoxicity (d) Neuropathy

15. Cyclophosphamide se haemorrhagic cystitis rokne ke liye –
(a) Fluids 3 L & mesna (b) Fluid restriction (c) Vit K (d) Leucovorin

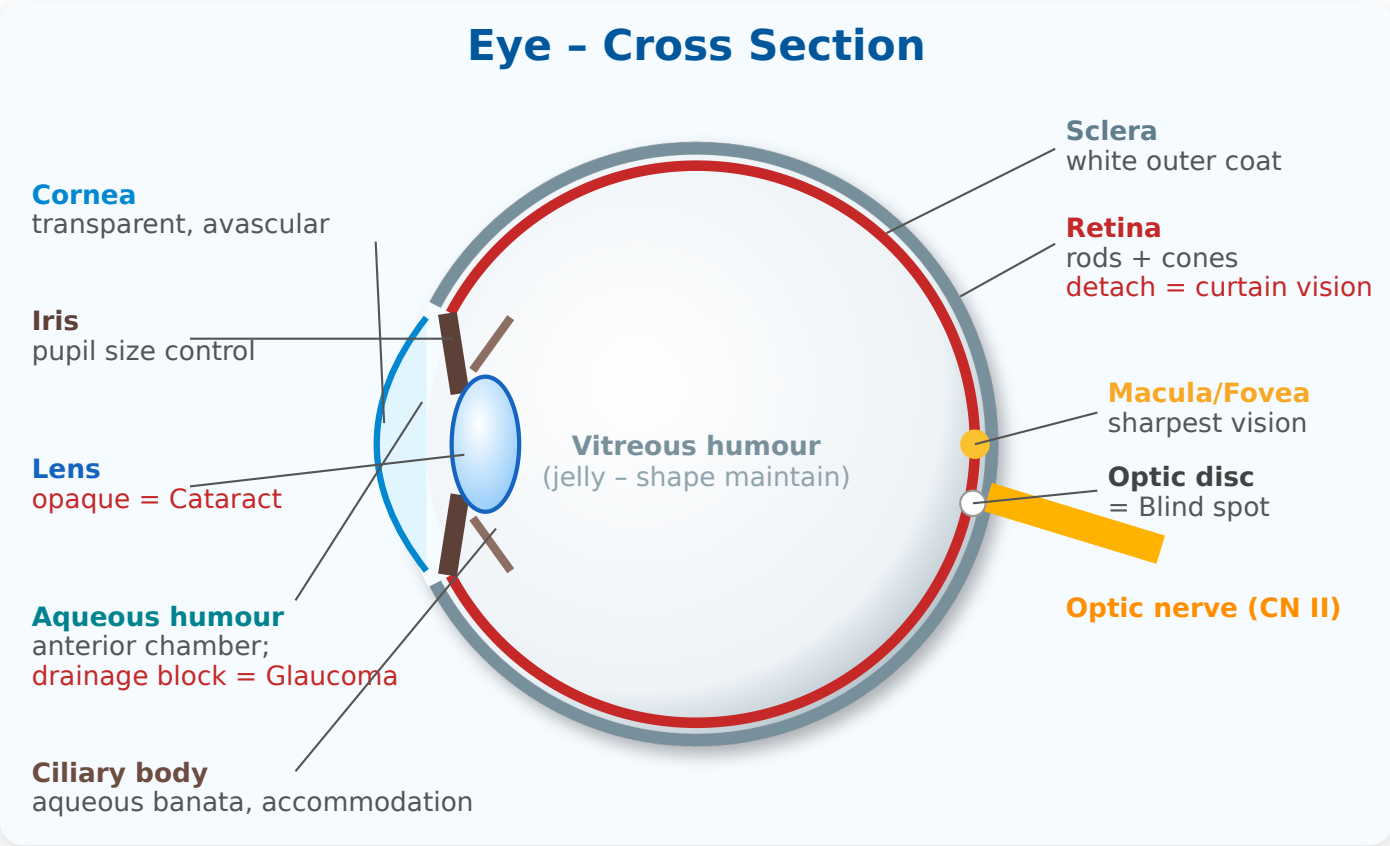
16. Laryngeal cancer ka earliest sign –
(a) Neck mass (b) Weight loss (c) Dysphagia (d) Persistent hoarseness

Answer Key – Chapter 16

1-d · 2-d · 3-d · 4-c · 5-b · 6-d · 7-b · 8-a · 9-d · 10-a · 11-c · 12-a · 13-c · 14-c · 15-a · 16-d

17 Eye & ENT Disorders

17.1 Eye – Applied Anatomy



Eye ke parts – kaunsa part kis disease se juda hai

Eye facts

- 3 layers: **Outer – sclera & cornea** · **Middle (uvea) – choroid, ciliary body, iris** · **Inner – retina**.
- **Cornea** – transparent, **avascular** (O_2 from air/aqueous); most refractive; **corneal reflex (CN V sensory, CN VII motor)**.
- **Aqueous humour** – made by **ciliary body** → posterior chamber → pupil → anterior chamber → drained via **trabecular meshwork & canal of Schlemm**. Normal **IOP 10–21 mmHg** (measured by **tonometer**).
- **Lens** – biconvex, avascular; **accommodation** (ciliary muscle).
- **Retina** – **rods (dim light, night vision – Vit A/rhodopsin)**, **cones (colour, bright light)**; **macula/fovea – sharpest central vision** (only cones); **optic disc – blind spot** (no rods/cones).
- Muscles: **6 extraocular muscles** – CN III (most), IV (superior oblique), VI (lateral rectus).
- **Lacrimal gland** – upper outer orbit; tears drain via **puncta → nasolacrimal duct → nose**.

Eye assessment

- **Snellen chart** – **20 feet (6 m)**; normal **6/6 (20/20)**; top number = distance of patient, bottom = distance a normal eye can read; **legal blindness ≤ 20/200 (6/60)** in better eye with correction.

- **Refractive errors**: **Myopia** (near-sighted; image in front of retina – **concave/minus lens**), **Hypermetropia/Hyperopia** (far-sighted; image behind retina – **convex/plus lens**), **Astigmatism** (irregular cornea – cylindrical lens), **Presbyopia** (age-related loss of accommodation, > 40 yrs – reading glasses/convex).
- **Ishihara chart** – colour vision. **Ophthalmoscope** – fundus. **Slit lamp** – anterior segment. **Fluorescein stain** – corneal abrasion/ulcer (green).
- **Mydriatics** (pupil dilate) – **atropine, tropicamide, phenylephrine** – **contraindicated in angle-closure glaucoma**; **Miotics** (pupil constrict) – **pilocarpine** (glaucoma).
- **National Programme for Control of Blindness & Visual Impairment (NPCB&VI, 1976)** – India first country to launch. Most common cause of blindness in India – **Cataract**; most common cause of visual impairment – **refractive errors**.

17.2 Cataract

- **Opacity (clouding) of the lens** – most common cause of blindness (**reversible**).
- Types: **Senile (age-related – most common, > 50)**, congenital (rubella), traumatic, secondary (**diabetes, steroids**), radiation, smoking, UV.
- S/S: **painless, gradual blurring of vision**, **glare/ halos at night**, faded colours, **"second sight"** (temporary improvement in near vision), **white/grey pupil (leukocoria)** in mature cataract, **frequent change of glasses**.
- Rx: **Surgery only** – **ECCE / SICS (manual small-incision – common in India camps)**, **Phacoemulsification (ultrasound – smallest incision)** + **IOL (intraocular lens) implant**; ICCE (old).

Post-cataract surgery care & teaching

- **Eye shield/patch** – especially **at night for 1–4 weeks**; dark glasses in day
- **Lie on back or on unoperated side** (not on operated side)
- **Avoid ↑ IOP**: **no bending (head below waist), lifting heavy objects (> 5 kg), straining, coughing, sneezing, vomiting**, rubbing eye, squeezing eyes; stool softener; antiemetics
- **Eye drops** – antibiotic + steroid as ordered; correct technique; hand hygiene
- **Report immediately**: **sudden severe eye pain** (↑IOP/haemorrhage), **↓ vision**, flashes/floaters, increasing redness, discharge (endophthalmitis)
- No eye make-up, swimming, dust; head bath (avoid water in eye) – 2 weeks
- Reading/TV allowed; spectacles after ~4–6 weeks

17.3 Glaucoma

Glaucoma

Group of eye diseases with **↑ intraocular pressure (IOP > 21 mmHg)** → **optic nerve damage (cupping)** → **irreversible peripheral vision loss** → blindness. "**Silent thief of sight**". Most common cause of **irreversible** blindness.

Feature	Open-angle (Chronic / POAG) – most common (~90 % west)	Angle-closure (Acute / Narrow-angle) – emergency
Mechanism	Trabecular meshwork slowly blocked (drainage ↓)	**Iris blocks angle suddenly** (pupil dilation – dark room, mydriatics, stress, anticholinergics)
Onset	**Slow, painless, gradual**	**Sudden**
Vision	**Gradual loss of peripheral vision → "tunnel vision"**	**Sudden blurred vision, halos (rainbow) around lights**
Pain	**Painless**	**Severe eye pain, headache**, nausea/vomiting
Eye	Normal looking	**Red eye, fixed mid-dilated pupil, cloudy (steamy) cornea**, rock-hard eye
IOP	Moderately ↑ (22–32)	**Very high (> 30–50)**
Rx	Lifelong eye drops; laser trabeculoplasty; **trabeculectomy**	**Emergency – IV acetazolamide/ mannitol, pilocarpine**, β-blocker drops → **laser peripheral iridotomy**

! Glaucoma drugs & teaching

- **Timolol** (β-blocker drop) – ↓ aqueous production; **systemic effects – bradycardia, bronchospasm (avoid in asthma/COPD)**
- **Pilocarpine** (miotic) – ↑ outflow; **poor night vision, blurred vision, brow ache**
- **Latanoprost** (prostaglandin) – ↑ outflow; **iris/eyelash darkening**, longer lashes
- **Acetazolamide** (oral/IV carbonic anhydrase inhibitor) – **hypokalaemia**, paresthesia, kidney stones; sulfa allergy
- **Lifelong treatment** – glaucoma is **controlled, not cured**; lost vision cannot be restored
- **Punctal occlusion** (press inner canthus 1–2 min) after drops → systemic absorption ↓
- **Avoid atropine/ mydriatics & anticholinergics** (antihistamines, decongestants); avoid ↑ IOP activities; yearly eye check after 40 (tonometry)

17.4 Other Eye Disorders

- **Retinal detachment** – separation of retina from choroid; S/S: **sudden painless flashes of light (photopsia), floaters ("spots/cobwebs"), "curtain/ shadow/ veil" coming over field of vision → emergency**. Rx: **laser photocoagulation, cryopexy, scleral buckling, vitrectomy, pneumatic retinopexy (gas bubble – specific head positioning, e.g. face-down, so bubble presses retina; no air travel/high altitude till gas absorbed)**; bed rest, eye patches, avoid jerky head movements.
- **Macular degeneration (AMD)** – elderly; **loss of central vision** (peripheral intact), **distorted straight lines (Amsler grid)**; **dry (most common)** – antioxidants, **wet** – anti-VEGF injections.
- **Diabetic retinopathy** – microaneurysms, haemorrhages, neovascularization; yearly fundus exam; laser.
- **Conjunctivitis ("pink eye")** – bacterial (purulent discharge, lids stuck in morning), viral (watery, highly contagious – adenovirus), allergic (itching); **don't patch**; hand hygiene, **separate towels**, warm/cold compress, antibiotic drops (bacterial); clean from **inner to outer canthus**.
- **Trachoma** – Chlamydia trachomatis; leading infectious cause of blindness; **SAFE strategy** (Surgery, Antibiotics – azithromycin, Facial cleanliness, Environmental improvement); India declared **trachoma-free (2024, WHO)**.

- **Hordeolum (stye)** – painful, acute infection of eyelash gland (external) → **warm compresses**; **Chalazion** – painless chronic granuloma (meibomian).
- **Blepharitis** – lid margin inflammation – lid hygiene.
- **Keratitis/ Corneal ulcer** – pain, photophobia; contact lens users; don't patch (bacterial).
- **Uveitis/ iritis** – painful red eye, photophobia, small pupil.
- **Xerophthalmia (Vitamin A deficiency)** – **night blindness (first symptom)** → conjunctival xerosis → **Bitot's spots** → corneal xerosis → keratomalacia. Prophylaxis: **Vit A 1 lakh IU at 9 months, then 2 lakh IU every 6 months till 5 yrs** (total 9 doses).
- **Strabismus (squint)** – patching good eye (amblyopia); **Ptosis** – CN III, MG.

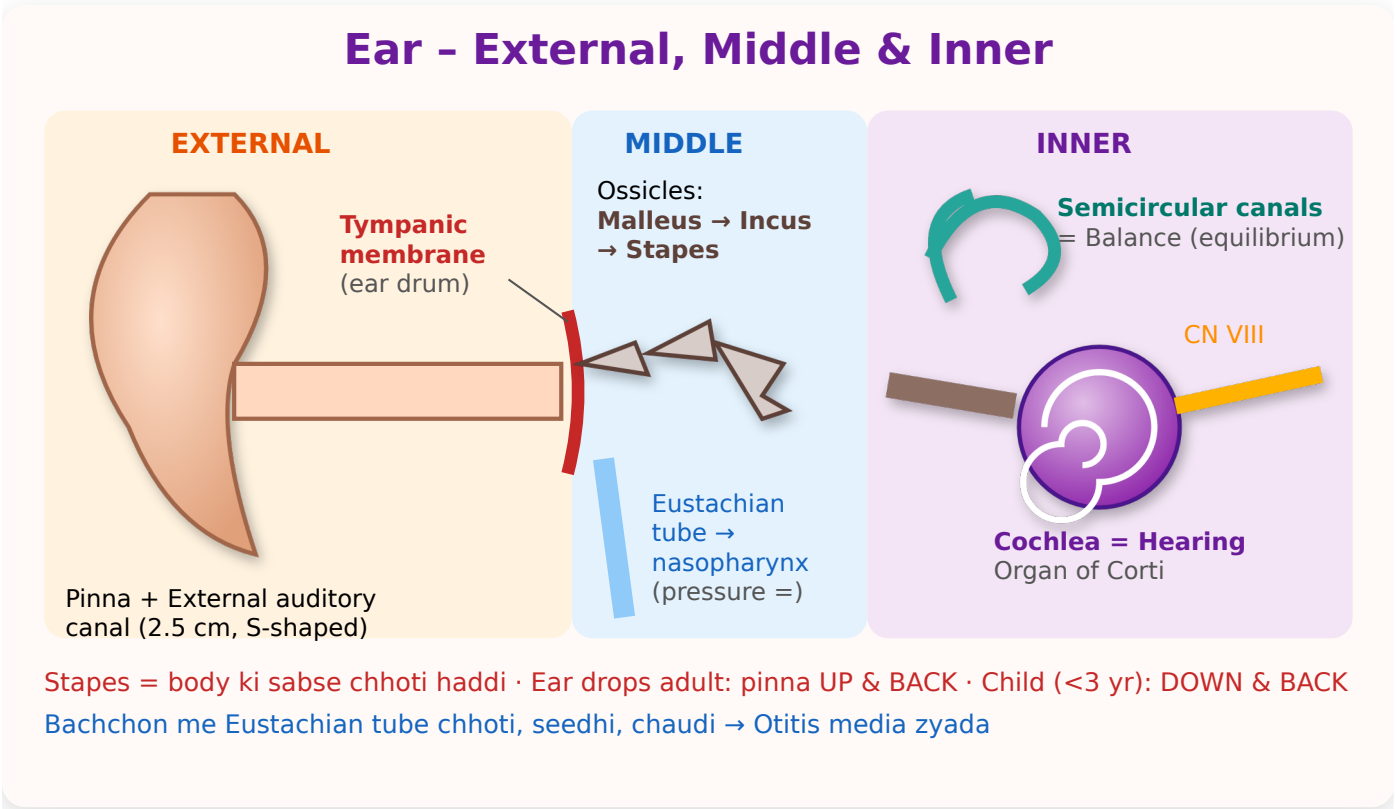
Eye injuries – first aid

- **Chemical burn** – **immediate copious irrigation with water/NS for at least 15–30 min** (most important – before anything else), irrigate **from inner to outer canthus**, check pH (7), evert lids.
- **Foreign body** – don't rub; flush; **penetrating object – DON'T remove**, **stabilize object (paper cup over it)**, **patch BOTH eyes** (sympathetic movement ↓), **no pressure**, HOB 30–45°, refer.
- **Hyphaema (blood in anterior chamber)** – bed rest, **HOB 30–45°**, eye shield, no aspirin.
- **Blunt injury** – ice packs (cold compress).

Eye drops/ointment instillation

- Hand hygiene; patient **head tilted back, looks up**; **pull lower lid down** → instil into **lower conjunctival sac (centre/outer third)** – **never directly on cornea**; dropper 1–2 cm above, don't touch eye.
- **Close eye gently (don't squeeze) & press inner canthus (nasolacrimal duct) 1–2 min**.
- Multiple drops – **wait 3–5 min between different drops**; **drops before ointment** (ointment last – wait 10 min).
- Ointment – thin ribbon from **inner to outer canthus**.
- **Contact lens** – remove before most drops; artificial eye – remove, clean.

17.5 Ear



Ear ke 3 parts – external, middle aur inner

Ear facts

- **External**: pinna, external auditory canal (2.5 cm, S-shaped; **cerumen/wax**), tympanic membrane (**pearly grey, cone of light at 5 o'clock (right) / 7 o'clock (left)**).
- **Middle**: ossicles – **malleus, incus, stapes** (stapes – smallest bone; **footplate in oval window**), **Eustachian tube** (pressure equalization; children – **short, straight, horizontal** → otitis media).
- **Inner (labyrinth)**: **cochlea (hearing – organ of Corti)**, **vestibule & semicircular canals (balance)**; CN VIII.
- Sound conduction: pinna → canal → TM → ossicles → oval window → cochlea → CN VIII → **temporal lobe**.

Hearing tests (tuning fork 512 Hz)

Test	Normal	Conductive loss	Sensorineural loss
Rinne (mastoid vs in front of ear)	**AC > BC (Rinne positive)**	**BC > AC (negative)** in affected ear	AC > BC (positive) but both reduced
Weber (vertex/forehead)	Heard equally (midline)	**Lateralizes to affected (bad) ear**	**Lateralizes to normal (good) ear**

- **Conductive** hearing loss – external/middle ear problem (wax, otitis media, otosclerosis, perforation). **Sensorineural** – inner ear/ CN VIII (presbycusis – age, **noise-induced**, **ototoxic drugs – aminoglycosides (gentamicin, streptomycin), loop diuretics (furosemide), aspirin (high dose – tinnitus), cisplatin, vancomycin, quinine**).
- **Audiometry** – frequency & intensity; **decibels**; **> 85 dB** prolonged → damage.

- **Presbycusis** – age-related, **high-frequency** loss first.

Communicating with hearing-impaired

Face the patient, **good light on your face**, **speak slowly & clearly in normal/low pitch (don't shout)**, reduce background noise, use gestures/writing, rephrase (don't just repeat), get attention first (touch), check hearing aid battery.

Ear disorders

- **Impacted cerumen** – **ear irrigation**: **body-temperature water (37 °C)** (cold/hot → **vertigo**), stream directed **towards upper/posterior canal wall** (not at TM), patient sitting, kidney tray under ear, **contraindicated if perforation/ infection**; softening drops (olive oil/ ceruminolytic) 2–3 days before. **No cotton buds** inside canal.
- **Ear drops**: warm to body temperature; **adult/child > 3 yrs – pinna UP & BACK**; **child < 3 yrs – DOWN & BACK**; lie on unaffected side **5 min** after.
- **Otitis externa (swimmer's ear)** – canal infection; **pain on moving pinna/tragus**; drops, keep dry.
- **Acute otitis media (AOM)** – middle ear infection after URI; **children** (Eustachian tube); **ear pain (ear pulling/rubbing in infants)**, fever, **red bulging TM**; rupture → discharge & pain relief. Rx: **amoxicillin**, analgesics; **Myringotomy + grommets (tympanostomy tubes)** for recurrent/ chronic effusion – keep ears dry (ear plugs).
- **Chronic suppurative otitis media (CSOM)** – persistent discharge through perforated TM; **cholesteatoma** (unsafe – erodes bone); **mastoiditis**; Rx – tympanoplasty, mastoidectomy.
- **Mastoiditis** – pain, swelling, redness behind ear, ear pushed forward; IV antibiotics, **mastoidectomy** (watch **facial nerve – CN VII**, vertigo).
- **Otosclerosis** – **stapes fixed (spongy bone)** → **progressive conductive hearing loss**, young women, hereditary; **Paracusis Willisii** (hears better in noise); Rx – **stapedectomy** (post-op: **avoid nose blowing (open mouth sneeze), bending, straining, air travel, head jerks** for weeks; vertigo common – move slowly; keep ear dry; lie flat with operated ear up as per surgeon).
- **Ménière's disease** – **↑ endolymph (endolymphatic hydrops)**; **TRIAD: episodic vertigo + tinnitus + fluctuating sensorineural hearing loss** + **aural fullness**; attacks hours. Rx: **low-sodium diet (< 2 g/day)**, diuretics, avoid **caffeine, alcohol, tobacco, MSG**; **meclizine, antiemetics**, diazepam during attack; **during attack – lie still in quiet dark room, keep eyes focused on fixed point**, side rails up, fall prevention; surgery – labyrinthectomy.
- **BPPV** – brief vertigo with head position change (otoconia); **Epley manoeuvre**.
- **Labyrinthitis** – vertigo + hearing loss after viral infection.
- **Acoustic neuroma (vestibular schwannoma)** – CN VIII benign tumour; unilateral sensorineural loss, tinnitus.

17.6 Nose & Throat

- **Epistaxis (nosebleed)** – most common site **anterior septum – Little's area (Kiesselbach's plexus)**; causes – trauma (nose picking), dry air, **hypertension**, anticoagulants, bleeding disorders.

Epistaxis first aid

- **Sit upright & lean FORWARD** (blood mouth me nahi jayega, aspiration/vomiting ↓) – **NOT head back**
- **Pinch soft lower part of nose (both nostrils) firmly for 10–15 min** continuously; breathe through mouth
- **Ice/cold compress** on nose bridge/ forehead; spit blood, don't swallow
- Persistent → **anterior nasal packing** (ribbon gauze with vaseline/ BIPP), cautery (silver nitrate); posterior bleed → posterior pack/ Foley balloon (**monitor airway & SpO_2**)
- After: **avoid nose blowing, picking, strenuous activity, hot drinks, bending, lifting for 1 week**; humidifier; no aspirin

- **Rhinitis** – allergic (sneezing, itching, watery discharge) – antihistamines, nasal steroids; **avoid overuse of decongestant sprays (> 3–5 days → rebound congestion – rhinitis medicamentosa)**.
- **Sinusitis** – facial pain/pressure (worse bending), purulent discharge, fever; **maxillary sinus most common** (adults); steam inhalation, decongestants, antibiotics, **FESS** (functional endoscopic sinus surgery).
- **Deviated nasal septum** – **septoplasty/ SMR** → nasal packing 24–48 hrs (mouth breathing – oral care, humidifier), **semi-Fowler's**, ice to nose, watch **frequent swallowing (posterior bleeding)**, **no nose blowing** 1–2 weeks, sneeze with mouth open.
- **Nasal polyps** – aspirin sensitivity (Samter's triad – asthma + polyps + aspirin sensitivity).
- **Nasal fracture** – ice, check septal haematoma, CSF leak.
- **Pharyngitis** – mostly viral; **strep throat (GABHS)** → rheumatic fever/ GN risk → full course penicillin.
- **Tonsillitis** – sore throat, fever, dysphagia, enlarged red tonsils with exudate; Rx antibiotics; **Tonsillectomy** – indications: recurrent (≥ 7/yr in 1 yr, ≥ 5/yr for 2 yrs, ≥ 3/yr for 3 yrs), **peritonsillar abscess (quinsy)**, obstructive sleep apnoea.

Post-tonsillectomy care

- Position: **prone/ side-lying (semi-prone) with head turned to side** until fully awake (drainage of secretions); then **semi-Fowler's**
- **Haemorrhage – most important complication** (first 24 hrs – primary/reactionary; **5–10 days – secondary, when scab/ slough separates**)
- **Earliest sign of bleeding = frequent swallowing** (also clearing throat, restlessness, ↑ pulse, vomiting bright red/ dark blood)
- **Ice collar**, cold fluids, **ice chips/ popsicles**, soft cool foods (ice cream, custard, jelly)
- **Avoid**: **red/brown coloured fluids** (confuse with blood), **citrus juices** (irritate), **hot/spicy/rough foods**, **milk products initially** (thicken mucus – controversial), **straws** (suction → clot dislodge), **gargling, vigorous coughing, clearing throat, nose blowing**
- Analgesics – **paracetamol (NO aspirin/ NSAIDs – bleeding)**
- Throat pain worse around day 4–8; **referred ear pain** common; white membrane/ scab normal; halitosis

- **Peritonsillar abscess (quinsy)** – **trismus, "hot potato voice"**, uvula deviation; drainage.
- **Laryngitis** – voice rest; **Acute epiglottitis** – **H. influenzae b**, children – **drooling, dysphagia, distressed, tripod position, high fever** → **don't examine throat with tongue depressor** (laryngospasm), keep calm, airway equipment ready (emergency intubation/ tracheostomy).

- **Obstructive sleep apnoea** – snoring, daytime sleepiness, obesity → **CPAP**, weight loss.
- **Laryngeal cancer & laryngectomy** – see oncology chapter.
- **Foreign body airway obstruction (choking)** – universal sign (hands at throat); conscious adult – **abdominal thrusts (Heimlich manoeuvre)**; pregnant/ obese – **chest thrusts**; **infant – 5 back blows + 5 chest thrusts** (no abdominal thrusts); unconscious – CPR (look in mouth before breaths; no blind finger sweep).

? Practice MCQs – Chapter 17

1. Cataract ka main symptom –
(a) Painful sudden vision loss (b) Painless gradual blurring (c) Curtain over vision (d) Red eye

2. Cataract surgery ke baad patient ko –
(a) Avoid bending, lifting, straining (b) Rub eyes (c) Bending allowed (d) Lie on operated side

3. Open-angle glaucoma me vision loss –
(a) Sudden total (b) Peripheral (tunnel vision) (c) Central (d) Colour only

4. Glaucoma me kaunsi drug contraindicated hai?
(a) Acetazolamide (b) Atropine (c) Timolol (d) Pilocarpine

5. Retinal detachment ka classic symptom –
(a) Itching (b) Flashes, floaters & curtain over vision (c) Halos (d) Watery discharge

6. Eye me chemical injury ka first action –
(a) Antibiotic drops (b) Patch eye (c) Refer to doctor first (d) Copious irrigation 15–30 min

7. Eye drops kahan daalte hain?
(a) Cornea par (b) Lower conjunctival sac (c) Upper lid (d) Inner canthus directly

8. Night blindness kis vitamin ki kami se?
(a) Vit D (b) Vit A (c) Vit K (d) Vit C

9. Adult me ear drops daalte waqt pinna –
(a) Down & back (b) Straight down (c) Up & back (d) Forward

10. Ménière's disease ka triad –
(a) Vertigo, tinnitus, hearing loss (b) Headache, vomiting, fever (c) Fever, pain, discharge (d) Pain, redness, swelling

11. Ménière's disease me diet –
(a) High sodium (b) Low sodium (c) High protein (d) High fat

12. Epistaxis me position –
(a) Supine flat (b) Sit up, lean forward (c) Trendelenburg (d) Head tilted back

13. Post-tonsillectomy bleeding ka earliest sign –
(a) Fever (b) Hypotension (c) Ear pain (d) Frequent swallowing

14. Ear irrigation ka water temperature –
(a) Body temperature (37 °C) (b) Ice cold (c) Hot (d) Room temperature always

15. Ototoxic drug –
(a) Amoxicillin (b) Insulin (c) Gentamicin (d) Paracetamol

16. Choking infant me first aid –
(a) Water pilana (b) Abdominal thrusts (c) Blind finger sweep (d) 5 back blows + 5 chest thrusts

Answer Key – Chapter 17

1-b · 2-a · 3-b · 4-b · 5-b · 6-d · 7-b · 8-b · 9-c · 10-a · 11-b · 12-b · 13-d · 14-a · 15-c · 16-d

18 Emergency, Critical Care & Disaster Nursing

18.1 Emergency Nursing – Principles

Triage

D Triage

French word **"trier" = to sort**. Patients ko **severity/ urgency** ke hisaab se priority dena – "**greatest good for the greatest number**" (mass casualty).

Colour (START / mass casualty)	Priority	Example
RED – Immediate (Priority I)	Life-threatening but **salvageable** – treat first	**Airway obstruction, tension pneumothorax, shock, severe bleeding**, RR > 30, cap refill > 2 sec, can't follow commands
YELLOW – Delayed/ Urgent (Priority II)	Serious but stable, can wait (hours)	Open fractures, stable abdominal injuries, moderate burns
GREEN – Minor ("walking wounded") (Priority III)	Can wait long	Sprains, minor cuts, minor fractures
BLACK – Expectant/ Dead (Priority IV)	Dead or **unsurvivable** injuries	No breathing after airway opened, massive head injury

- **START triage** (Simple Triage And Rapid Treatment) – **RPM**: **R**espiration (> 30 → red), **P**erfusion (cap refill > 2 sec/ no radial pulse → red), **M**ental status (can't follow simple commands → red); walking → green. Pediatric – **JumpSTART**.
- Hospital ED triage: **ESI (Emergency Severity Index) – 5 levels**; ESI 1 = resuscitation (immediate).

Primary & Secondary survey (ATLS)

- **Primary survey – ABCDE**:
  - **A** – **Airway with cervical spine protection** (jaw thrust)
  - **B** – Breathing & ventilation (tension pneumothorax, flail chest, open pneumothorax)
  - **C** – Circulation & **haemorrhage control** (direct pressure; tourniquet for limb), 2 large-bore IVs
  - **D** – Disability (neuro – **AVPU**: Alert, Voice, Pain, Unresponsive; GCS; pupils)
  - **E** – **Exposure & Environment** (undress fully, then **prevent hypothermia**)
- **Secondary survey** – head-to-toe exam, **SAMPLE history** (**S**igns/symptoms, **A**llergies, **M**edications, **P**ast history, **L**ast meal, **E**vents).
- **Golden hour** – first 60 min after trauma (maximum survival if definitive care). **Platinum 10 minutes** (on-scene).
- **Trauma deaths – trimodal**: (1) seconds–minutes (immediate), (2) **minutes–hours (golden hour – preventable)**, (3) days–weeks (sepsis, MODS).

- **Medico-legal cases (MLC)** – RTA, burns, poisoning, assault, suicide attempt, unknown/unconscious → inform police; **preserve evidence** (clothes in paper bag, gastric lavage sample), document accurately; **treatment first, formalities later** (Supreme Court – Parmanand Katara case 1989).
- **108** – ambulance (GVK EMRI in Gujarat, launched 2007); **112** – national emergency; **1098** – childline; **181** – women helpline (Abhayam in Gujarat).

18.2 Poisoning

General management

1. **ABC** & vital signs first (treat patient, not the poison)
2. **Identify poison** – container, history, smell, symptoms (toxidrome)
3. **Decontamination**:
   - **Skin** – remove clothes, wash with soap & water (organophosphates – staff wear gloves)
   - **Eyes** – irrigate 15–20 min
   - **GI – Activated charcoal 1 g/kg (adult 50 g)** within **1 hr** (not for **acids/alkalis, alcohols, hydrocarbons/kerosene, iron, lithium, heavy metals**)
   - **Gastric lavage** – within 1 hr of life-threatening ingestion; position **left lateral with head low (left lateral, 15–20° head down)**; tube size **36–40 Fr** (adult); **200–300 ml warm water/NS** each cycle; save first sample for **forensic analysis**; **contraindicated in corrosives (acid/alkali), hydrocarbons (kerosene – aspiration pneumonitis), unprotected airway, convulsions**
   - **Induced vomiting (ipecac) – NOT recommended**
4. **Enhance elimination** – urinary alkalinization (salicylates), **haemodialysis** (methanol, ethylene glycol, lithium, salicylate)
5. **Antidote** if available; supportive care; **psychiatric evaluation** (suicidal)

Common poisons & antidotes

Poison / Drug	Antidote
Organophosphorus (OP) insecticides (most common poisoning in rural India), carbamates	**Atropine** (till atropinization – **dry secretions/ clear chest**) + **Pralidoxime (2-PAM)** (OP only; within 24–48 hrs)
Paracetamol (acetaminophen)	**N-acetylcysteine (NAC)** – within 8–10 hrs best (hepatotoxicity)
Opioids (morphine, heroin)	**Naloxone**
Benzodiazepines	**Flumazenil**
Heparin	**Protamine sulphate**
Warfarin	**Vitamin K** (+ FFP / PCC)
Digoxin	**Digibind (Digoxin-specific Fab)**
Iron	**Deferoxamine (Desferrioxamine)**
Lead	**EDTA (Calcium disodium), Dimercaprol (BAL), Succimer (DMSA)**, Penicillamine
Arsenic, Mercury, Gold	**Dimercaprol (BAL)**
Copper (Wilson's)	**Penicillamine**

Poison / Drug	Antidote
Methanol, Ethylene glycol	**Fomepizole / Ethanol** + dialysis
Cyanide	**Hydroxocobalamin**; amyl nitrite + sodium nitrite + sodium thiosulfate
Carbon monoxide (CO)	**100 % O_2 / Hyperbaric O_2**
Atropine / anticholinergics	**Physostigmine**
Beta-blockers	**Glucagon**
Calcium channel blockers	**Calcium gluconate/ chloride**, insulin-glucose
Magnesium sulphate	**Calcium gluconate**
Insulin/ sulfonylurea (hypoglycaemia)	**Dextrose**, glucagon; **Octreotide** (sulfonylurea)
Methotrexate	**Leucovorin (folinic acid)**
Isoniazid (INH)	**Pyridoxine (Vit B6)**
Snake bite (neurotoxic/ haemotoxic)	**Anti-snake venom (ASV – polyvalent)**
Scorpion sting	**Prazosin**, scorpion antivenom
Tricyclic antidepressants	**Sodium bicarbonate**
Neuromuscular blockers (non-depolarizing)	**Neostigmine** (+ glycopyrrolate), Sugammadex (rocuronium)
Malignant hyperthermia	**Dantrolene**
Aluminium phosphide (celphos – wheat pill)	**No specific antidote**; supportive, $MgSO_4$, no water/ lavage with water; $KMnO_4$ (1:10,000) lavage – debated
Paraquat	**No antidote**; **avoid O_2** unless hypoxic (↑ toxicity), Fuller's earth
Kerosene	No lavage, no emesis; O_2, observe for pneumonitis

Organophosphorus poisoning – DUMBELS/ SLUDGE

Muscarinic: **D**iarrhoea, **U**rination, **M**iosis (**pin-point pupils**), **B**radycardia/ **B**ronchorrhoea/ **B**ronchospasm, **E**mesis, **L**acrimation, **S**alivation/ **S**weating – plus **garlic/kerosene smell**.
Nicotinic: muscle fasciculations, weakness, paralysis (respiratory failure), tachycardia, HTN.
Rx: decontamination (remove clothes, wash skin), airway & O_2, **Atropine 1.8–3 mg IV bolus, double dose every 5 min until atropinization** (clear chest on auscultation, HR > 80, SBP > 80, dry axillae, pupils no longer pin-point), then infusion; **Pralidoxime**; watch **intermediate syndrome** (24–96 hrs – proximal muscle weakness, respiratory failure).

- **Corrosive (acid/alkali)** – burns of mouth, drooling, dysphagia, stridor → **no emesis, no lavage, no neutralization, no charcoal**; small sips of water/milk only if no perforation & within minutes (controversial); endoscopy within 12–24 hrs; NPO, airway.
- **Alcohol (ethanol) intoxication** – hypoglycaemia, aspiration risk; thiamine before glucose (Wernicke's).
- **Datura** (anticholinergic) – "**dry as a bone, red as a beet, blind as a bat, hot as a hare, mad as a hatter**" → physostigmine.

18.3 Bites & Stings

Snake bite

- India – **"Big Four"**: **Common cobra, Common krait** (neurotoxic), **Russell's viper, Saw-scaled viper** (haemotoxic/ vasculotoxic).
- **Neurotoxic (cobra, krait)**: **ptosis (first sign)**, diplopia, dysphagia, slurred speech, descending paralysis, **respiratory failure**; krait – painless bite at night, abdominal pain.
- **Haemotoxic (vipers)**: local pain, swelling, blistering, **bleeding (gums, bite site), haematuria**, **20WBCT (20-minute whole blood clotting test) – blood doesn't clot = coagulopathy → give ASV**, AKI (Russell's viper).

Snake bite first aid – "Do It R.I.G.H.T."

R – **Reassure** the patient (70 % bites non-venomous; fear ↑ heart rate)
I – **Immobilize** the bitten limb like a fracture (splint), keep at/below heart level; patient lies still
G.H. – **Get to Hospital** immediately (108)
T – **Tell** the doctor about symptoms
DON'T: **tourniquet/ tight bandage**, **cut, suck**, ice, electric shock, herbal/ traditional remedies, alcohol, painkillers like aspirin; remove rings/ tight items (swelling).
ASV – **polyvalent (covers Big Four)**, given **IV** (10 vials initial – 100 ml; no test dose as per national protocol) **only if signs of envenomation**; watch **anaphylaxis (keep adrenaline ready)**; neurotoxic – **neostigmine + atropine**, ventilation.

- **Scorpion sting** – red scorpion (Mesobuthus tamulus) – **autonomic storm**: severe pain, sweating, salivation, priapism, **hypertension, pulmonary oedema, myocarditis** → **Prazosin** (drug of choice), antivenom, local anaesthetic infiltration.
- **Dog bite** – see Rabies (Ch 15).
- **Bee sting** – remove sting by **scraping** (don't squeeze), ice; anaphylaxis watch.

18.4 Environmental Emergencies

Condition	Features	Management
Heat cramps	Painful muscle spasms, sweating	Rest in cool place, **oral salt-water/ ORS**
Heat exhaustion	Heavy sweating, weakness, dizziness, nausea, headache, temp **< 40 °C**, normal mental status, **cool clammy skin**	Cool place, lie down, legs up, remove extra clothes, **oral fluids/ ORS**, IV NS
Heat stroke – EMERGENCY	**Core temp > 40 °C (104 °F)**, **hot, dry, flushed skin (sweating may stop)**, **CNS dysfunction – confusion, seizures, coma**, tachycardia, hypotension	**Rapid cooling** (cold water immersion – exertional; **evaporative cooling – spray tepid water + fan**; ice packs to **neck, axilla, groin**), cool IV fluids; **stop cooling at ~38.5–39 °C** (overshoot); **antipyretics (paracetamol) NOT effective**; watch rhabdomyolysis, DIC, AKI
Hypothermia	Core temp **< 35 °C**; shivering (stops < 32 °C), confusion, bradycardia, arrhythmias (VF)	**Remove wet clothes**, **passive rewarming (blankets)** mild; **active external (warm air)**, **active internal (warm IV fluids 40–42 °C, warm humidified O_2)** severe; handle gently (VF); "**not dead until warm and dead**"

Condition	Features	Management
Frostbite	Freezing of tissue – white, hard, numb	**Rapid rewarming in warm water (37–40 °C) 15–30 min**; **don't rub/ massage**, no dry heat; analgesia (very painful); elevate; don't rupture blisters
Drowning	Hypoxia – main problem	Remove from water safely, **start with 5 rescue breaths** then CPR (A-B-C emphasis), cervical spine care only if diving injury, O_2, **all near-drowning admit for observation 24 hrs** (secondary drowning – pulmonary oedema), warm
Electric shock	Entry/exit burns, arrhythmias (VF – AC current), muscle damage, fractures	**Switch off power**, CPR, ECG monitor, fluids (myoglobinuria), C-spine
Lightning	Asystole/ respiratory arrest	**Reverse triage** – treat "apparently dead" first (they may recover with CPR)

18.5 Critical Care (ICU) Nursing

ICU basics

- **Nurse : patient ratio – 1 : 1** (ventilated/ critical), 1 : 2 (step-down).
- **Glasgow coma scale**, **RASS** (Richmond Agitation-Sedation Scale, −5 to +4; target 0 to −2), **CAM-ICU** (delirium), **APACHE II** (severity), **SOFA** (organ failure).
- **Invasive monitoring**: **Arterial line** (radial – **Allen's test**; continuous BP & ABG; transducer at **phlebostatic axis**), **CVP**, pulmonary artery (Swan-Ganz) catheter – **PCWP (pulmonary capillary wedge pressure) normal 6–12 mmHg** (↑ in LV failure).
- **ICU psychosis/ delirium** – sensory overload/ deprivation, sleep deprivation; orientation, clock, day-night cycle, family, minimize noise.
- **Sepsis** – life-threatening organ dysfunction due to dysregulated host response to infection; **qSOFA (≥ 2)**: **RR ≥ 22, altered mentation (GCS < 15), SBP ≤ 100**. **Septic shock** – vasopressors needed to keep **MAP ≥ 65** + **lactate > 2 mmol/L** despite fluids.
- **Hour-1 sepsis bundle**: measure **lactate**, **blood cultures before antibiotics**, **broad-spectrum antibiotics**, **30 ml/kg crystalloid** for hypotension or lactate ≥ 4, **vasopressors (noradrenaline)** to keep MAP ≥ 65.
- **MODS** – multiple organ dysfunction syndrome.

Mechanical ventilation

Term	Meaning
Tidal volume (VT)	**6–8 ml/kg ideal body weight** (lung-protective; ARDS 6 ml/kg)
RR (f)	12–20 breaths/min
FiO_2	21–100 %; keep lowest to maintain SpO_2 92–96 % (avoid O_2 toxicity > 60 % long)
PEEP	**Positive End-Expiratory Pressure** – **5–10 cmH_2O** (alveoli open, oxygenation ↑); high PEEP → **↓ venous return → hypotension**, barotrauma
I : E ratio	Normally **1 : 2**

Term	Meaning
Modes	**Controlled (CMV)**, **Assist-Control (AC)**, **SIMV** (weaning), **Pressure support (PSV)**, **CPAP** (spontaneous, weaning), **BiPAP** (non-invasive – COPD exacerbation)

Ventilator alarms – causes

HIGH pressure alarm ("obstruction")	LOW pressure/ volume alarm ("leak/ disconnection")
Secretions (suction), coughing, **patient biting tube**, kinked tubing, water in tubing, bronchospasm, **patient fighting ventilator**, pneumothorax, ↓ compliance (ARDS, pulmonary oedema)	**Disconnection (most common)**, **cuff leak** (low cuff pressure), extubation/ tube displaced, leak in circuit/ chest tube
Rule: if cause can't be found quickly → **disconnect patient & manually ventilate with Ambu bag (100 % O_2)** & call for help. **Never turn off alarms.**	

VAP prevention bundle

HOB 30–45° · **daily sedation interruption ("sedation vacation") & spontaneous breathing trial** · **oral care with chlorhexidine (0.12 %)** · **subglottic secretion drainage** · **cuff pressure 20–30 cmH_2O** · DVT prophylaxis · **peptic ulcer (stress ulcer) prophylaxis** · hand hygiene · avoid unnecessary circuit changes · drain condensate away from patient.

- **ET tube**: placement check – **end-tidal CO_2 (capnography – most reliable)**, bilateral chest rise & breath sounds, **chest X-ray (tip 2–4 cm above carina)**; note **cm mark at lips/ teeth** (21 cm women, 23 cm men approx.); secure; prevent self-extubation.
- **Weaning criteria**: underlying cause resolved, **FiO_2 ≤ 40–50 %, PEEP ≤ 5–8**, SpO_2 > 90 %, hemodynamically stable, alert, **RSBI (RR/VT in L) < 105**, adequate cough; trials of **T-piece/ CPAP/ PSV**. Extubate – **high Fowler's**, suction, deflate cuff, remove at peak inspiration, O_2, watch **stridor** (laryngeal oedema).

Shock, cardiac & other ICU drugs

Drug	Use	Point
Noradrenaline	Septic shock (first-line vasopressor)	**Central line** preferred; extravasation → **phentolamine**
Dopamine	Low dose renal (old), β (cardiac), α (vasopressor)	Extravasation – phentolamine
Dobutamine	Cardiogenic shock, HF (inotrope)	Tachycardia
Adrenaline	Cardiac arrest, anaphylaxis	
Atropine	Symptomatic bradycardia (1 mg IV every 3–5 min, max 3 mg), OP poisoning	
Amiodarone	VF/pVT, AF	Pulmonary fibrosis, thyroid, hypotension
Adenosine	**SVT** (6 mg rapid IV push + flush, then 12 mg)	Brief asystole – warn patient; **rapid push near heart (antecubital)**

Drug	Use	Point
Sodium bicarbonate	Severe acidosis, hyperkalaemia, TCA overdose	Don't mix with calcium
Mannitol	↑ ICP	Filter needle
Nitroglycerin IV	MI, hypertensive emergency	Glass bottle/ non-PVC tubing

18.6 Disaster Nursing

D Disaster (WHO)

"Any occurrence that causes **damage, ecological disruption, loss of human life, deterioration of health & health services on a scale sufficient to warrant an extraordinary response** from outside the affected community or area."

- Types: **Natural** (earthquake – **Bhuj, Gujarat 26 Jan 2001**, cyclone, flood, drought, tsunami 2004, pandemic) & **Man-made** (**Bhopal gas tragedy 2–3 Dec 1984 – Methyl isocyanate (MIC)**, industrial, war, terrorism, fire, RTA, stampede).
- **Disaster cycle**: **Pre-disaster** (prevention/ **mitigation**, **preparedness**) → **Impact/ Response** (rescue, triage, first aid, evacuation) → **Post-disaster** (**recovery – rehabilitation & reconstruction**).
- **Disaster Management Act 2005**; **NDMA** (Chairman – **Prime Minister**); **SDMA** (Chief Minister); **DDMA** (District Collector); **NDRF** (National Disaster Response Force, 2006). **GSDMA** – Gujarat State Disaster Management Authority (2001, after Bhuj earthquake).
- **International Day for Disaster Risk Reduction – 13 October**.
- **Post-disaster health problems**: injuries, **communicable diseases (water-borne – cholera, typhoid, hepatitis A/E; vector-borne – malaria, dengue; measles in camps)**, malnutrition, **PTSD** & mental health issues; dead body management (identification, respectful handling – **dead bodies of disaster victims rarely cause epidemics**).
- Nurse's role: triage, first aid, emergency care, **shelter health (water chlorination, sanitation, immunization – measles, vitamin A)**, psychological first aid, record keeping, health education, coordination.
- **Mass casualty – hospital disaster plan**, incident command system, **Code Red (fire)**, **Code Blue (cardiac arrest)**, Code Pink (infant abduction).
- **Fire safety – RACE**: **R**escue → **A**larm → **C**ontain (close doors) → **E**xtinguish/ Evacuate. **Fire extinguisher – PASS**: **P**ull pin → **A**im at base of fire → **S**queeze handle → **S**weep side to side. Fire classes: **A** – ordinary combustibles (water), **B** – flammable liquids (foam/ CO_2/ dry powder), **C** – electrical (CO_2/ dry powder – **never water**), D – metals.

? Practice MCQs – Chapter 18

1. START triage me "red tag" ka matlab –
   (a) Dead (b) Delayed (c) Minor (d) Immediate treatment
2. Mass casualty me black tag –
   (a) Immediate (b) Minor (c) Urgent (d) Dead / expectant
3. Trauma primary survey me sabse pehle –
   (a) Circulation (b) Disability (c) Breathing (d) Airway with C-spine protection

4. OP poisoning ka antidote –
(a) Flumazenil (b) Naloxone (c) Vitamin K (d) Atropine + Pralidoxime

5. Paracetamol overdose ka antidote –
(a) Deferoxamine (b) BAL (c) Protamine (d) N-acetylcysteine

6. Iron poisoning antidote –
(a) Penicillamine (b) EDTA (c) Atropine (d) Deferoxamine

7. Gastric lavage kis poisoning me contraindicated hai?
(a) Kerosene / corrosives (b) OP (c) Paracetamol (d) Barbiturates

8. Snake bite first aid me kya NAHI karna chahiye?
(a) Reassure (b) Hospital le jana (c) Tight tourniquet & cutting (d) Immobilize

9. Neurotoxic snake bite ka first sign –
(a) Haematuria (b) Bleeding (c) Ptosis (d) Swelling

10. Scorpion sting ka drug of choice –
(a) Prazosin (b) Atropine (c) Adrenaline (d) Steroid

11. Heat stroke ka characteristic –
(a) Temp > 40 °C with altered mental status (b) Temp < 38 °C (c) Shivering (d) Only cramps

12. Ventilator high pressure alarm ka common cause –
(a) Extubation (b) Secretions / patient biting tube (c) Disconnection (d) Cuff leak

13. VAP prevention me head of bed –
(a) 30–45° (b) 0° (c) 90° (d) Trendelenburg

14. Normal tidal volume setting (lung protective) –
(a) 2–4 ml/kg (b) 6–8 ml/kg (c) 12–15 ml/kg (d) 20 ml/kg

15. Fire safety ka sequence –
(a) PASS (b) ABCD (c) SAMPLE (d) RACE

16. Bhopal gas tragedy me kaunsi gas leak hui thi?
(a) Methyl isocyanate (b) Ammonia (c) CO (d) Chlorine

17. Benzodiazepine overdose ka antidote –
(a) Flumazenil (b) Physostigmine (c) Naloxone (d) Glucagon

18. Electrical fire par kya use nahi karte?
(a) Switch off mains (b) CO_2 extinguisher (c) Dry powder (d) Water

Answer Key – Chapter 18

1-d · 2-d · 3-d · 4-d · 5-d · 6-d · 7-a · 8-c · 9-c · 10-a · 11-a · 12-b · 13-a · 14-b · 15-d · 16-a · 17-a · 18-d

19 Geriatric Nursing & Quick Revision

19.1 Geriatric Nursing

- **Geriatrics** = medical care of elderly; **Gerontology** = study of ageing. India – elderly = **60+ yrs** (MWPSC Act 2007 – Maintenance and Welfare of Parents & Senior Citizens). **NPHCE (2010–11)** – National Programme for Health Care of the Elderly. **1 October = International Day of Older Persons**.

System	Normal ageing changes	Nursing implication
Cardiovascular	↓ CO, stiff arteries (↑ SBP), ↓ baroreceptor response	**Orthostatic hypotension** – rise slowly
Respiratory	↓ vital capacity, ↓ cough reflex, ↑ residual volume	Pneumonia risk – deep breathing, vaccines
GI	↓ saliva, ↓ taste, ↓ peristalsis	**Constipation** – fibre, fluids
Renal	↓ GFR, ↓ bladder capacity	**Nocturia**, drug toxicity – dose adjust
Musculoskeletal	↓ bone mass, ↓ muscle mass (**sarcopenia**)	**Falls & fractures**
Neuro/ Sensory	Slower reaction, **presbyopia, presbycusis**, ↓ pain & temperature sensation	Safety, burns risk, good lighting
Skin	Thin, dry, ↓ subcutaneous fat, ↓ sweat	Pressure ulcers, **hypothermia**; less frequent bathing, moisturize
Immune	↓ immunity; **fever may be absent** in infection	**Confusion may be first sign of infection** (UTI, pneumonia)

- **Polypharmacy** – ≥ 5 drugs → adverse effects; **Beers criteria** (potentially inappropriate drugs in elderly); "**start low, go slow**".
- **Falls** – most common cause of injury/ accidental death in elderly → **Morse Fall Scale**; prevention: **non-slip footwear, good lighting/ night light, remove rugs/ clutter, grab bars, bed in lowest position, call bell within reach**, regular toileting, review sedatives, hip protectors, exercise (tai chi).
- **Elder abuse** – physical, psychological, financial, neglect (most common).

Delirium vs Dementia vs Depression

	Delirium	Dementia	Depression
Onset	**Acute/ sudden (hours–days)**	**Insidious, gradual (months–years)**	Recent, may be sudden
Course	**Fluctuating, worse at night**	Progressive, stable day-to-day	Diurnal (worse morning)
Consciousness/ attention	**Altered/ ↓ attention**	Normal till late	Normal
Reversible?	**Usually reversible** (treat cause – infection, drugs, hypoxia, dehydration, electrolyte)	**Irreversible** (Alzheimer's)	Treatable

	Delirium	Dementia	Depression
Assessment tool	**CAM** (Confusion Assessment Method)	**MMSE** (Mini-Mental State – 30; < 24 impaired)	GDS (Geriatric Depression Scale)

19.2 Master Table – Normal Lab Values

Test	Normal value
Hb – Male / Female	13–17 / 12–15 g/dL
WBC	4,000–11,000 /mm^3
Platelets	1.5–4.5 lakh /mm^3
ESR – M / F	0–15 / 0–20 mm/hr
Fasting blood sugar	70–100 mg/dL
PPBS (2 hr)	< 140 mg/dL
HbA1c	< 5.7 % (diabetic target < 7 %)
Sodium	135–145 mEq/L
Potassium	3.5–5.0 mEq/L
Chloride	96–106 mEq/L
Calcium	8.5–10.5 mg/dL
Magnesium	1.5–2.5 mEq/L
Phosphate	2.5–4.5 mg/dL
Blood urea	15–40 mg/dL
BUN	7–20 mg/dL
Serum creatinine	0.6–1.2 mg/dL
Uric acid	3.5–7 mg/dL
Total bilirubin	0.3–1.2 mg/dL
SGPT / SGOT	7–56 / 10–40 U/L
Albumin	3.5–5.0 g/dL
Total cholesterol	< 200 mg/dL
LDL / HDL / TG	< 100 / > 40 (M), > 50 (F) / < 150 mg/dL
Serum amylase / lipase	30–110 / 0–160 U/L
PT / INR	11–13.5 s / 0.8–1.2
aPTT	25–40 s
ABG: pH / $PaCO_2$ / HCO_3^- / PaO_2	7.35–7.45 / 35–45 / 22–26 / 80–100
SpO_2	95–100 %
Serum osmolality	275–295 mOsm/kg
Urine output	0.5–1 ml/kg/hr (≥ 30 ml/hr)
Urine specific gravity	1.005–1.030

Test	Normal value
CSF pressure	60–180 mmH_2O
ICP	5–15 mmHg
CVP	2–8 mmHg (3–8 cmH_2O)
MAP	70–100 mmHg
IOP	10–21 mmHg
CD4 count	500–1500 /mm^3
TSH	0.4–4.0 mIU/L
Digoxin level	0.5–2.0 ng/ml
Lithium level	0.6–1.2 mEq/L
Phenytoin level	10–20 µg/ml
Theophylline level	10–20 µg/ml
Troponin I	< 0.04 ng/ml
BNP	< 100 pg/ml
PSA	< 4 ng/ml

19.3 Master Table – Diseases & Diagnostic Tests / Signs

Disease / Condition	Test / Sign
MI	**Troponin**, ST elevation, Q wave
Heart failure	**BNP**
DVT	**Doppler USG**, D-dimer
Pulmonary embolism	**CT pulmonary angiography**
TB	**CBNAAT**, sputum AFB, Mantoux
Diabetes (long term control)	**HbA1c**
Pancreatitis	**Serum lipase**, Cullen's & Grey Turner's sign
Appendicitis	**McBurney's**, Rovsing's, Psoas, Obturator
Cholecystitis	**Murphy's sign**, USG
Meningitis	**Kernig's, Brudzinski's**, CSF
Hypocalcaemia	**Chvostek's, Trousseau's**
Basilar skull fracture	**Battle's sign, Raccoon eyes**, halo sign
Cardiac tamponade	**Beck's triad**, pulsus paradoxus
↑ ICP (late)	**Cushing's triad**
Myasthenia gravis	**Tensilon (edrophonium) test**
Carpal tunnel	**Phalen's, Tinel's**
Psoriasis	**Auspitz sign**, Koebner

Disease / Condition	Test / Sign
Pemphigus	**Nikolsky sign**
Hip dysplasia (child)	Ortolani, Barlow
Varicose veins	**Trendelenburg test**
Ulnar artery patency	**Allen's test**
Hearing	**Rinne, Weber**
Glaucoma	**Tonometry**
Colour blindness	**Ishihara chart**
Typhoid	**Blood culture (1st wk), Widal**
Malaria	**Peripheral smear**, RDT
Dengue	**NS1 antigen**, IgM, tourniquet test
HIV infant	**DNA PCR**
Sickle cell	**Sickling test, Hb electrophoresis**
Thalassaemia	**Hb electrophoresis (↑ HbF)**
Rheumatic fever	**ASO titre**, Jones criteria
RA	**Anti-CCP**, RA factor
SLE	**ANA, anti-dsDNA**
Pheochromocytoma	**Urine VMA/ metanephrines**
Cushing's	Dexamethasone suppression test
Osteoporosis	**DEXA scan**
Bladder / urinary retention	Bladder scan (PVR)
Cystic fibrosis	Sweat chloride test

19.4 Master Table – Important Drugs & Key Side Effects

Drug	Key side effect / Nursing point
Digoxin	Check apical pulse (< 60 hold), **yellow-green halos**, hypokalaemia ↑ toxicity
Furosemide	**Hypokalaemia**, ototoxicity, give in morning
Spironolactone	**Hyperkalaemia**, gynaecomastia
ACE inhibitors	**Dry cough**, angioedema, hyperkalaemia
Beta-blockers	Bradycardia, bronchospasm, mask hypoglycaemia
CCB (amlodipine)	Ankle oedema, headache
Nitroglycerin	**Headache**, hypotension; amber bottle
Heparin	Bleeding; aPTT; antidote **protamine**
Warfarin	Bleeding; PT/INR; antidote **Vit K**
Aspirin	GI bleed, **Reye's syndrome in children**, tinnitus

Drug	Key side effect / Nursing point
Insulin	**Hypoglycaemia**, lipodystrophy
Metformin	**Lactic acidosis**, stop before contrast
Sulfonylurea	Hypoglycaemia
Levothyroxine	Empty stomach morning; tachycardia (overdose)
Methimazole/ PTU	**Agranulocytosis** (sore throat, fever)
Steroids (prednisolone)	Hyperglycaemia, infection, osteoporosis, Cushingoid; **don't stop suddenly**
Isoniazid	Peripheral neuropathy (**give B6**), hepatitis
Rifampicin	**Orange-red body fluids**, hepatotoxic, ↓ OCP
Ethambutol	**Optic neuritis**
Pyrazinamide	Hyperuricaemia
Streptomycin/ Gentamicin	**Ototoxicity, nephrotoxicity**
Phenytoin	**Gingival hyperplasia**, hirsutism, ataxia; IV with NS only
Valproate	Hepatotoxicity, pancreatitis, teratogenic
Carbamazepine	Agranulocytosis, SJS
Levodopa	Dyskinesia, orthostatic hypotension; avoid B6
Morphine/ opioids	**Respiratory depression**, constipation; **naloxone**
Paracetamol	Hepatotoxicity (overdose); **NAC**
NSAIDs	GI ulcer, renal damage, bleeding
Allopurinol	Rash; fluids 2–3 L
Colchicine	Diarrhoea
Methotrexate	Bone marrow suppression, hepatotoxic; **leucovorin**; folic acid
Cyclophosphamide	**Haemorrhagic cystitis** (mesna, fluids)
Doxorubicin	**Cardiotoxicity**, red urine
Cisplatin	**Nephrotoxicity, ototoxicity**
Vincristine	**Neuropathy**, constipation
Bleomycin	**Pulmonary fibrosis**
Tamoxifen	Hot flashes, endometrial cancer, DVT
Amiodarone	Pulmonary fibrosis, thyroid dysfunction, photosensitivity, blue-grey skin
Lithium	Narrow range 0.6–1.2; tremors; maintain Na & fluids
Chloramphenicol	**Aplastic anaemia**, **gray baby syndrome**
Tetracycline	**Tooth discoloration** (avoid < 8 yrs & pregnancy), photosensitivity; not with milk/antacids
Fluoroquinolones	Tendon rupture; avoid in children
Vancomycin	**Red man syndrome** (infuse slowly over ≥ 60 min), nephro/ototoxic
Metronidazole	**Disulfiram-like reaction with alcohol**, metallic taste

Drug	Key side effect / Nursing point
Iron	Black stools, constipation; Z-track IM
Potassium chloride IV	**Never IV push**; diluted, ≤ 10 mEq/hr
Mannitol	Use filter; fluid & electrolyte imbalance
Magnesium sulphate	Loss of knee jerk, resp. depression; **calcium gluconate**
Oxytocin	Water intoxication, uterine hyperstimulation
Atropine	Dry mouth, tachycardia, urinary retention, mydriasis
Neostigmine	SLUDGE effects
Ondansetron	Headache, QT prolongation
Sucralfate	Before meals, separate from other drugs
Alendronate	Upright 30 min, full glass of water

19.5 Important Days (Health) – Exam Favourite

Date	Day
30 January	Anti-Leprosy Day (Gandhiji's death anniversary)
4 February	World Cancer Day
2nd Thursday of March	World Kidney Day
24 March	**World TB Day** (Koch 1882)
7 April	**World Health Day** (WHO founded 1948)
25 April	**World Malaria Day**
8 May	World Red Cross Day & **World Thalassaemia Day**
12 May	**International Nurses Day** (Florence Nightingale)
16 May	National Dengue Day
17 May	World Hypertension Day
31 May	**World No Tobacco Day**
14 June	**World Blood Donor Day**
21 June	International Yoga Day
1 July	**National Doctors' Day** (Dr. B.C. Roy)
11 July	World Population Day
28 July	**World Hepatitis Day**
1–7 August	**World Breastfeeding Week**
1–7 September	National Nutrition Week
10 September	World Suicide Prevention Day
28 September	**World Rabies Day**
29 September	**World Heart Day**

Date	Day
1 October	International Day of Older Persons; National Voluntary Blood Donation Day
10 October	**World Mental Health Day**
2nd Thursday of October	World Sight Day
12 October	World Arthritis Day
21 October	Global Iodine Deficiency Disorders Day
7 November	National Cancer Awareness Day
14 November	**World Diabetes Day** (Banting's birthday)
1 December	**World AIDS Day**
3 December	International Day of Persons with Disabilities

19.6 Top 100 One-Liners (MSN) – Revision before exam

One-liners 1–25

1. Normal adult pulse – **60–100/min**; RR – **12–20/min**.
2. Pain is the **5th vital sign**.
3. Nursing process – **ADPIE**; nursing diagnosis – **NANDA**.
4. Abdominal assessment order – **Inspection, Auscultation, Percussion, Palpation**.
5. Most accurate indicator of fluid balance – **daily weight** (1 kg = 1 L).
6. Main ICF cation – **K^+**; main ECF cation – **Na^+**.
7. Fluid compatible with blood – **0.9 % NS**.
8. IV KCl – **never IV push**.
9. Tall peaked T wave – **hyperkalaemia**; U wave – **hypokalaemia**.
10. Trousseau & Chvostek – **hypocalcaemia**.
11. Antidote of $MgSO_4$ – **10 % calcium gluconate**.
12. ROME – Respiratory Opposite, Metabolic Equal.
13. Vomiting → **metabolic alkalosis**; diarrhoea → **metabolic acidosis**.
14. Kussmaul breathing – **DKA**.
15. Neurogenic shock – **hypotension + bradycardia**.
16. Anaphylaxis drug of choice – **adrenaline IM 1:1000**.
17. Shock position – **modified Trendelenburg (legs raised)**.
18. Opioid antidote – **naloxone**.
19. Most effective infection control – **hand hygiene**.
20. Airborne precautions – **TB, measles, chickenpox (N95, negative pressure)**.
21. Sharps – **white puncture-proof container**; glass – **blue**.
22. Autoclave – **121 °C, 15 psi, 15–20 min**.
23. Wound evisceration – **cover with sterile saline-soaked dressing**.
24. Pre-op hair removal – **clipper, just before surgery**.
25. Post-spinal headache – **flat position**, fluids.

One-liners 26–50

26. Malignant hyperthermia – **dantrolene**.
27. Post-op fever 24–48 hrs – **atelectasis**.
28. Most common cause of MI – **coronary thrombosis/ atherosclerosis**; artery – **LAD**.
29. Most specific MI marker – **troponin**.
30. MONA order of giving – **O_2, Aspirin, Nitrate, Morphine**.
31. NTG – **max 3 tablets, 5 min apart**.
32. Digoxin toxicity – **yellow-green vision**, nausea; antidote **Digibind**.
33. Left HF – **lungs (crackles, orthopnoea)**; Right HF – **body (oedema, JVD)**.
34. ACE inhibitor side effect – **dry cough**.
35. Heparin – **aPTT, protamine**; Warfarin – **PT/INR, Vit K**.
36. Pericarditis pain relieved by – **sitting up & leaning forward**.
37. Beck's triad – **cardiac tamponade**.
38. Arterial ulcer – legs **dependent**; venous ulcer – legs **elevated**.
39. Adult CPR – **30:2, 100–120/min, 5–6 cm**.
40. Shockable rhythms – **VF, pulseless VT**.
41. COPD O_2 – **low flow 1–2 L / Venturi**, SpO_2 88–92 %.
42. Pink puffer – **emphysema**; blue bloater – **chronic bronchitis**.
43. Asthma rescue drug – **salbutamol**.
44. Mantoux read at – **48–72 hrs**; positive **≥ 10 mm**.
45. Rifampicin – **orange urine**; INH – **B6**; Ethambutol – **eye**.
46. Tension pneumothorax – **trachea shifts to opposite side**; needle at **2nd ICS MCL**.
47. Chest tube continuous bubbling in water seal – **air leak**.
48. Tracheal suction – **max 10–15 sec**, hyperoxygenate.
49. Duodenal ulcer pain – **relieved by food**; gastric – **worsened by food**.
50. Most common cause of peptic ulcer – **H. pylori**.

One-liners 51–75

51. Appendicitis pain point – **McBurney's**; Rovsing's – LLQ pressure → RLQ pain.
52. Suspected appendicitis – **no heat, no enema/laxative**.
53. Dumping syndrome – **no fluids with meals, lie down after meals, low carb**.
54. NG tube placement – **X-ray (gold standard)**, pH ≤ 5.
55. Healthy stoma – **beefy red, moist**.
56. Colostomy irrigation height – **45–50 cm**.
57. Hepatitis A & E – **faeco-oral**; B, C, D – **blood**.
58. Hepatic encephalopathy – **lactulose**; sign – **asterixis**.
59. Varices bleeding – **band ligation**, octreotide, SB tube.
60. Cholecystitis – **Murphy's sign**; pain to **right shoulder**.
61. Pancreatitis – **lipase**; position – **sitting leaning forward/ fetal**.
62. Most common cause of CKD – **diabetes**.
63. AV fistula patency – **thrill & bruit**; no BP on that arm.
64. PD complication – **peritonitis (cloudy effluent)**.
65. HD complication – **hypotension**; disequilibrium syndrome.
66. Calcium oxalate stones – avoid **spinach, tomato, chocolate, tea**.
67. Nephrotic syndrome – **massive proteinuria, hypoalbuminaemia, oedema, hyperlipidaemia**.
68. AGN – **cola-coloured urine, HTN, periorbital oedema**.
69. TURP – **CBI**, TUR syndrome = **hyponatraemia**.
70. Foley balloon – **sterile water**.
71. Diabetes FBS – **≥ 126 mg/dL**; HbA1c – **≥ 6.5 %**.
72. Only IV insulin – **regular**; mix – **clear before cloudy**.
73. Hypoglycaemia – **rule of 15**; unconscious – **D50 / glucagon**.
74. DKA first – **IV 0.9 % NS**, then insulin; watch **K^+**.
75. Somogyi – **night hypo → morning hyper**.

One-liners 76–100

76. Graves' – **exophthalmos**; drug – methimazole/ PTU.
77. Post-thyroidectomy bedside – **tracheostomy set, calcium gluconate**.
78. Addison's – **bronze skin, ↓ Na, ↑ K, hypotension**.
79. Cushing's – **moon face, buffalo hump, striae, ↑ sugar**.
80. DI – **polyuria, ↑ Na, desmopressin**; SIADH – **↓ Na, fluid restriction**.
81. GCS – **3–15**; ≤ 8 – **intubate**.
82. Earliest sign of ↑ ICP – **change in LOC**; late – **Cushing's triad**.
83. ↑ ICP position – **HOB 30°, neck neutral**.
84. Epidural haematoma – **lucid interval** (middle meningeal artery).
85. Stroke thrombolysis – **within 4.5 hrs** after CT rules out bleed.
86. Seizure – **never put anything in mouth**; status – **IV lorazepam**.
87. Phenytoin – **gingival hyperplasia**.
88. Autonomic dysreflexia – **sit up, check bladder**.
89. Parkinson's – **↓ dopamine, pill-rolling tremor**; MG – **ptosis, pyridostigmine**.
90. Compartment syndrome – **pain on passive stretch**, limb at heart level, fasciotomy.
91. Fat embolism – **long bone fracture, petechiae, hypoxia**.
92. THR – **no hip flexion > 90°, no crossing legs, abduction pillow**.
93. Crutches – weight on **hands**; stairs **up with good, down with bad**.
94. Cane – **opposite (strong) side**.
95. Parkland – **4 ml × kg × %TBSA (RL)**, half in **first 8 hrs**.
96. Burns – adequacy of fluid – **urine output 0.5 ml/kg/hr**.
97. Transfusion reaction – **stop transfusion first**, keep line open with NS.
98. Platelets storage – **20–24 °C with agitation**, 5 days.
99. Glaucoma – **atropine contraindicated**; open-angle – **tunnel vision**.
100. Epistaxis – **sit up, lean forward, pinch nose 10–15 min**.



Exam strategy – Asha Didi ki last advice

- **GSSSB Staff Nurse** paper me Nursing subject ka sabse bada hissa **MSN + Fundamentals** se aata hai – inhe roz revise karo.
- Har question me socho: **Patient safety → ABC → Assessment pehle → Least invasive action pehle**.
- "**First / priority / immediate**" word dekho – wahi question ki chaabi hai.
- Normal values aur antidotes ki tables **5 baar** likh ke yaad karo.
- Har chapter ke MCQs time laga ke karo (**1 question = 36–45 sec**), galti wale questions alag notebook me likho.

20 MSN Grand Test – 60 Mixed MCQs



Test kaise do?

60 questions – 45 minutes. Timer lagao, notes band karo, OMR ki tarah answers alag kagaz par likho. Phir answer key se check karo. **GSSSB pattern me negative marking (0.25)** ho sakti hai – guess soch-samajh ke karo.

Score guide: **50+ Excellent · 40–49 Good · 30–39 Revise weak chapters · < 30 Dobara padho**

? Grand Test – Part A (1–30)

1. Nursing diagnosis ka format hai –
(a) SOAP (b) PES (c) SBAR (d) ADPIE

2. Blood transfusion ke liye preferred cannula size (adult) –
(a) 24 G (b) 26 G (c) 18–20 G (d) 14 G only

3. Hypokalaemia ka ECG sign –
(a) Wide QRS (b) U wave (c) Peaked T (d) ST elevation

4. pH 7.25, $PaCO_2$ 40, HCO_3^- 16 –
(a) Metabolic alkalosis (b) Normal (c) Respiratory acidosis (d) Metabolic acidosis

5. Septic shock ka first-line vasopressor –
(a) Dobutamine (b) Dopamine (c) Vasopressin (d) Noradrenaline

6. Post-op urinary retention – patient ko first void kitne ghante me karna chahiye?
(a) 2 hrs (b) 6–8 hrs (c) 24 hrs (d) 48 hrs

7. WHO Surgical Safety Checklist me "Sign Out" kab?
(a) In ward (b) Before induction (c) Before incision (d) Before patient leaves OT

8. Angina aur MI me main difference –
(a) Pain location (b) MI me myocardial necrosis hota hai (c) Koi farak nahi (d) Angina me ST elevation permanent

9. Thrombolytic therapy ka main risk –
(a) Hypoglycaemia (b) Constipation (c) Bleeding (d) Hypertension

10. Orthopnoea kis condition ka sign hai?
(a) Right HF (b) Anaemia (c) Hypothyroidism (d) Left HF

11. Mitral stenosis ka common cause –
(a) MI (b) Hypertension (c) Syphilis (d) Rheumatic heart disease

12. Warfarin patient ko kaunsa food consistent quantity me lena chahiye?
(a) Rice (b) Bread (c) Milk (d) Green leafy vegetables

13. Non-rebreather mask ka flow rate –
(a) 1–2 L (b) 4–6 L (c) 10–15 L (d) 20–30 L

14. TB ke liye rapid molecular test (rifampicin resistance) –
(a) ELISA (b) Widal (c) CBNAAT (d) VDRL

15. Chest drainage bottle ka level –
(a) Bed par (b) Chest level (c) Chest se upar (d) Chest se neeche

16. Sputum culture ke liye best sample –
(a) Evening (b) After mouthwash (c) Early morning (d) After meals

17. Gastric ulcer me pain –
(a) Khane se relief (b) Kabhi nahi (c) Raat 2 baje only (d) Khana khane ke ½–1 hr baad

18. Hiatus hernia patient ko kya avoid karna chahiye?
(a) Elevating HOB (b) Small meals (c) Lying down right after meals (d) Weight loss

19. Paracentesis se pehle patient ko –
(a) NPO 12 hrs (b) Enema do (c) Fluid pilao (d) Bladder empty karao

20. Hepatitis B neonatal prophylaxis –
(a) BCG (b) Antibiotics (c) HBIG + vaccine within 12 hrs (d) Only vaccine at 6 weeks

21. Lactulose ka expected effect –
(a) 2–3 soft stools/day (b) Polyuria (c) Vomiting (d) Constipation

22. Pancreatitis me pain kahan radiate karta hai?
(a) Right shoulder (b) Jaw (c) Left arm (d) Back

23. Steatorrhoea kis organ ki exocrine failure ka sign hai?
(a) Kidney (b) Liver (c) Pancreas (d) Spleen

24. Oliguria ka matlab urine output –
(a) < 100 ml/day (b) < 400 ml/day (c) > 3 L/day (d) < 1 L/day

25. Pyelonephritis ka sign –
(a) Jaundice (b) Haematemesis (c) CVA (flank) tenderness & high fever (d) Only dysuria

26. ESWL kis condition ka treatment hai?
(a) Gallstones only (b) UTI (c) Renal calculi (d) BPH

27. CKD patient me kaunsa laxative avoid karna chahiye?
(a) Psyllium (b) Bisacodyl (c) Milk of magnesia (d) Lactulose

28. Kidney transplant rejection ka sign –
(a) Hypotension only (b) Weight loss (c) ↑ urine output (d) Fever, graft tenderness, ↓ urine output

29. Long-acting peakless insulin –
(a) NPH (b) Glargine (c) Lispro (d) Regular

30. Insulin injection ka fastest absorption site –
(a) Thigh (b) Arm (c) Buttock (d) Abdomen

? Grand Test – Part B (31–60)

31. Myxoedema coma me patient ko garam karne ka tareeka –
(a) Heating pad (b) Hot water bath (c) Passive rewarming – blankets (d) Kuch nahi

32. Hyperparathyroidism me kidney stones rokne ke liye –
(a) Bed rest (b) 3–4 L fluids/day (c) High calcium diet (d) Fluid restriction

33. Pheochromocytoma patient me avoid karna chahiye –
(a) Abdomen palpation (b) Fluids (c) Rest (d) BP monitoring

34. Decerebrate posturing ka matlab –
(a) Brainstem damage (b) Spinal shock (c) Cerebral cortex damage (d) Normal

35. Stroke patient ki homonymous hemianopia me food tray kahan rakhein?
(a) Unaffected side / teach scanning (b) Peeche (c) Upar (d) Affected side

36. Lumbar puncture kis condition me contraindicated hai?
(a) Meningitis suspicion (b) ↑ ICP (c) Fever (d) Headache

37. Myasthenia gravis drug pyridostigmine kab dena chahiye?
(a) Kabhi bhi (b) Khane ke 30–60 min pehle (c) Khane ke 2 hrs baad (d) Raat ko only

38. GBS me paralysis ka pattern –
(a) Descending (b) Only face (c) Ascending symmetrical (d) One side only

39. Trigeminal neuralgia ka drug of choice –
(a) Levodopa (b) Carbamazepine (c) Phenytoin (d) Morphine

40. Plaster cast me "hot spot" ka matlab –
(a) Good circulation (b) Infection under cast (c) Fracture healed (d) Normal drying

41. Buck's traction kis type ka hai?
(a) Manual (b) Skeletal (c) Halo (d) Skin

42. Osteoporosis ke liye best exercise –
(a) Bed rest (b) Weight-bearing walking (c) Swimming (d) Cycling only

43. Amputation stump bandaging technique –
(a) Spiral loose (b) Figure-of-8, distal to proximal (c) Not needed (d) Circular proximal to distal

44. Superficial (1st degree) burn ka example –
(a) Electric burn (b) Sunburn (c) Bone exposed (d) Charred skin

45. Burns me "Curling's ulcer" kahan hota hai?
(a) Eye (b) Mouth (c) Stomach / duodenum (d) Skin

46. Electrical burn me main danger –
(a) Cardiac arrhythmia (b) Itching (c) Alopecia (d) Hypothermia

47. Severe anaemia patient ke transfusion ke liye best component –
(a) Platelets (b) FFP (c) Whole blood (d) Packed RBCs

48. ITP me kya kam hota hai?
(a) WBC (b) RBC (c) Plasma (d) Platelets

49. Haemophilia patient ke liye safe painkiller –
(a) Ibuprofen (b) Diclofenac IM (c) Aspirin (d) Paracetamol

50. HIV ka window period –
(a) 1 day (b) 1 year (c) 5 years (d) ~3–12 weeks

51. Leprosy me MDT ka kaunsa drug skin ko reddish-brown karta hai?
(a) Rifampicin (b) Clofazimine (c) Dapsone (d) Isoniazid

52. Malaria me paroxysm ka sequence –
(a) Only fever (b) Sweating → hot → cold (c) Cold → hot → sweating (d) Hot → cold → sweating

53. CAUTION signs me "N" ka matlab –
(a) Numbness (b) Nagging cough / hoarseness (c) Night sweats (d) Nausea

54. Chemotherapy se alopecia –
(a) Never happens (b) Usually temporary (c) Permanent always (d) Only in eyebrows

55. Brachytherapy patient ke paas pregnant nurse –
(a) Allowed (b) Only at night (c) Allowed with gloves (d) Not allowed

56. Cataract ka first-choice treatment –
(a) Laser for all (b) Eye drops (c) Glasses only (d) Surgery with IOL

57. Acute angle-closure glaucoma ka sign –
(a) Floaters (b) Severe eye pain, halos, red eye (c) Painless gradual loss (d) Itching

58. Conductive deafness me Weber test lateralize hota hai –
(a) Midline (b) Affected ear (c) Normal ear (d) Both

59. OP poisoning me pupils –
(a) Pin-point (miosis) (b) Dilated (c) Unequal (d) Normal

60. Heat stroke ka immediate management –
(a) Paracetamol (b) Warm blankets (c) Hot drinks (d) Rapid cooling

Answer Key – Grand Test

1-b · 2-c · 3-b · 4-d · 5-d · 6-b · 7-d · 8-b · 9-c · 10-d · 11-d · 12-d · 13-c · 14-c · 15-d · 16-c · 17-d · 18-c · 19-d · 20-c · 21-a · 22-d · 23-c · 24-b · 25-c · 26-c · 27-c · 28-d · 29-b · 30-d · 31-c · 32-b · 33-a · 34-a · 35-a · 36-b · 37-b · 38-c · 39-b · 40-b · 41-d · 42-b · 43-b · 44-b · 45-c · 46-a · 47-d · 48-d · 49-d · 50-d · 51-b · 52-c · 53-b · 54-b · 55-d · 56-d · 57-b · 58-b · 59-a · 60-d



All the best!

"**Hard work + Daily revision + MCQ practice = Selection pakka.**"
Ye notes baar-baar padho, galtiyon ko note karo, aur exam me shaant dimaag se paper do.
– Sister Asha

— MSN Notes · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —